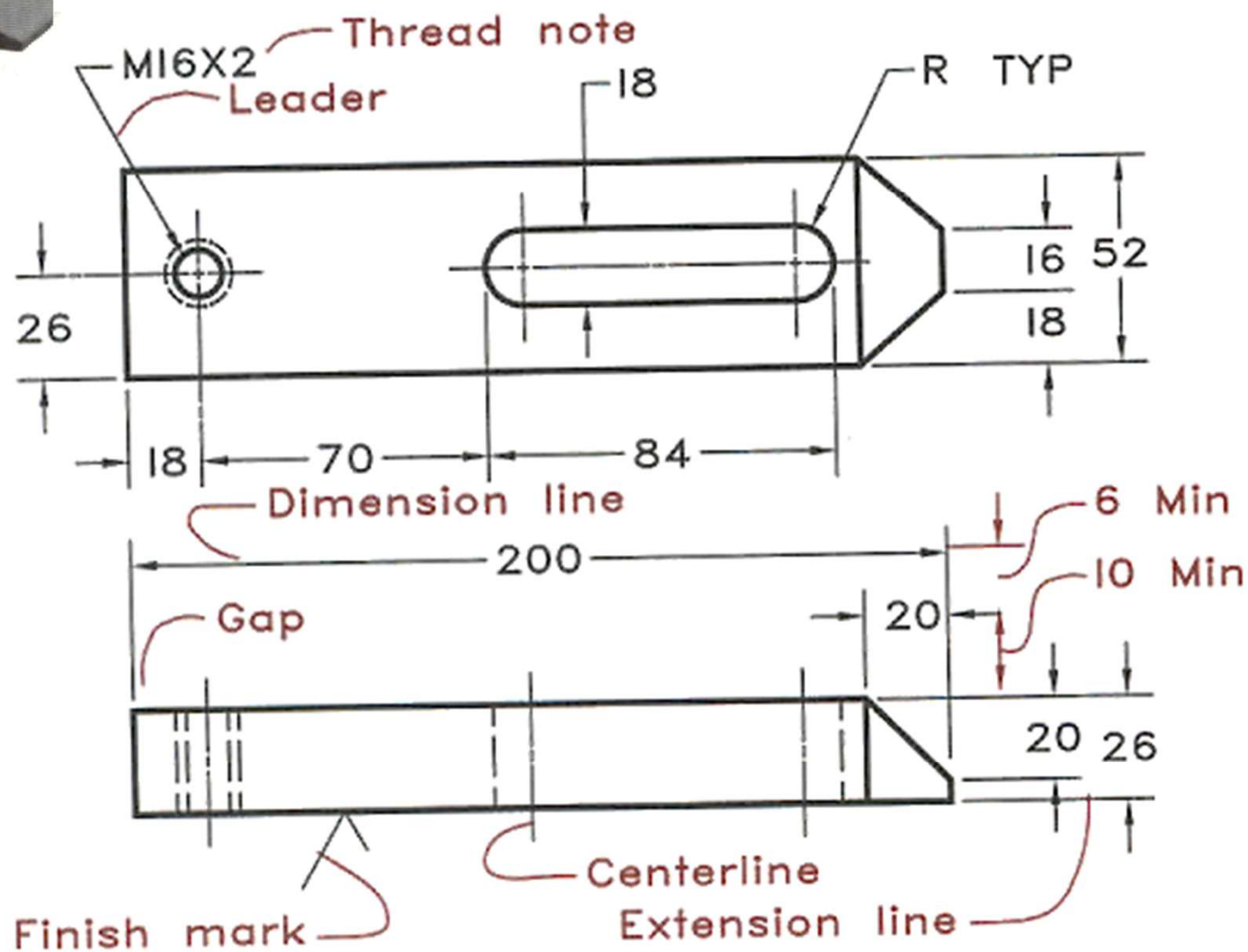
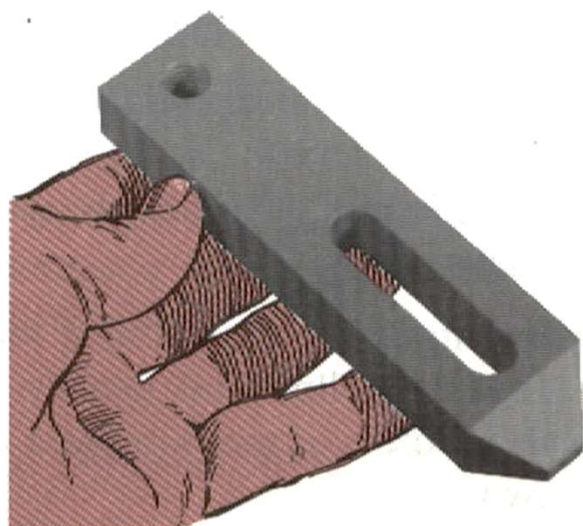
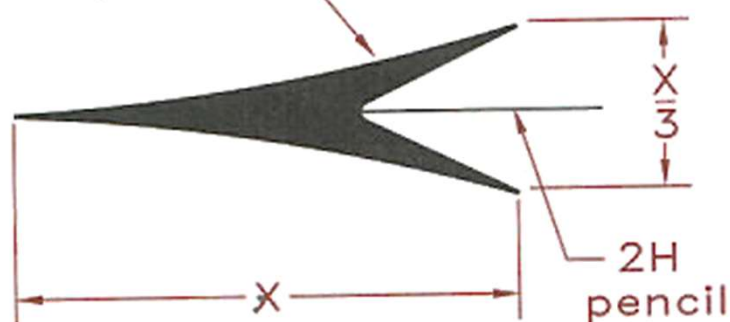




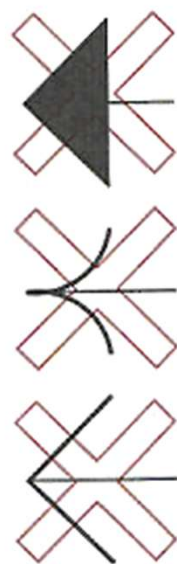
Use $\frac{1}{8}$ in./ (3 mm) letters & numerals

ARROWHEADS

Arrow size = Letter height (X)
Usually .125" or 3 mm
F pencil

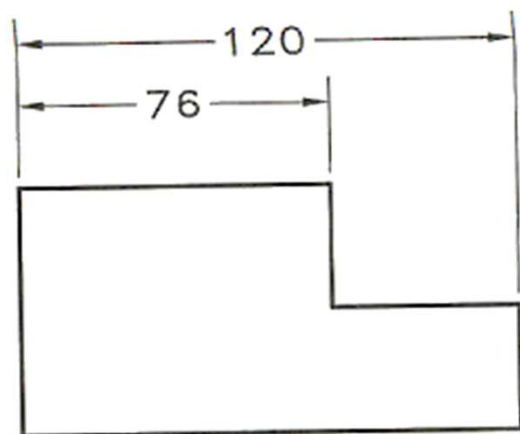


Poor arrowheads



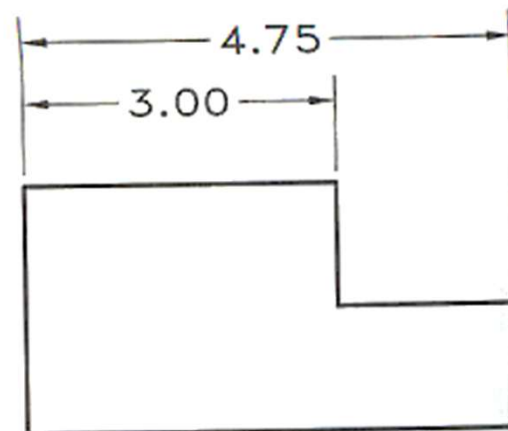
MILLIMETERS VS. INCHES

Millimeters are whole numbers, preferably even numbers



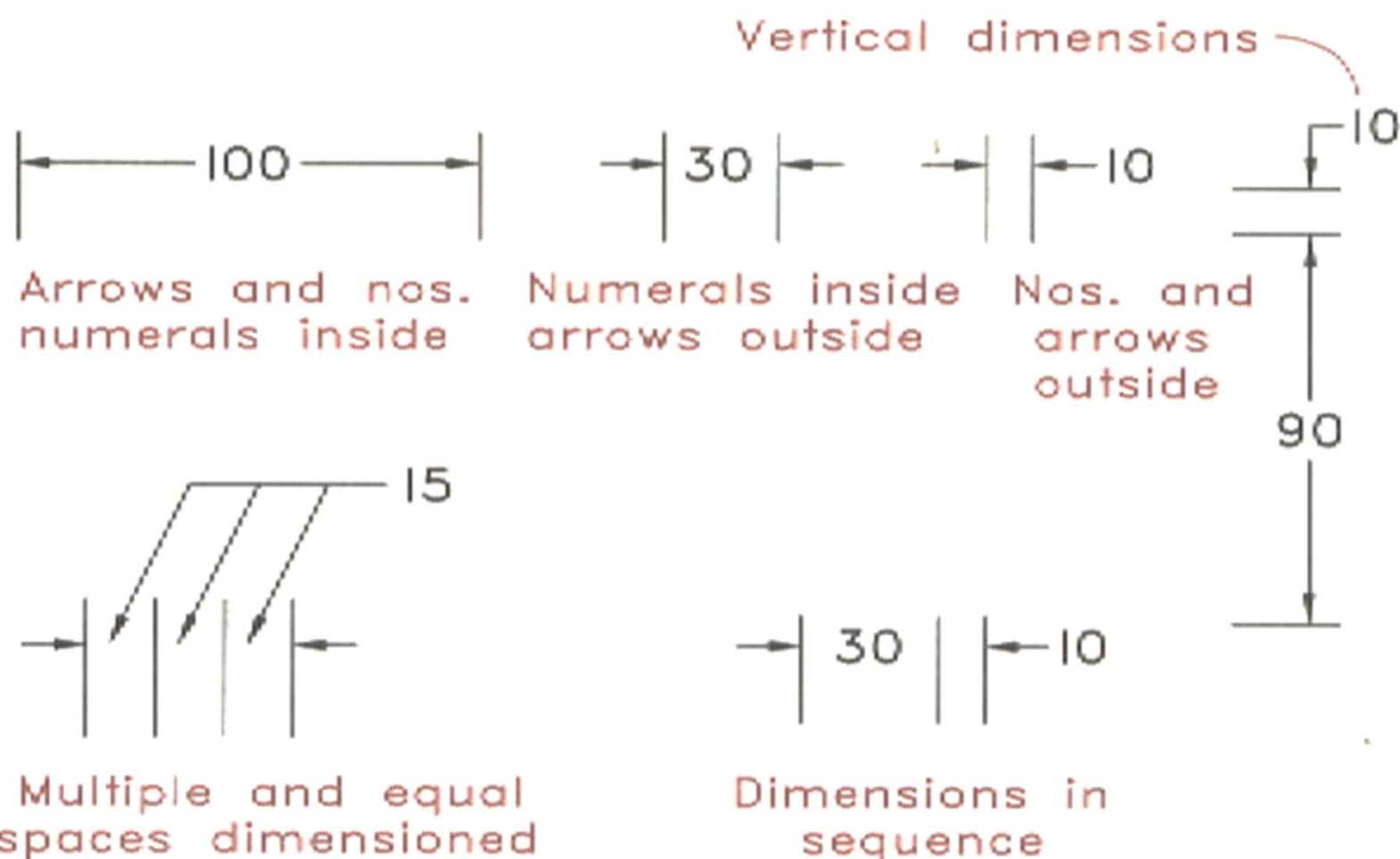
A. METRIC UNITS

Inches have two decimal places



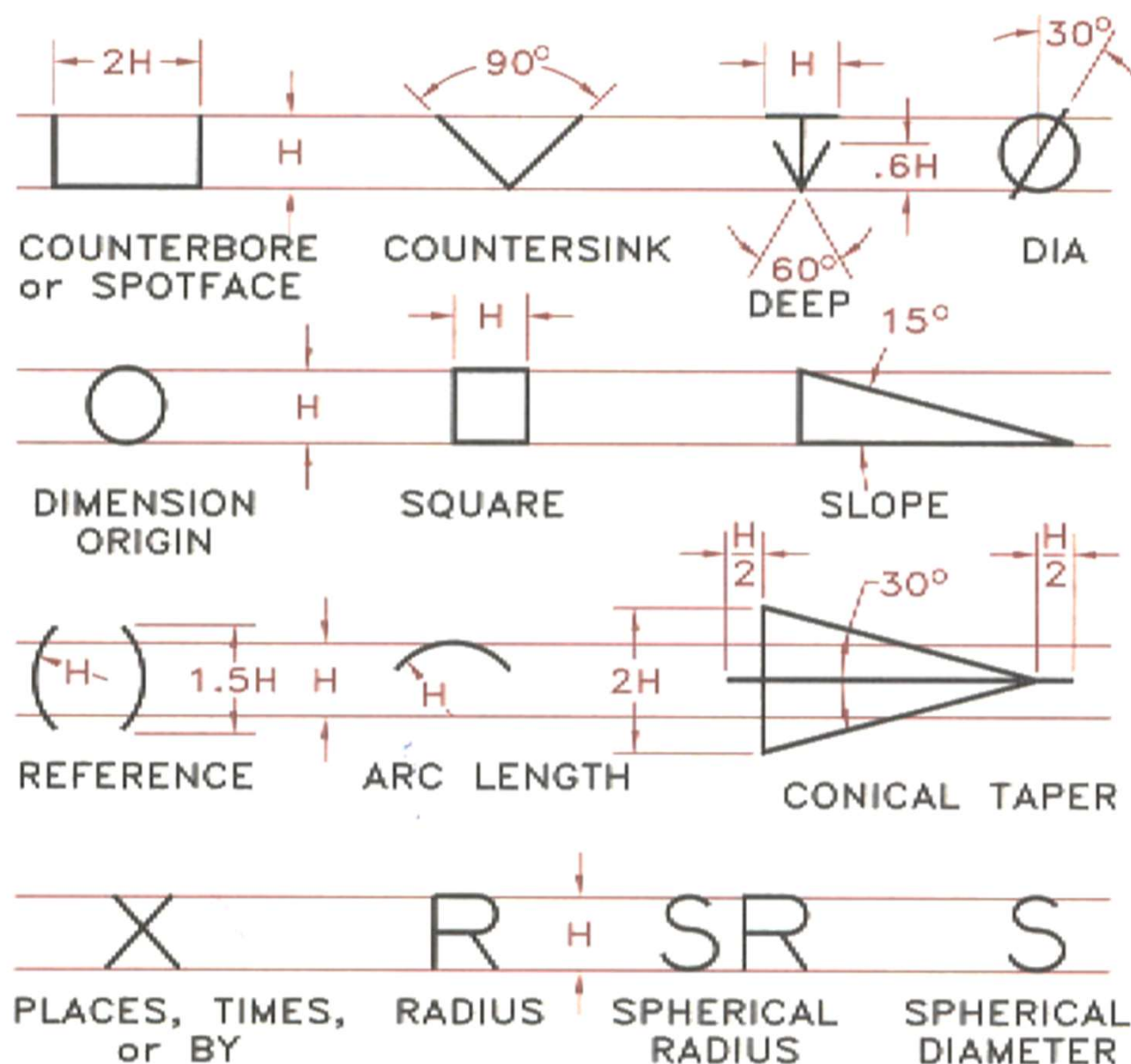
B. ENGLISH UNITS

DIMENSIONS IN TIGHT SPACES



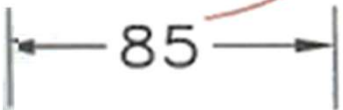
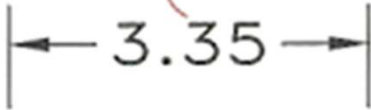
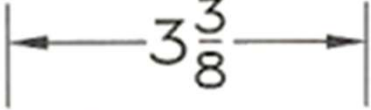
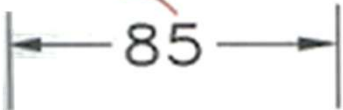
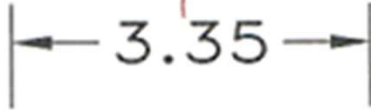
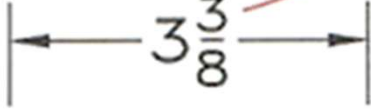
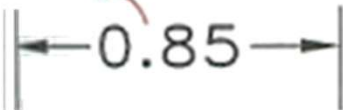
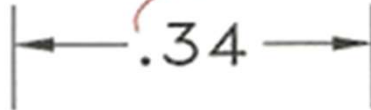
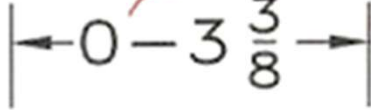
When space permits, place numerals and arrows inside extension lines. For smaller spaces use other placements, as shown.

DIMENSIONING SYMBOLS

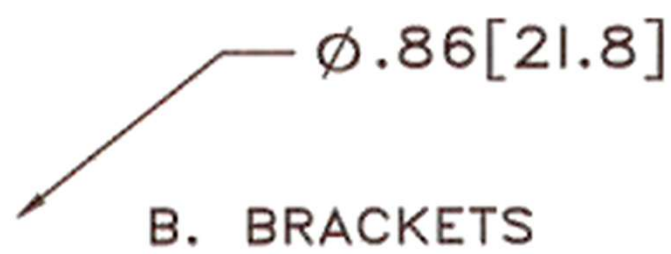
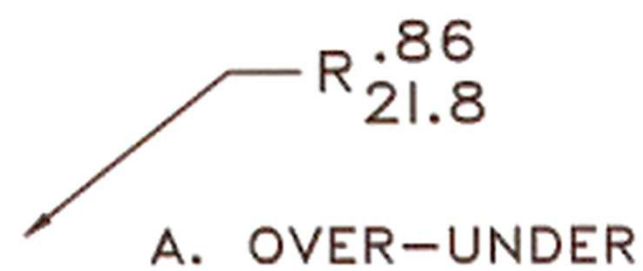
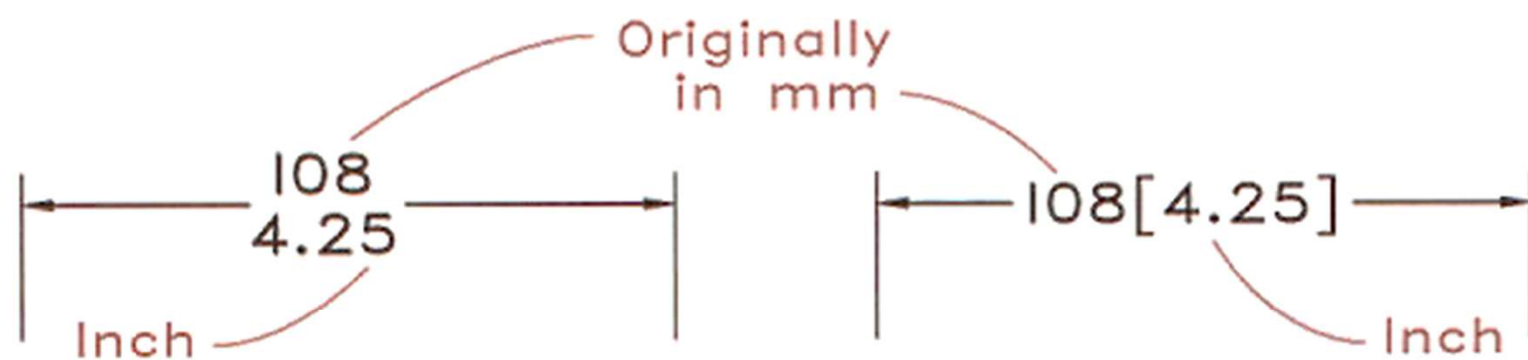
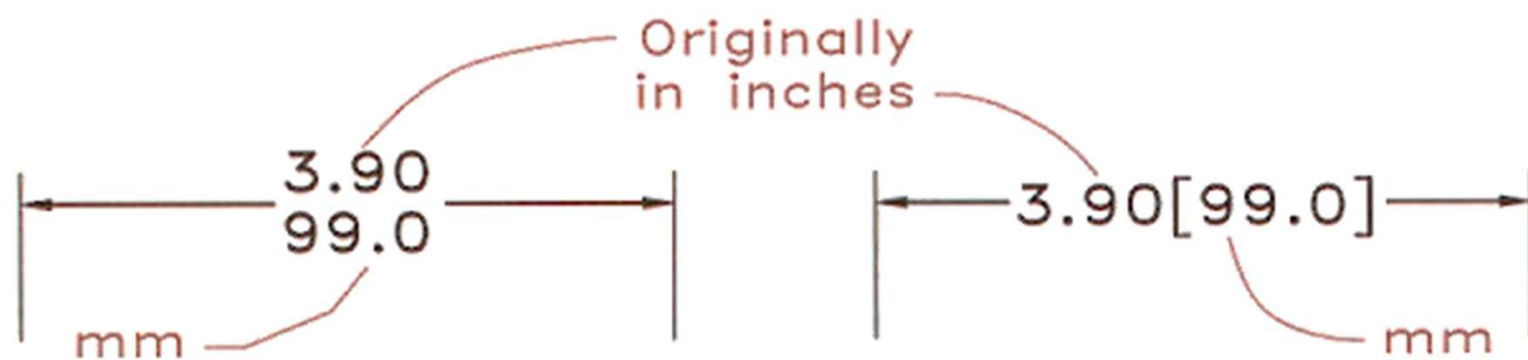


These symbols can be used instead of words to dimension parts. Their proportions are based on the letter height, H , which usually is $\frac{1}{8}$ inch.

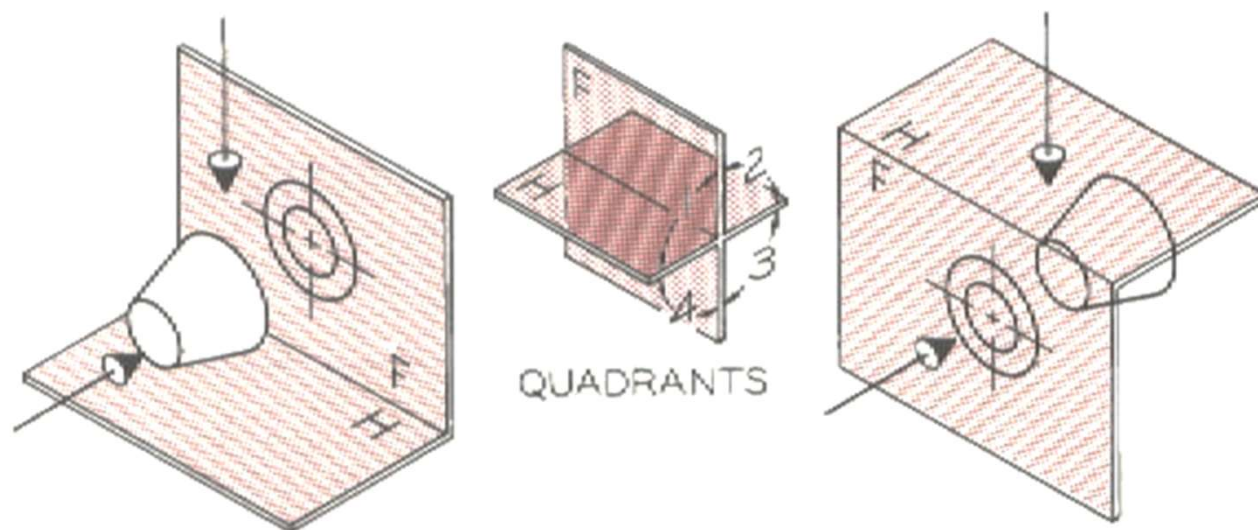
DIMENSIONING UNITS

Decimal fractions	Poor 85.2	Poor 3.4	Common fractions twice as tall
			
Whole nos.	2 decimals	Common fractions	
Poor 85 mm	Poor 3.35"	Poor 3 3/8"	
			
Omit mm	Omit inches	Omit inches	
Poor .85	Poor 0.34	Optional 3 3/8	
			
Give zero	Omit zero	Zero optional	
MILLIMETERS	DECIMAL IN.	FRACTIONAL IN.	

DUAL DIMENSIONING



FIRST- AND THIRD-ANGLE PROJECTION

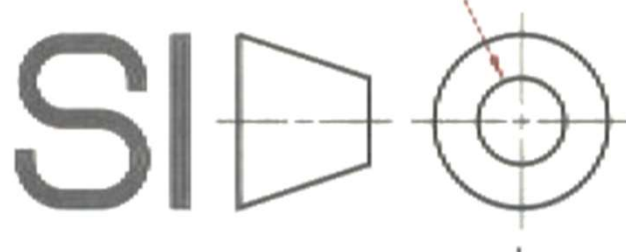


A 1ST ANGLE PROJECTION B 3RD ANGLE PROJECTION

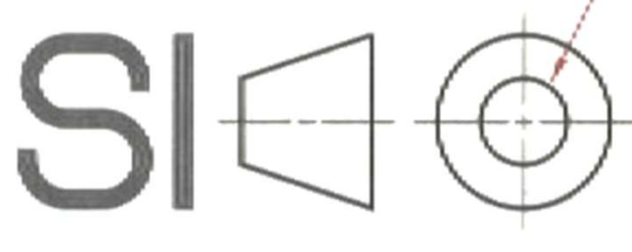
THE SI SYMBOL

U.S. projection,
where circle is
visible here

First angle, where circle
is visible, but would be
hidden in U.S. system

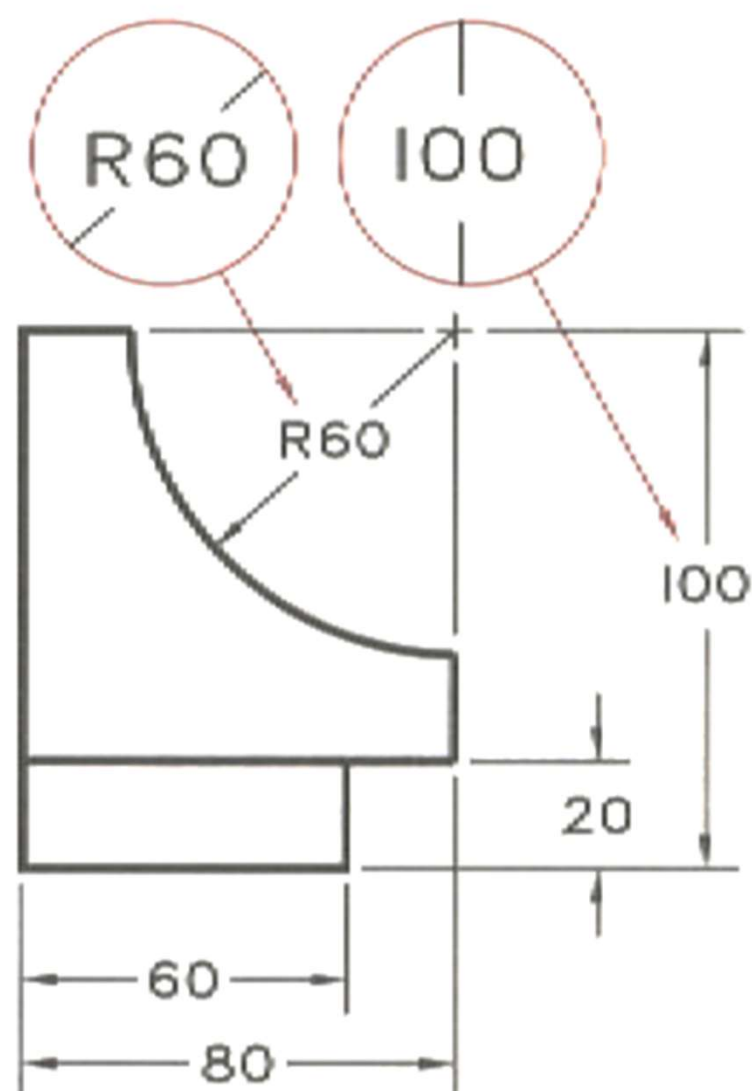


A. METRIC UNITS &
THIRD-ANGLE OF
PROJECTION

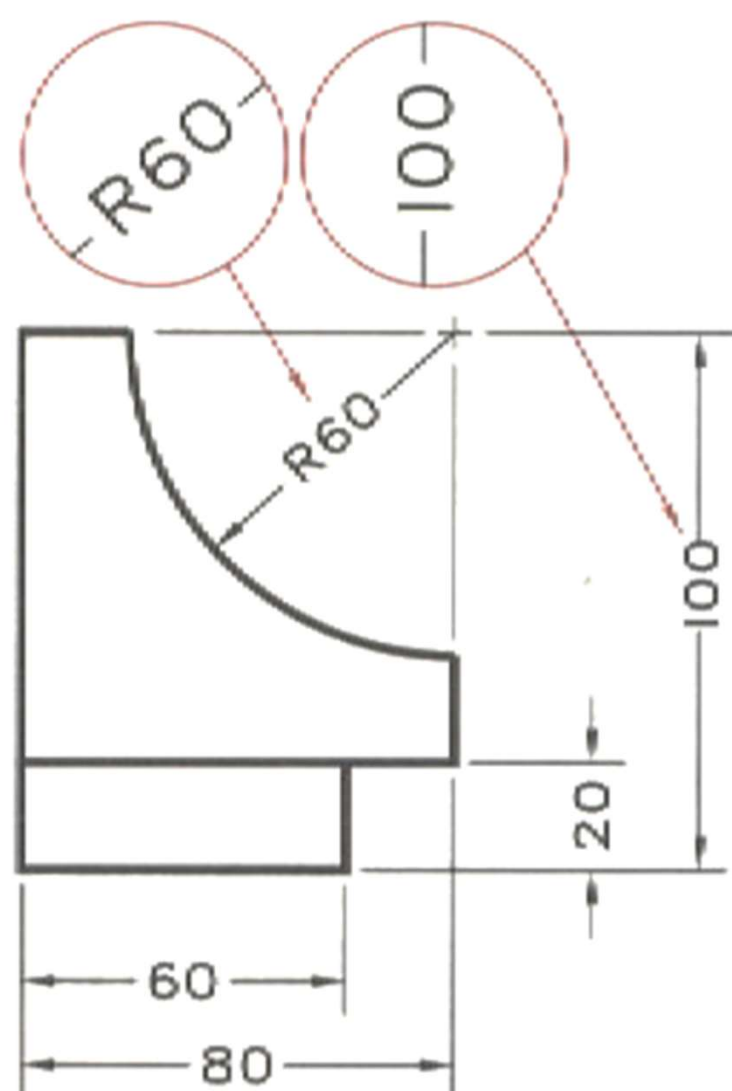


B. METRIC UNITS &
FIRST-ANGLE OF
PROJECTION

UNIDIRECTIONAL AND ALIGNED DIMENSIONS

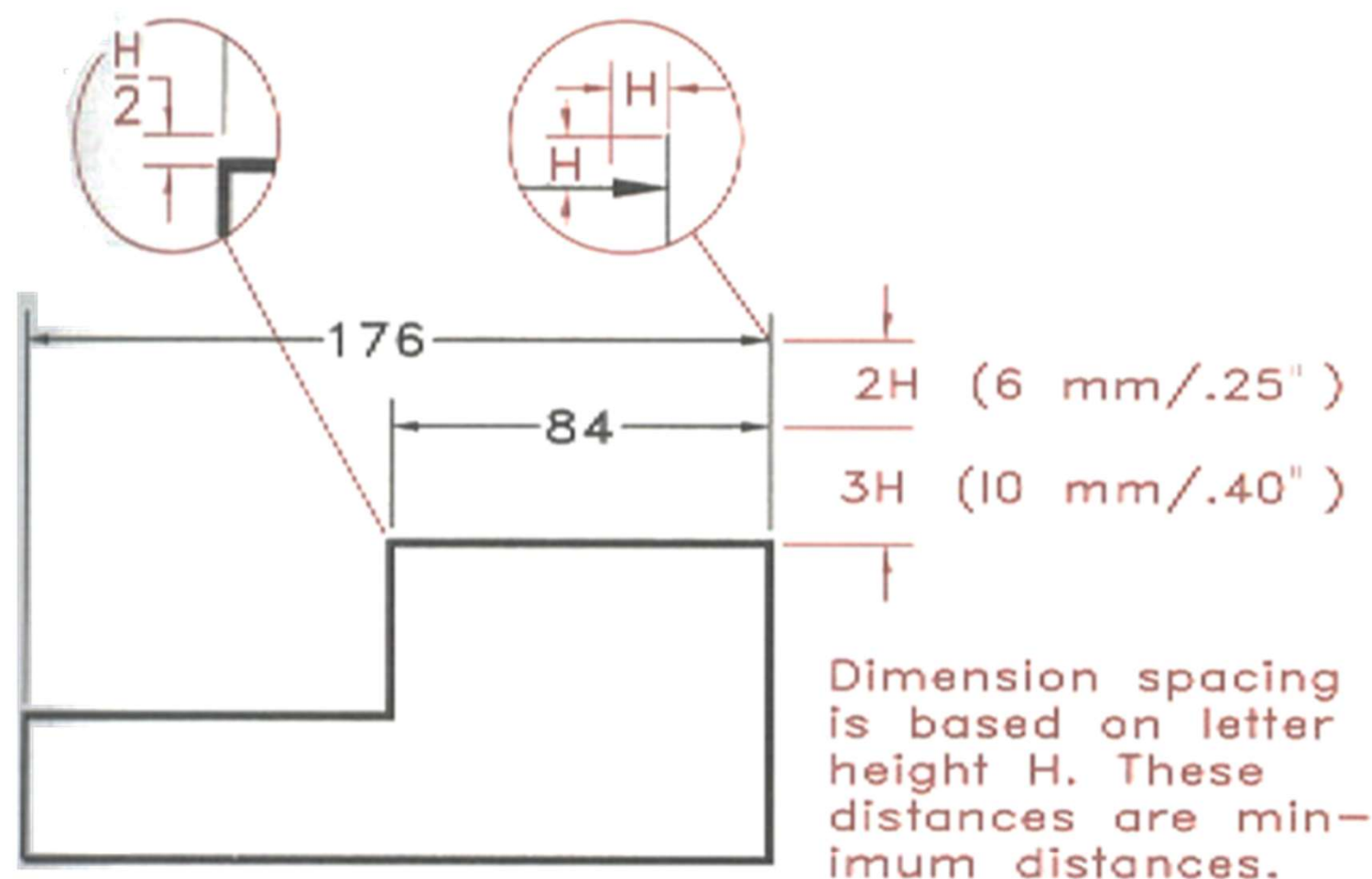


A. UNIDIRECTIONAL



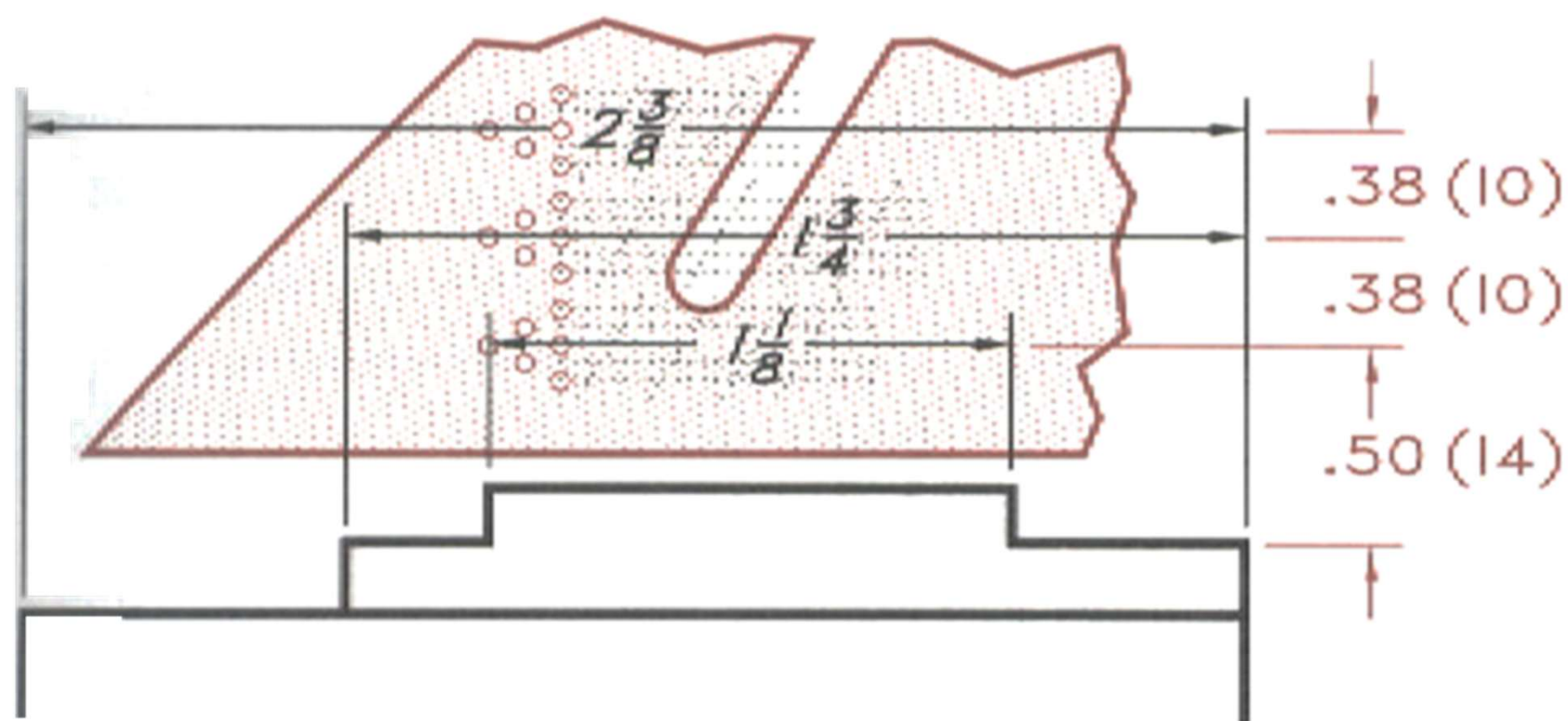
B. ALIGNED

PLACEMENT OF DIMENSIONS



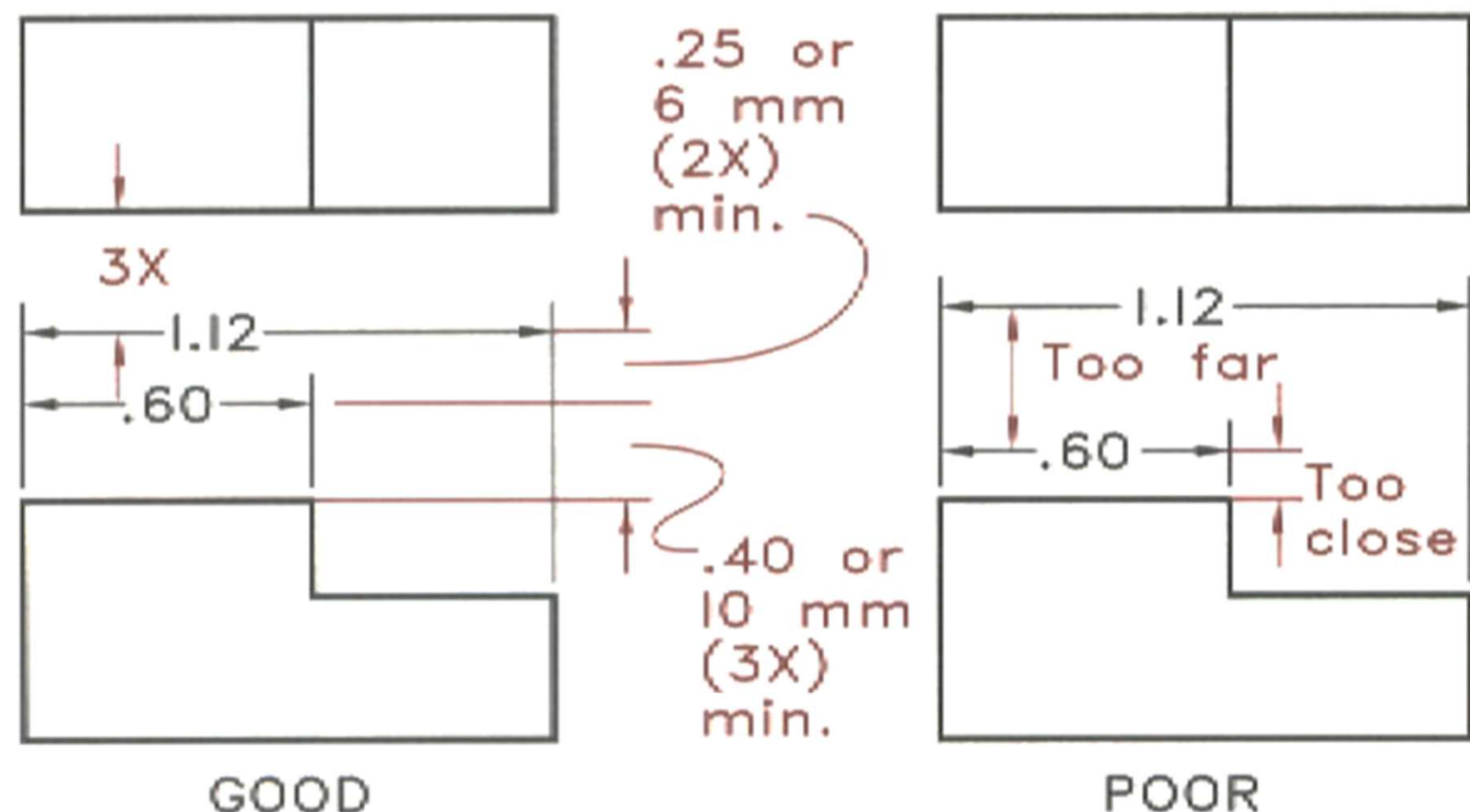
Place dimensions on a view as shown here, where all dimensioning geometry is based on the letter height (H) used.

COMMON-FRACTION DIMENSIONS



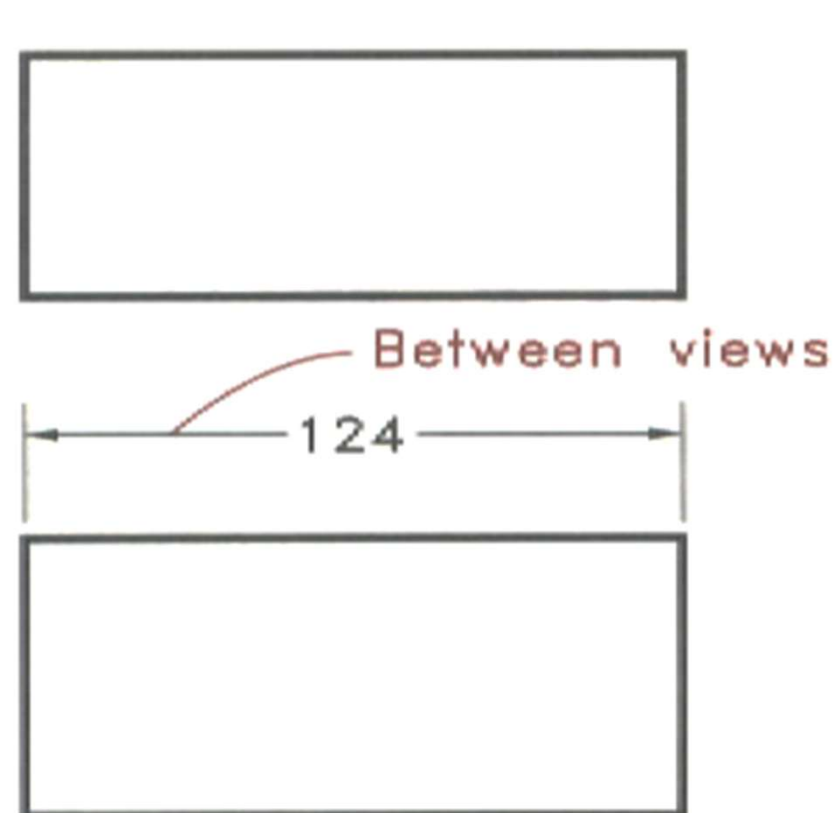
Draw guidelines for common-fraction dimensions by aligning the center holes in the Braddock-Rowe triangle with the dimension line.

RULE 1: First row 3 times letter height from part, minimum. (X=Letter height)

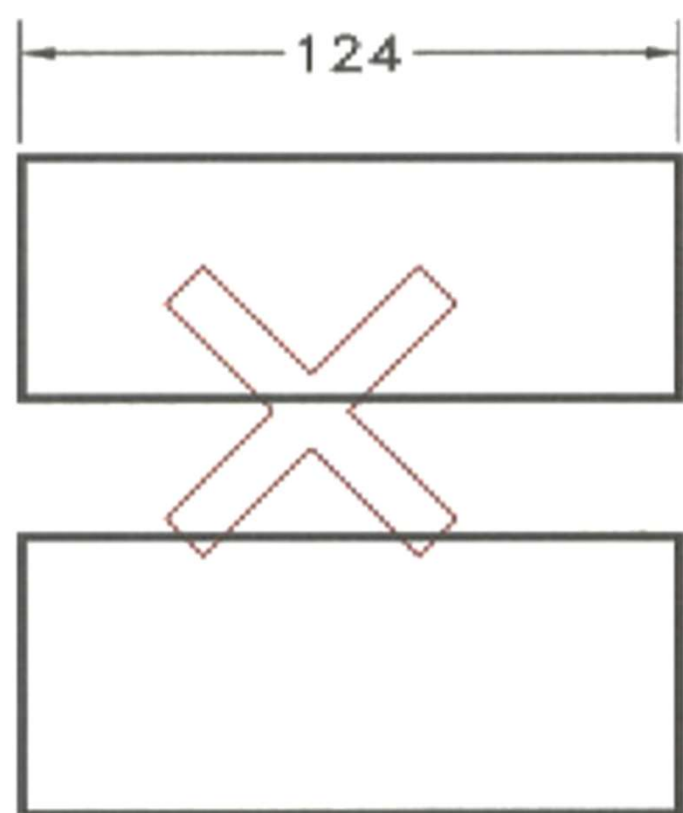


Place the first row of dimensions at least three times the letter height from the object. Successive rows should be at least two times the letter height apart.

RULE 2: Place dimensions between the views



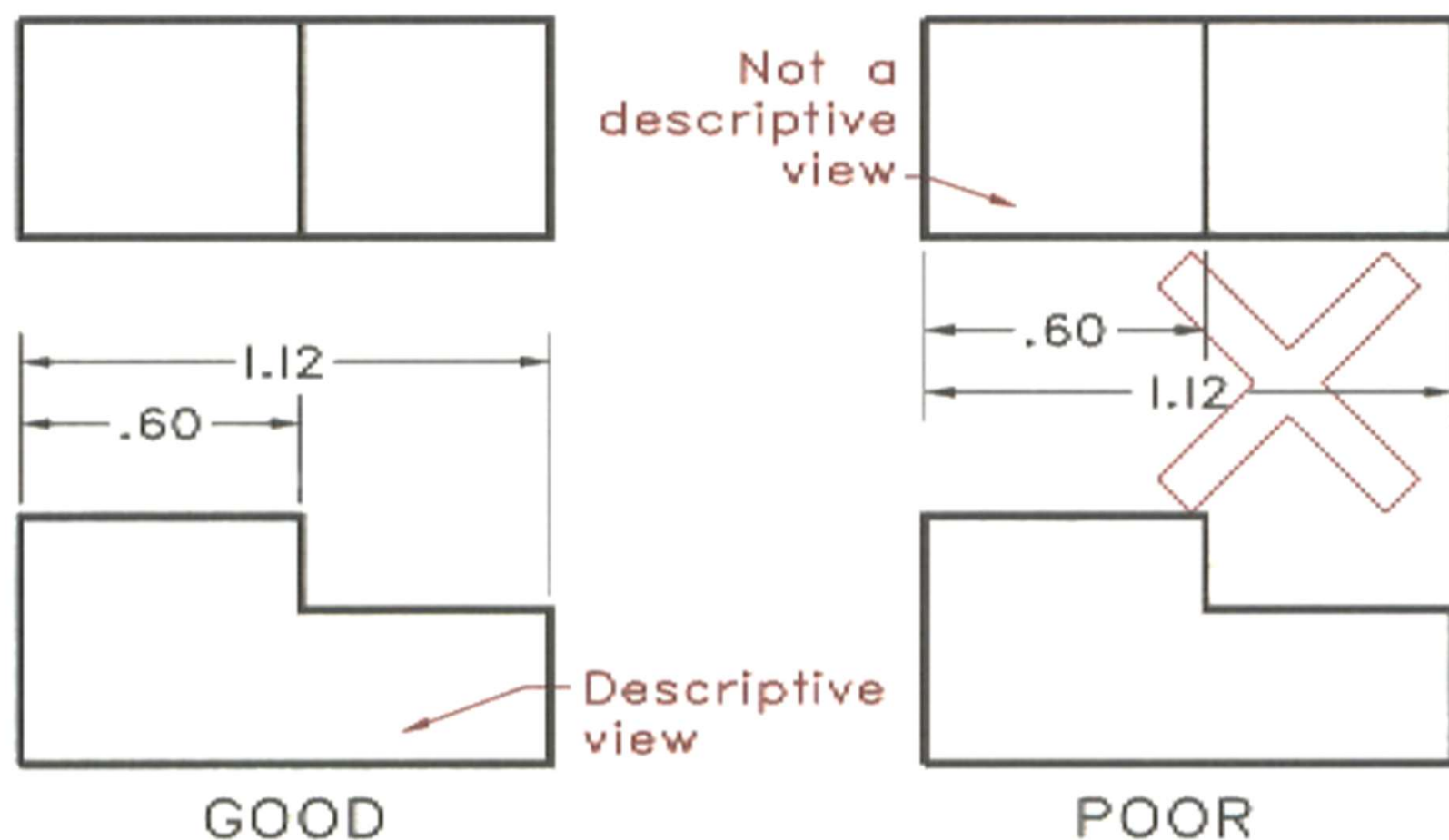
GOOD



POOR

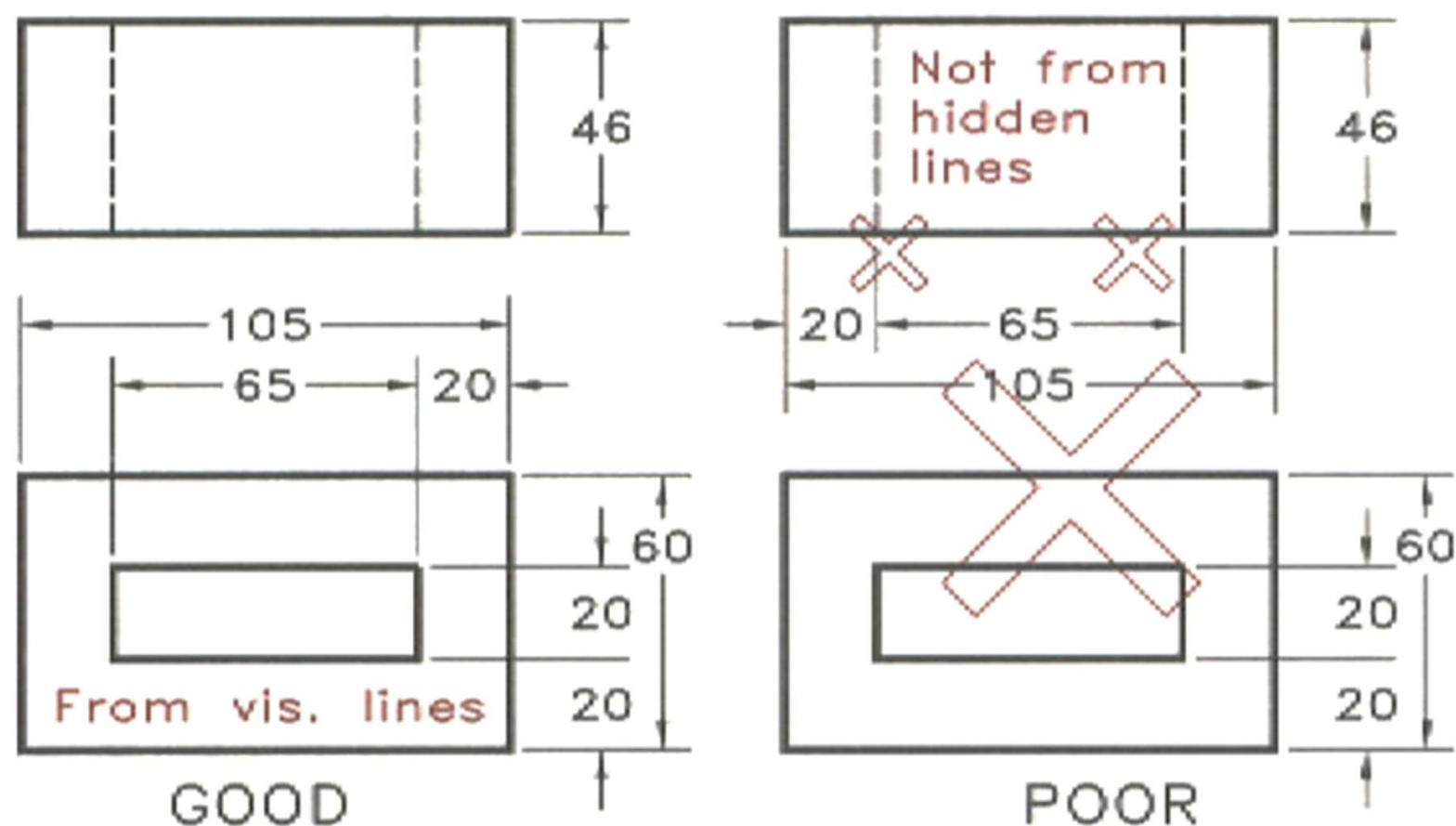
Place dimensions between the views sharing these dimensions.

RULE 3: Dimension the most descriptive views



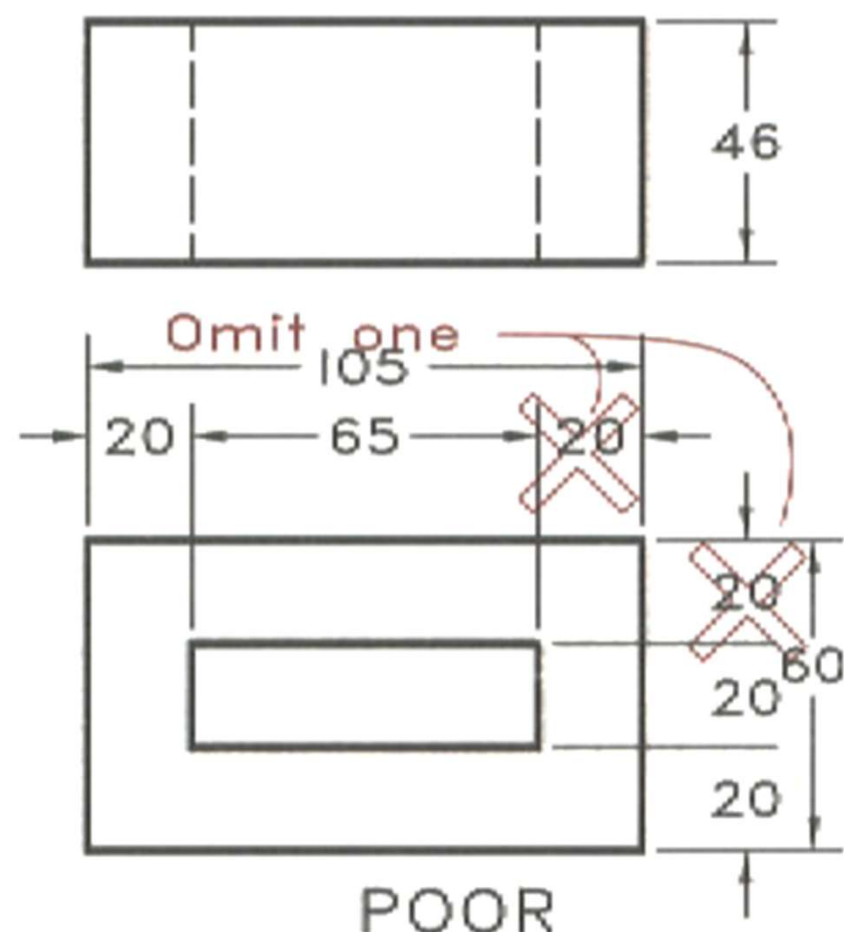
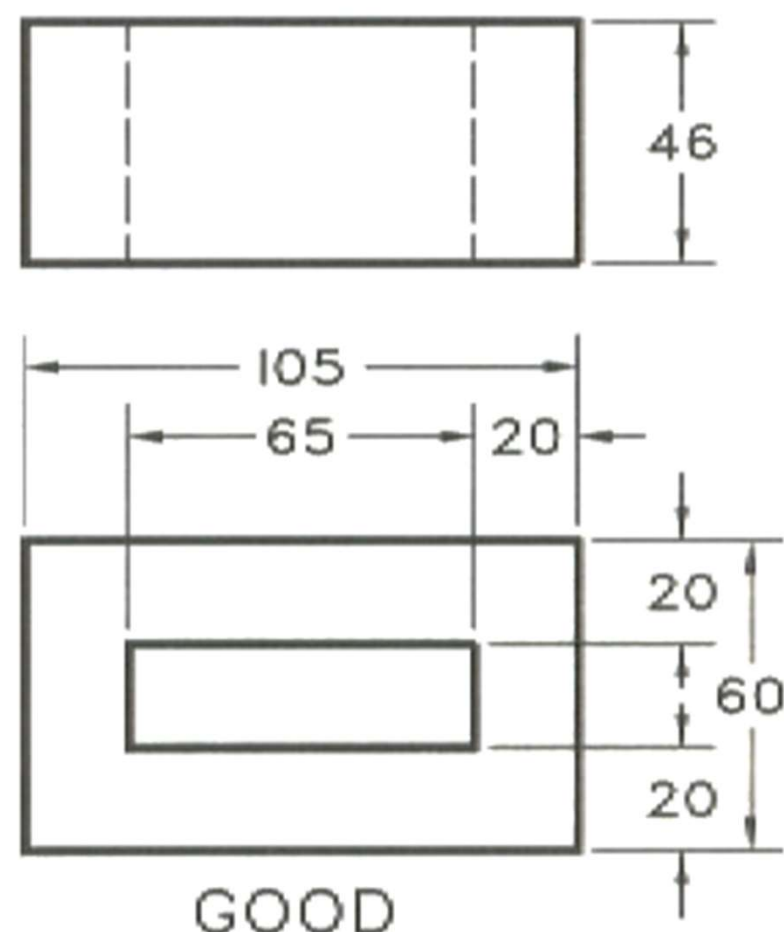
Place dimensions on the most descriptive views of an object.

RULE 4: Dimension from visible lines, not hidden lines



Dimension visible features, not hidden features.

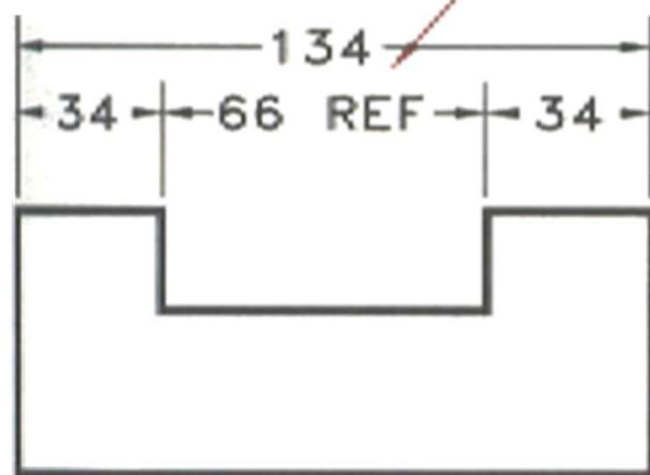
RULE 5: Give an overall dimension and omit one of the chain dimensions



Leave the last dimension blank in a chain of dimensions when you also give an overall dimension.

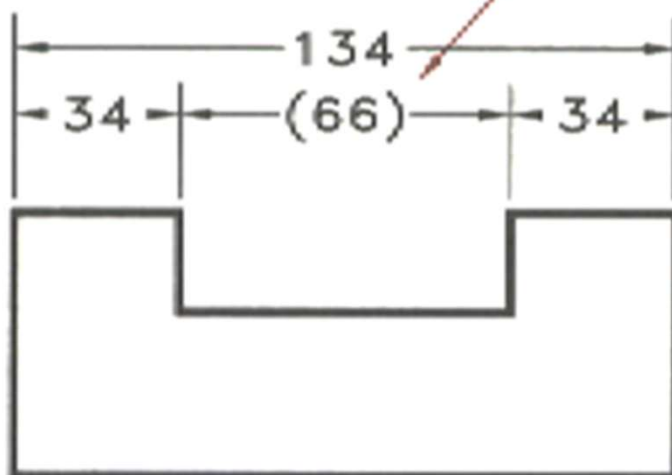
RULE 5 (DEVIATION): When one chain dimension is not omitted, mark one dimension as a reference dimension

Reference dimension marked "REF"



GOOD

Reference dimension in parentheses

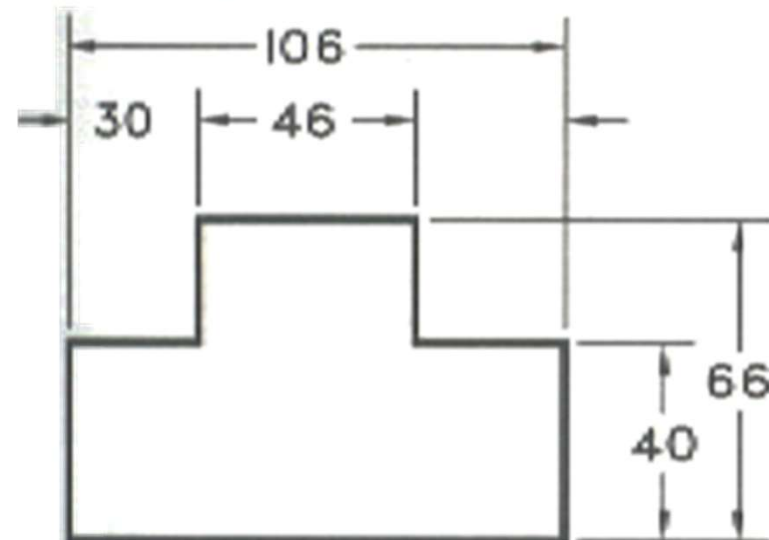


GOOD

If you give all dimensions in a chain, mark the reference dimension (the one that would be omitted) with REF or place it in parentheses. Giving a reference dimension is a way of eliminating mathematical calculations in the shop.

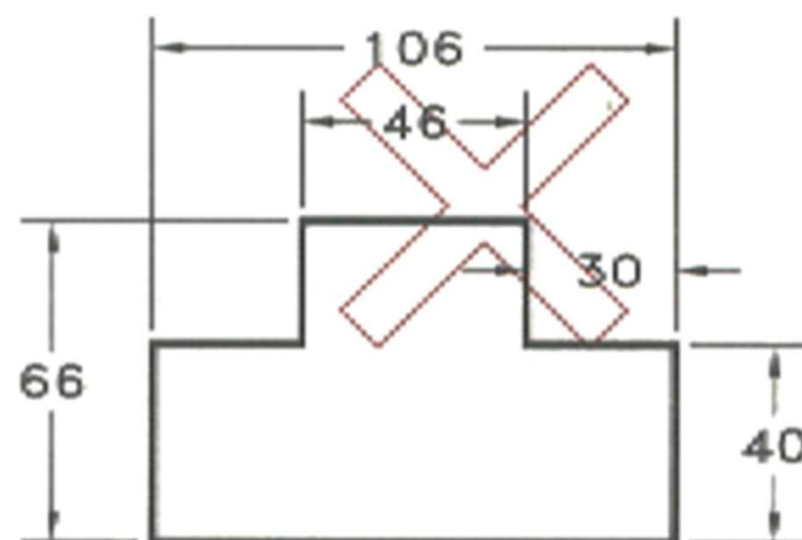
RULE 6: Organize and align dimensions for ease of reading

Dimensions aligned and grouped



GOOD

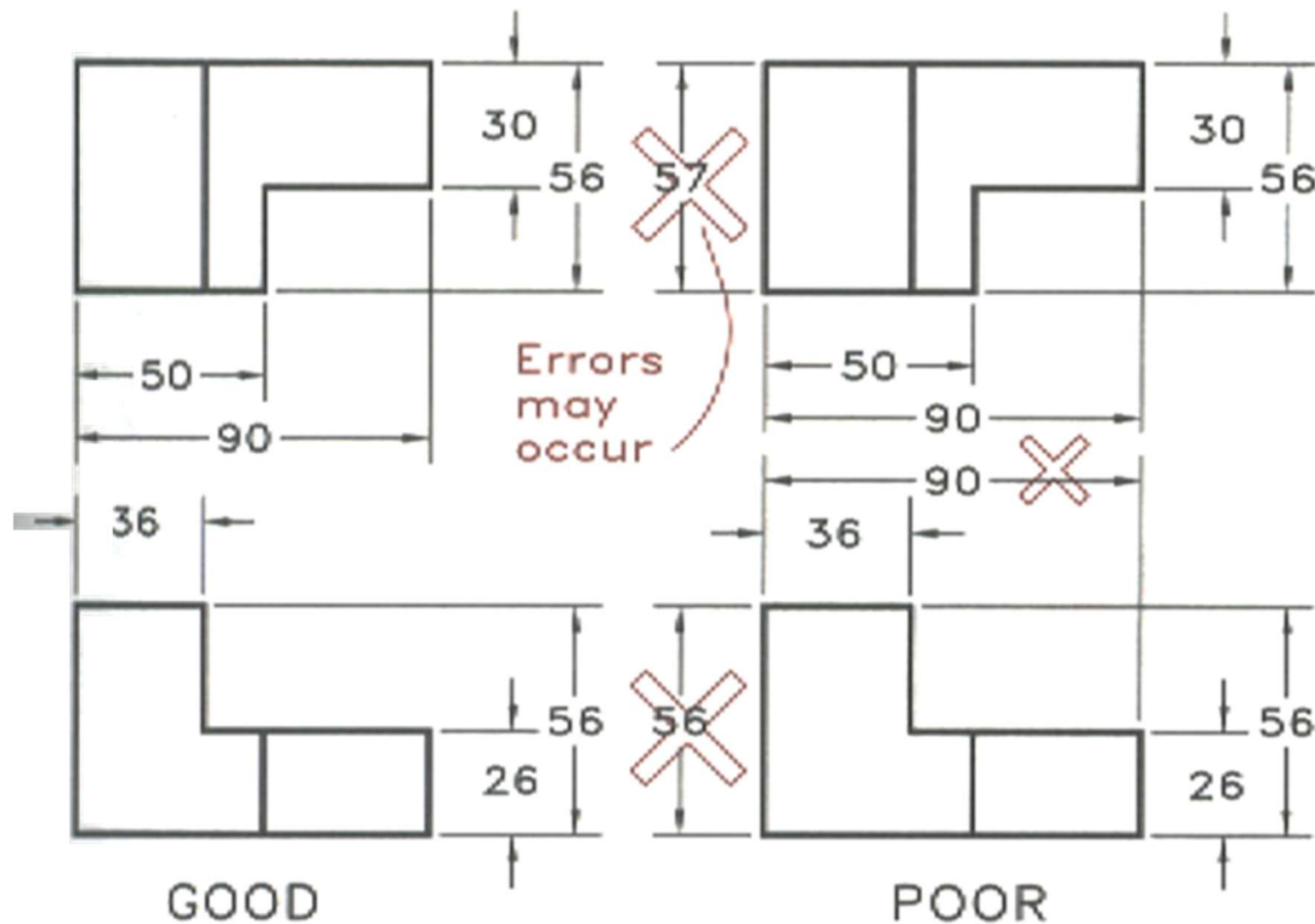
Dimensions not grouped or aligned



POOR

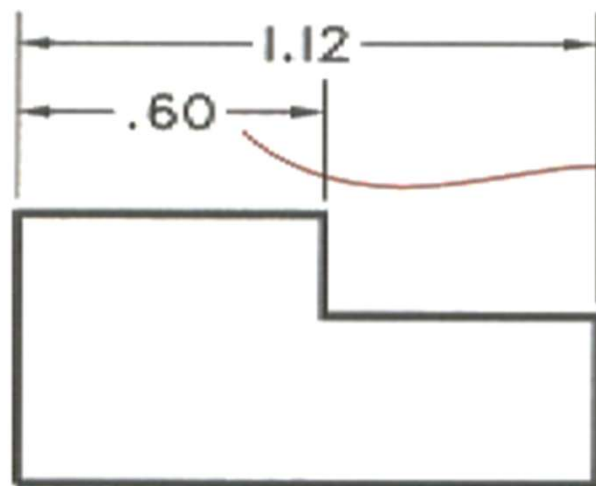
Place dimensions in well-organized lines for uncluttered drawings.

RULE 7: Do not repeat dimensions



To avoid errors or confusion, do not duplicate dimensions on a drawing.

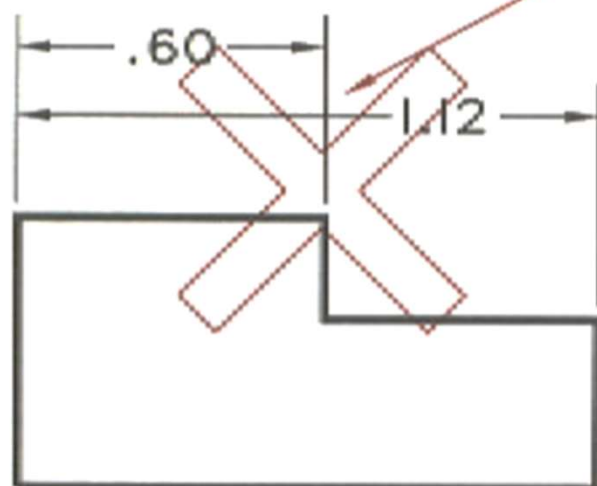
RULE 8: Dimension lines should not cross other lines



Start
with
short
dimen-
sion

GOOD

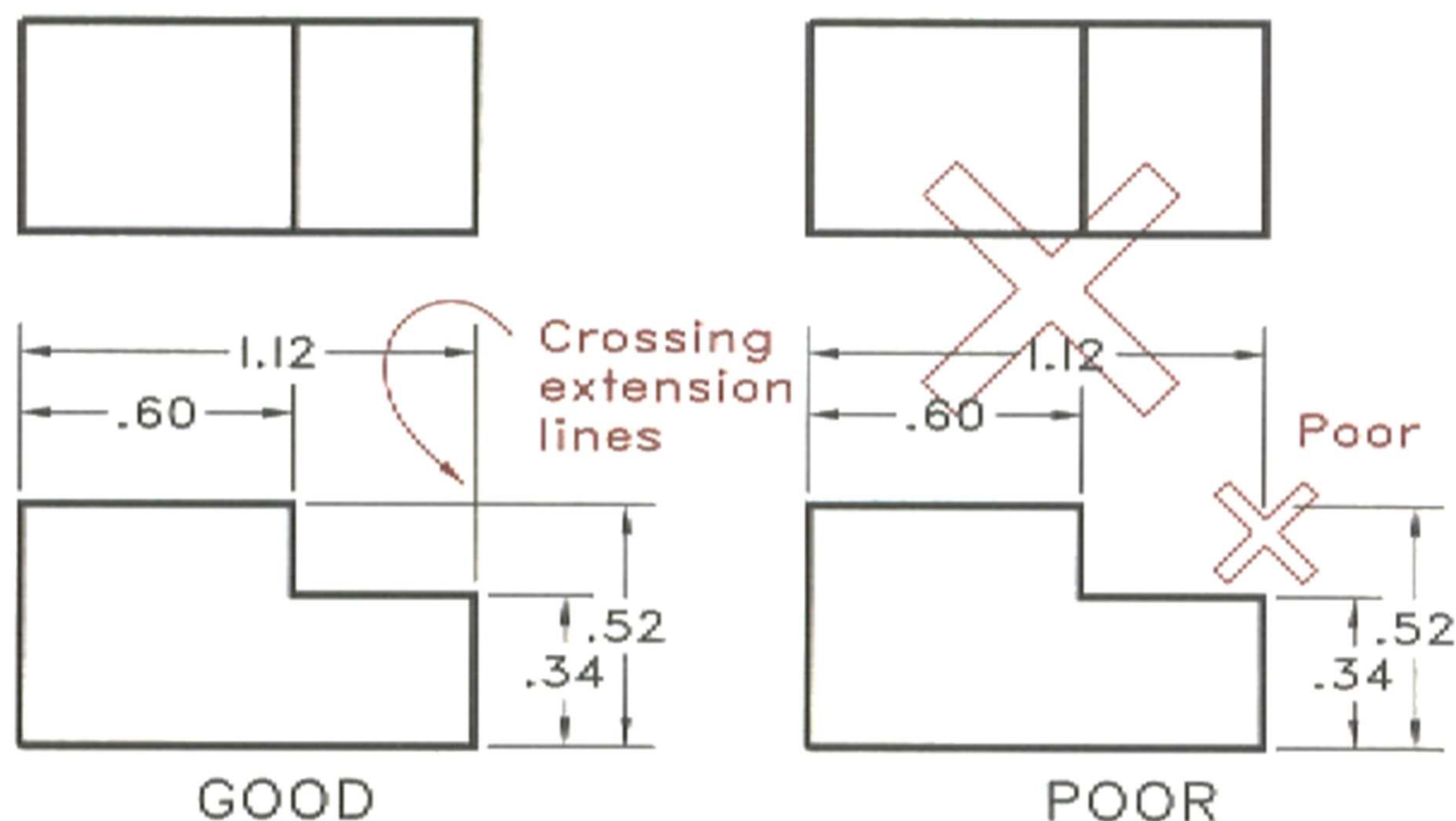
Dimension line crosses



POOR

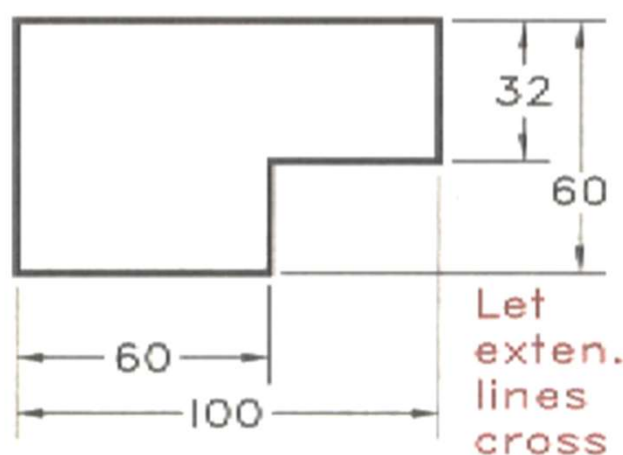
Dimension lines should not cross any other lines unless absolutely necessary.

RULE 9: Extension lines may cross other lines if they must

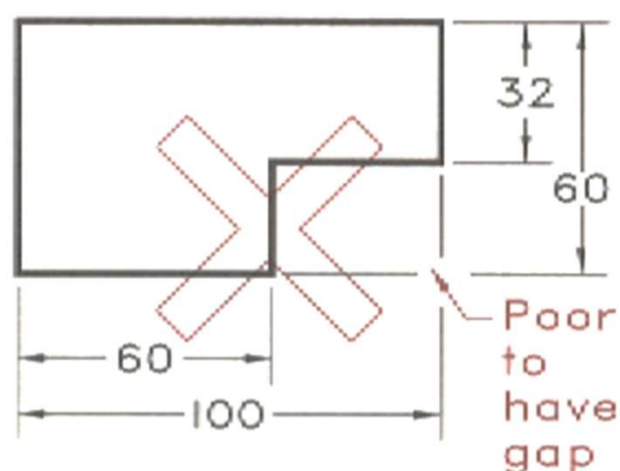


Extension lines may cross other extension lines or object lines if necessary.

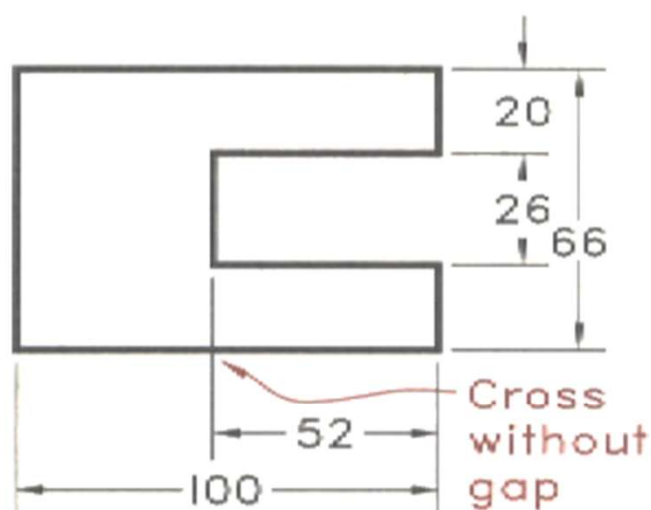
RULE 9 (Cont.): Crossing extension lines



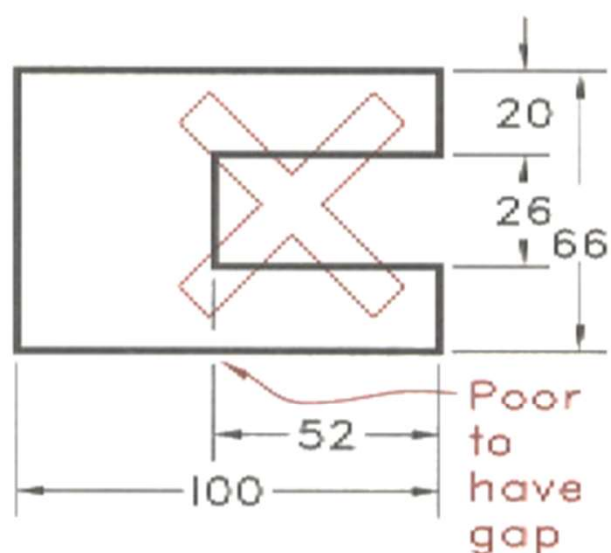
A. GOOD



B. POOR



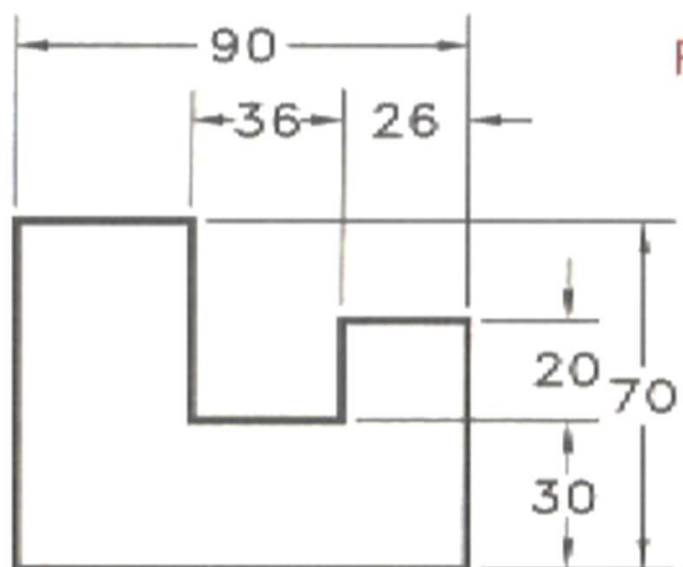
C. GOOD



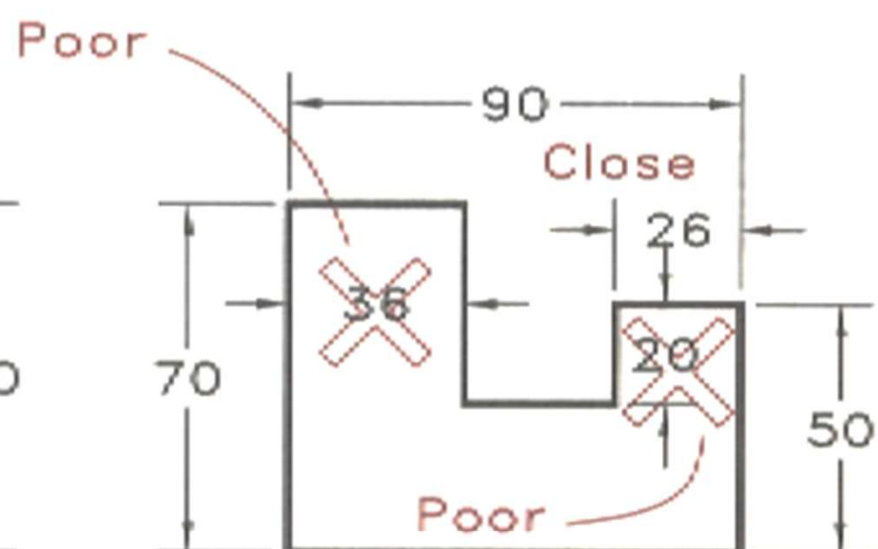
D. POOR

Leave a small gap from the edges of an object to extension lines that extend from them. Do not leave gaps where extension lines cross object lines or other extension lines.

RULE 10: Do not place dimensions within the views unless necessary



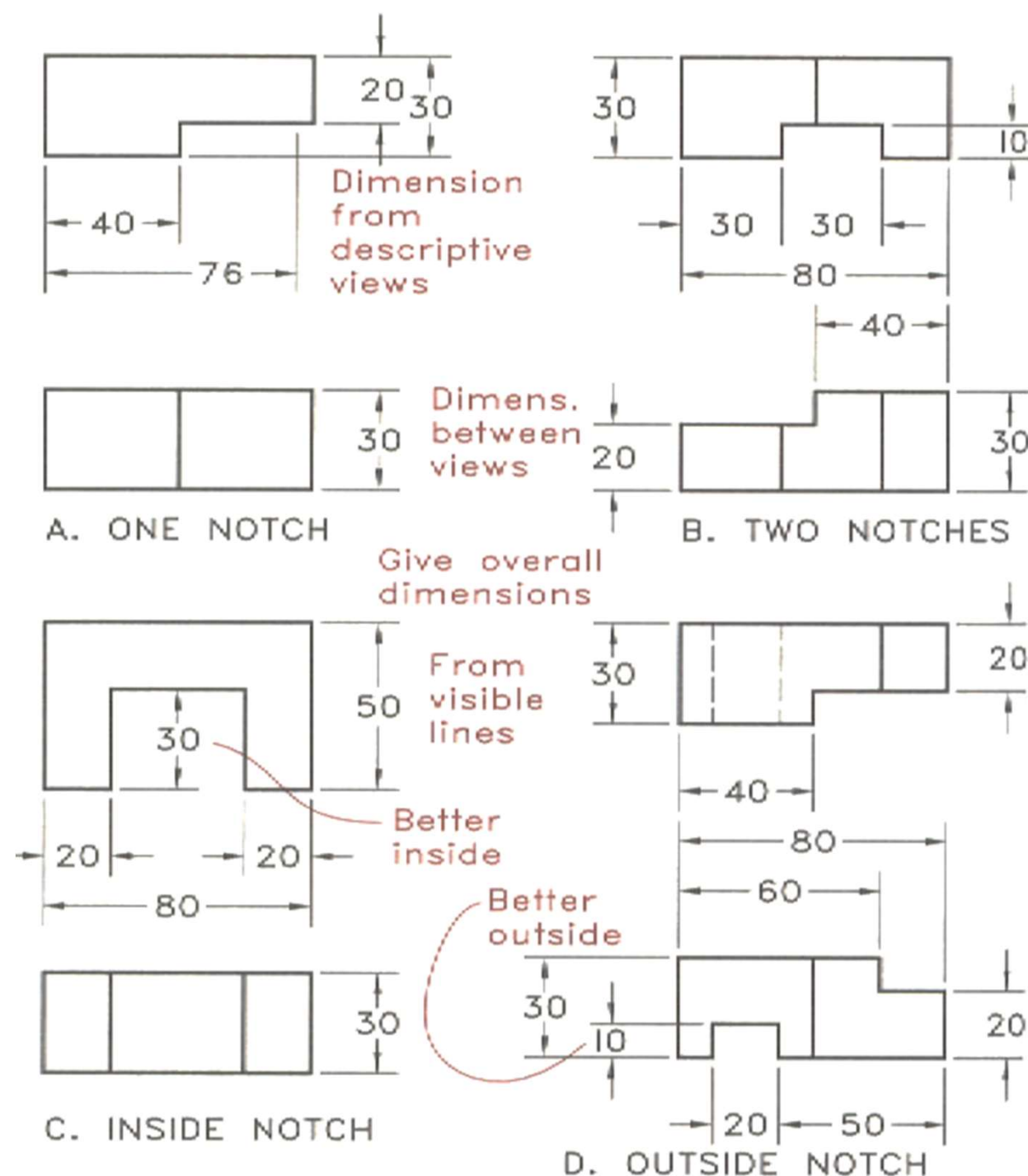
GOOD



POOR

Whenever possible, place dimensions outside objects rather than inside their outlines.

DIMENSIONING PRISMS

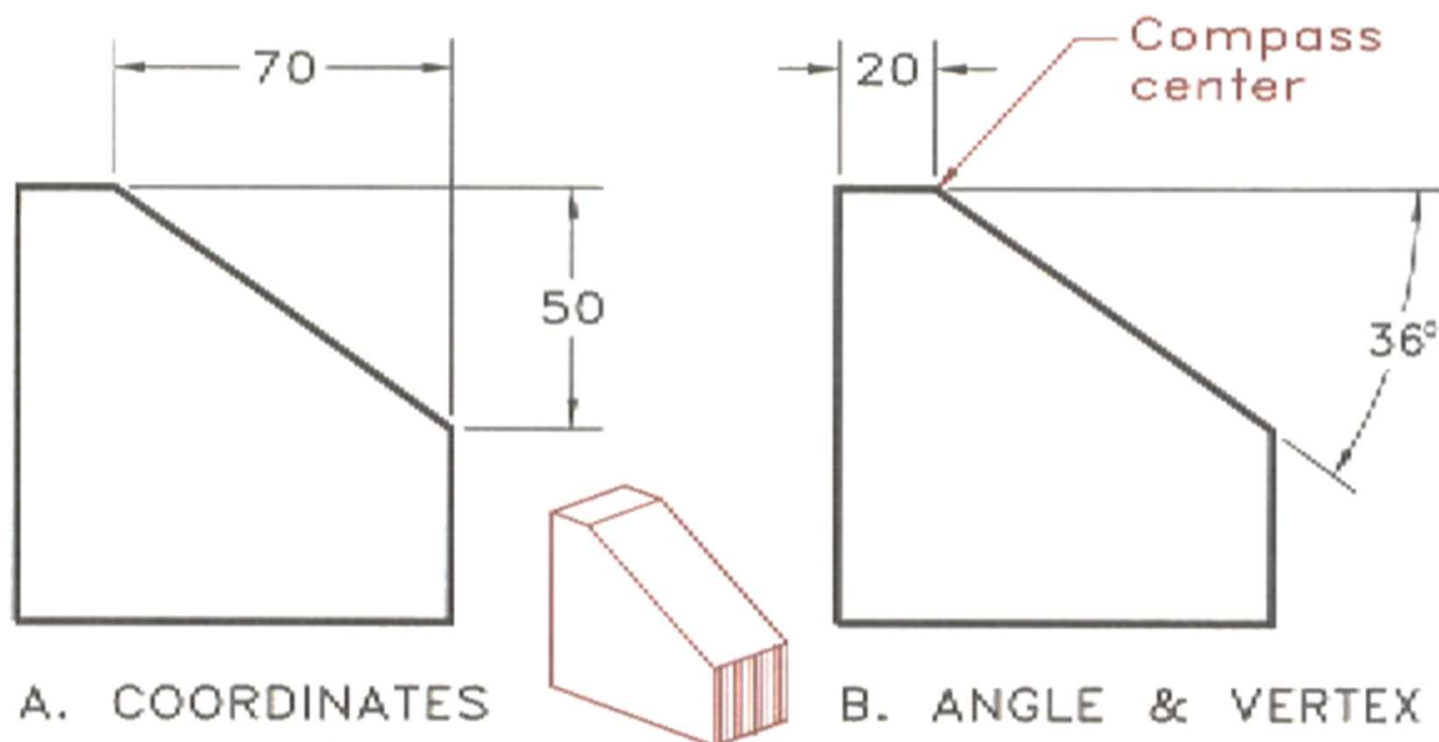


A and B Dimension prisms from descriptive views and between views.

C You may dimension a notch inside the object if doing so improves clarity.

D Dimension visible lines, not hidden lines.

ANGULAR DIMENSIONS

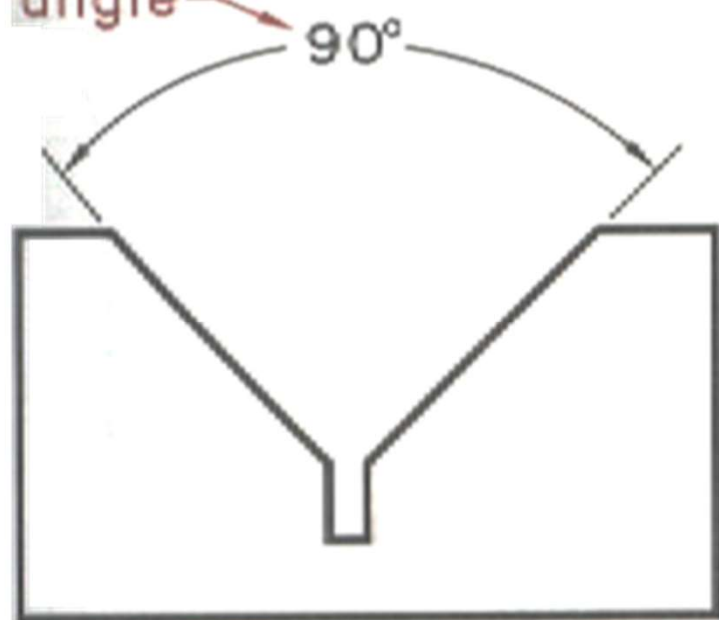


A Dimension angular planes by using coordinates.

B Measure angles by locating the vertex and measuring the angle in degrees. When accuracy is essential, specify angles in degrees, minutes, and seconds.

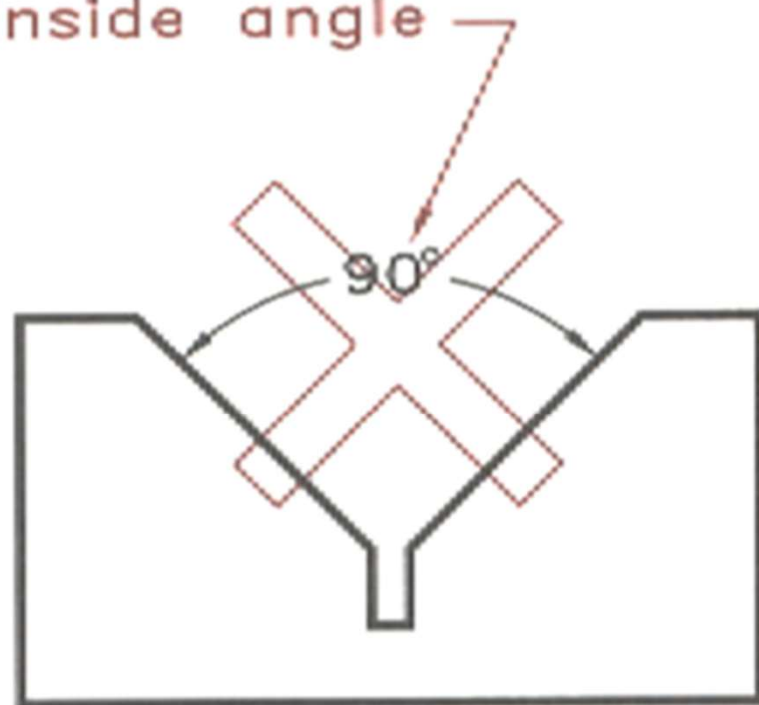
RULE 11: Place angular dimensions outside the angle

Use extension lines to place arc outside angle



GOOD

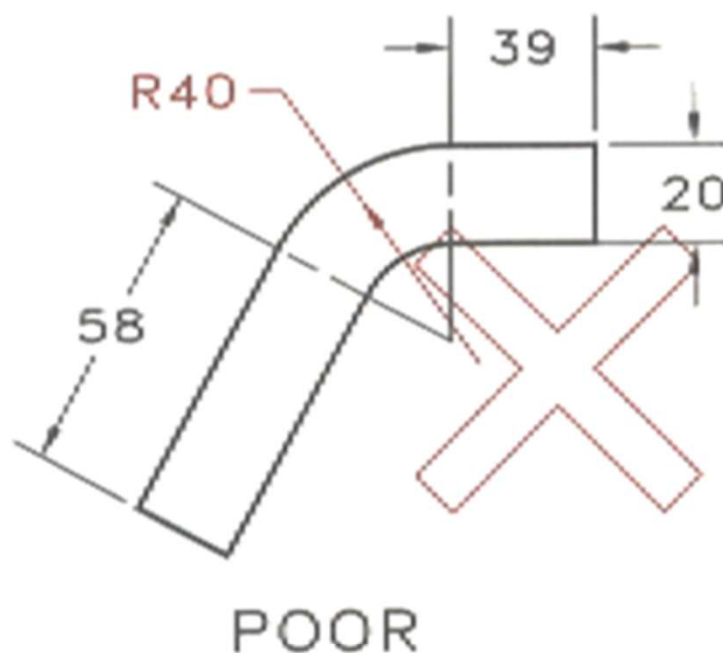
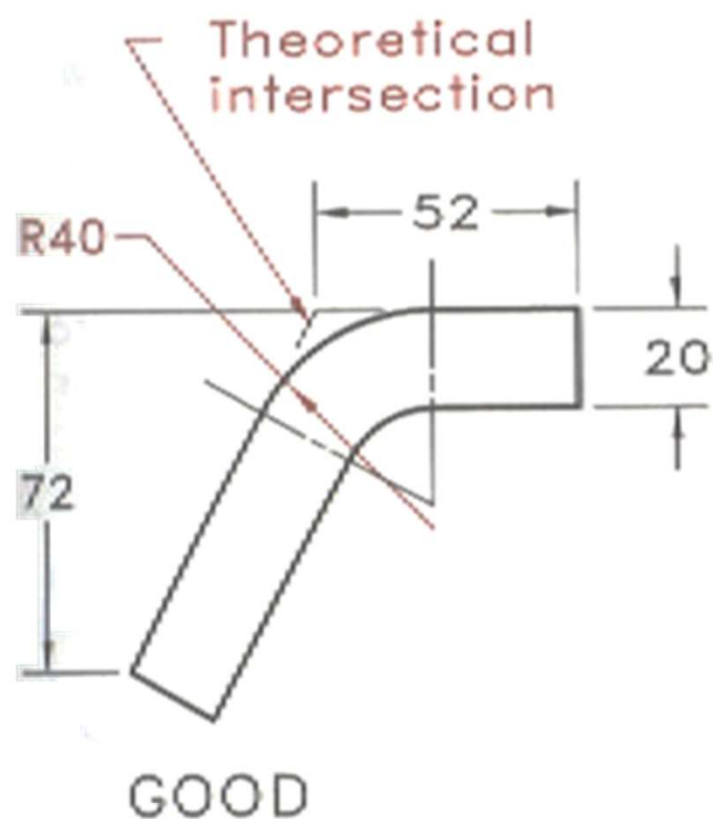
Dimension placed inside angle



POOR

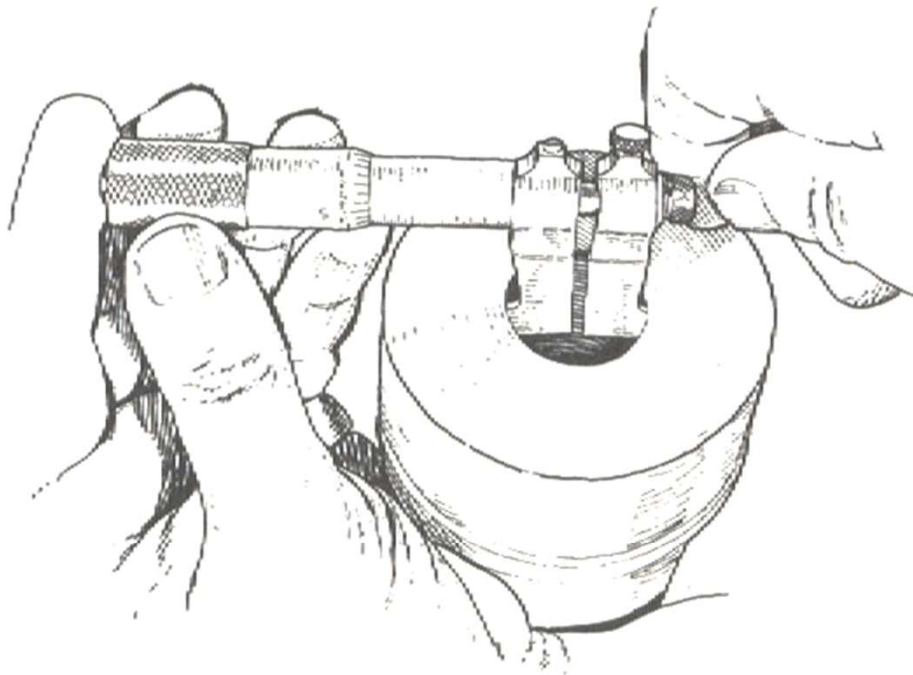
Place angular dimensions outside angular notches by using extension lines.

RULE 12: Dimension rounded corners to the theoretical intersection



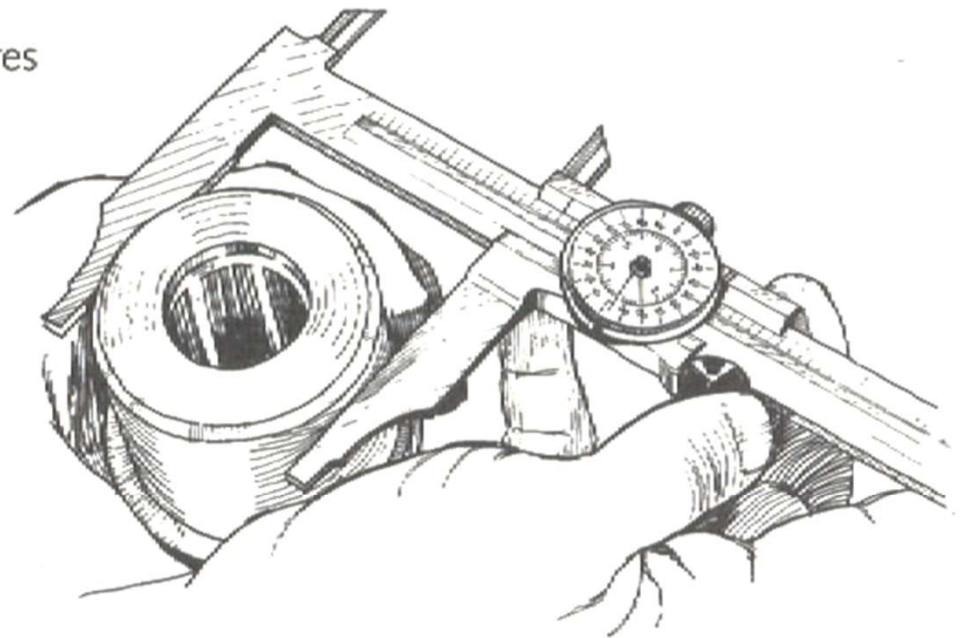
Dimension a bent surface rounded corner by locating its theoretical point of intersection with extension lines.

MEASURING INTERNAL DIAMETER



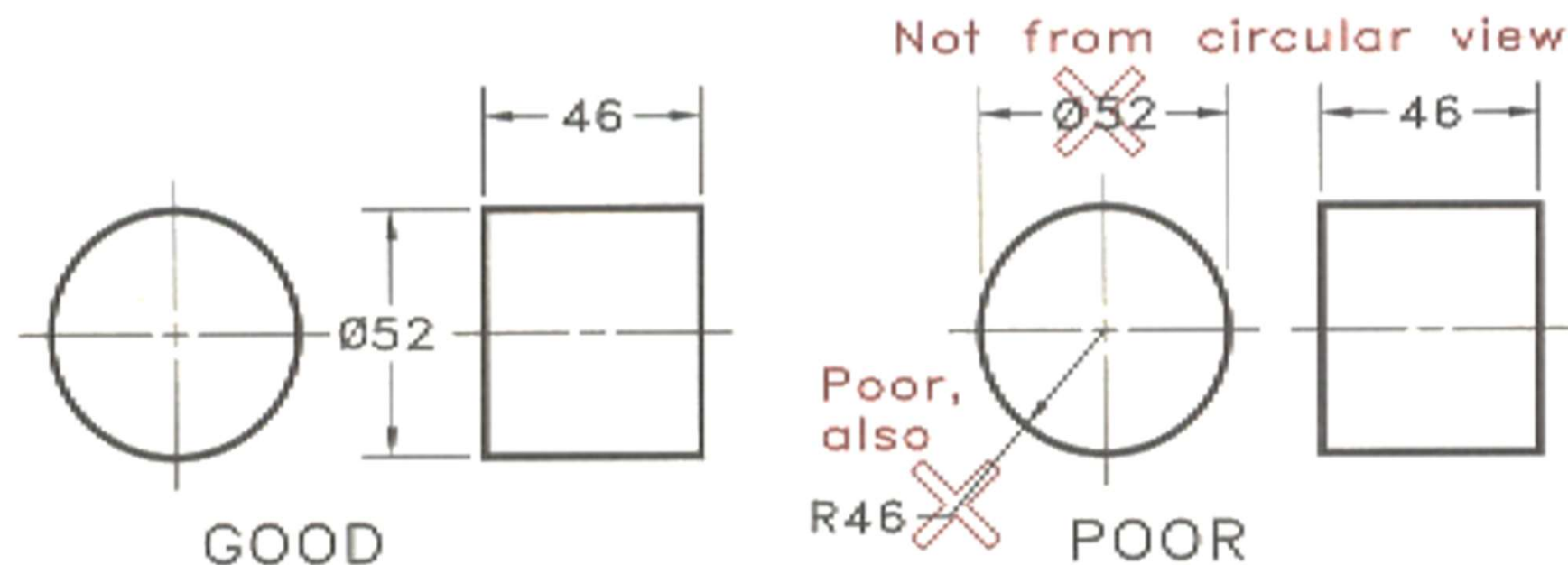
This internal micrometer caliper measures internal cylindrical diameters (radii cannot be measured).

MEASURING EXTERNAL DIAMETER



This external micrometer caliper measures the diameter of a cylinder.

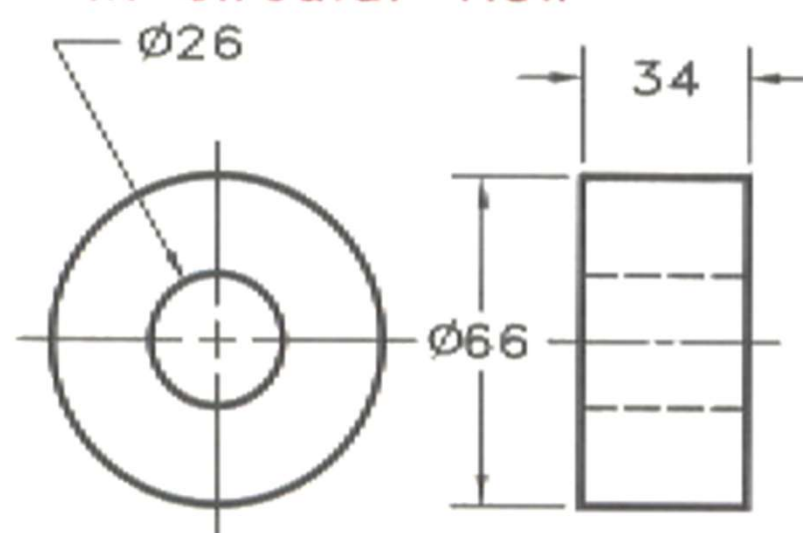
RULE 13: Dimension cylinders in their rectangular views with diameters



Dimension the diameter (not the radius) of a cylinder in the rectangular view.

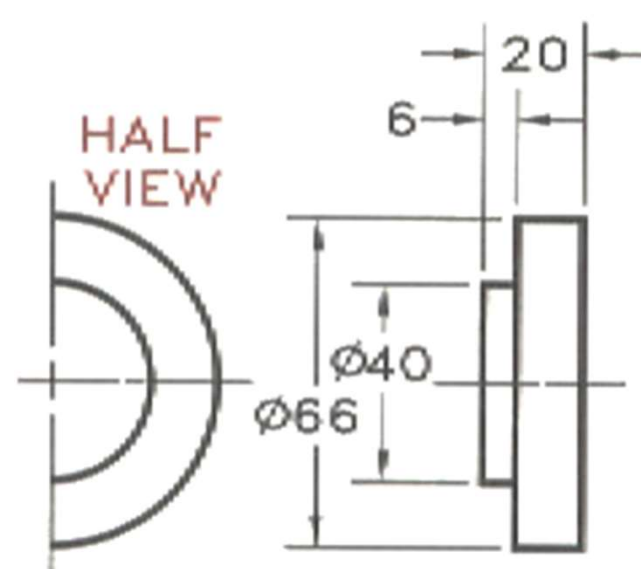
RULE 13 (Cont.):
Dimensioning
cylinders

Dimension holes
in circular view



GOOD

Dimension solid
cylinders in rectan-
gular views with ϕ

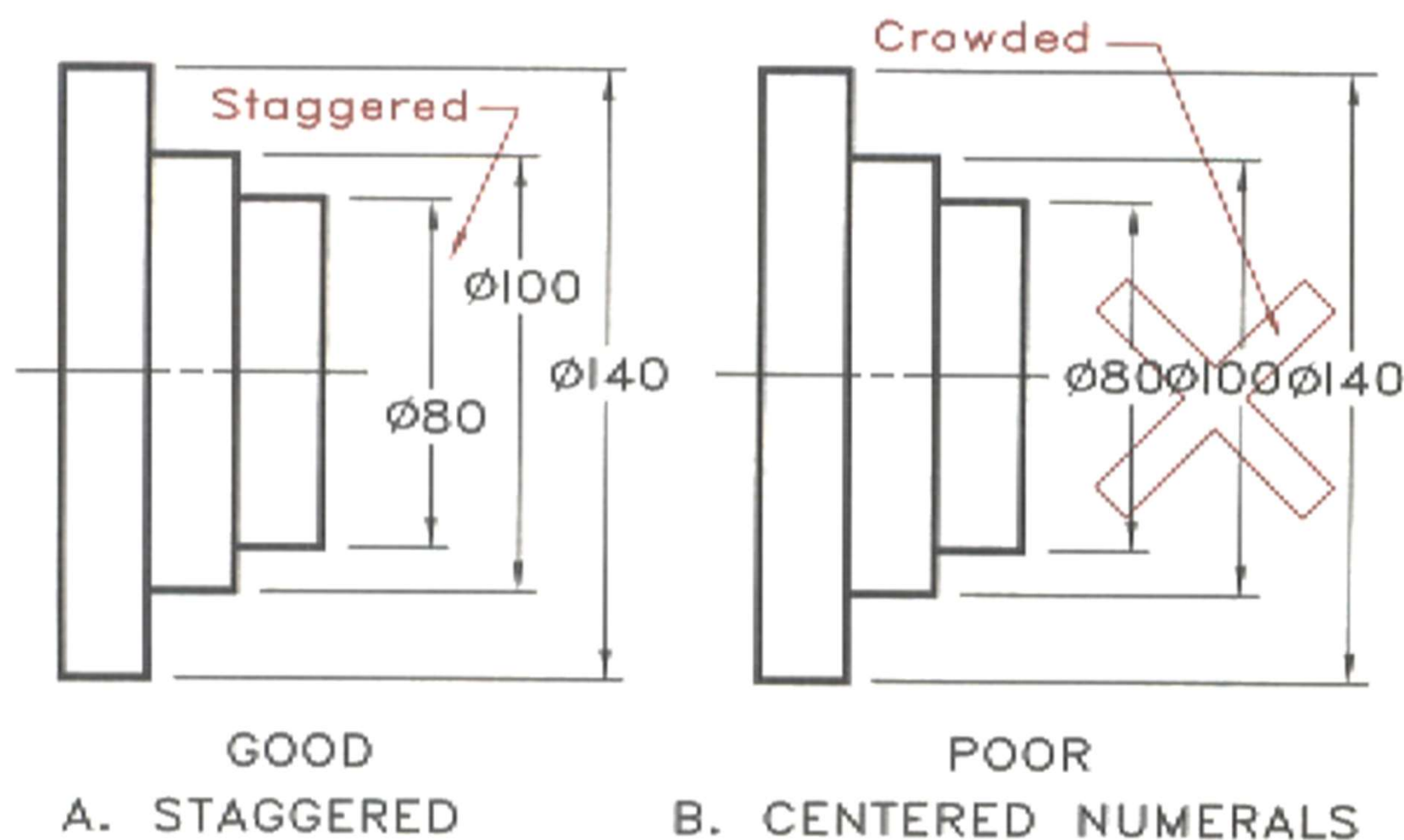


GOOD

Do not use radii on cylinders

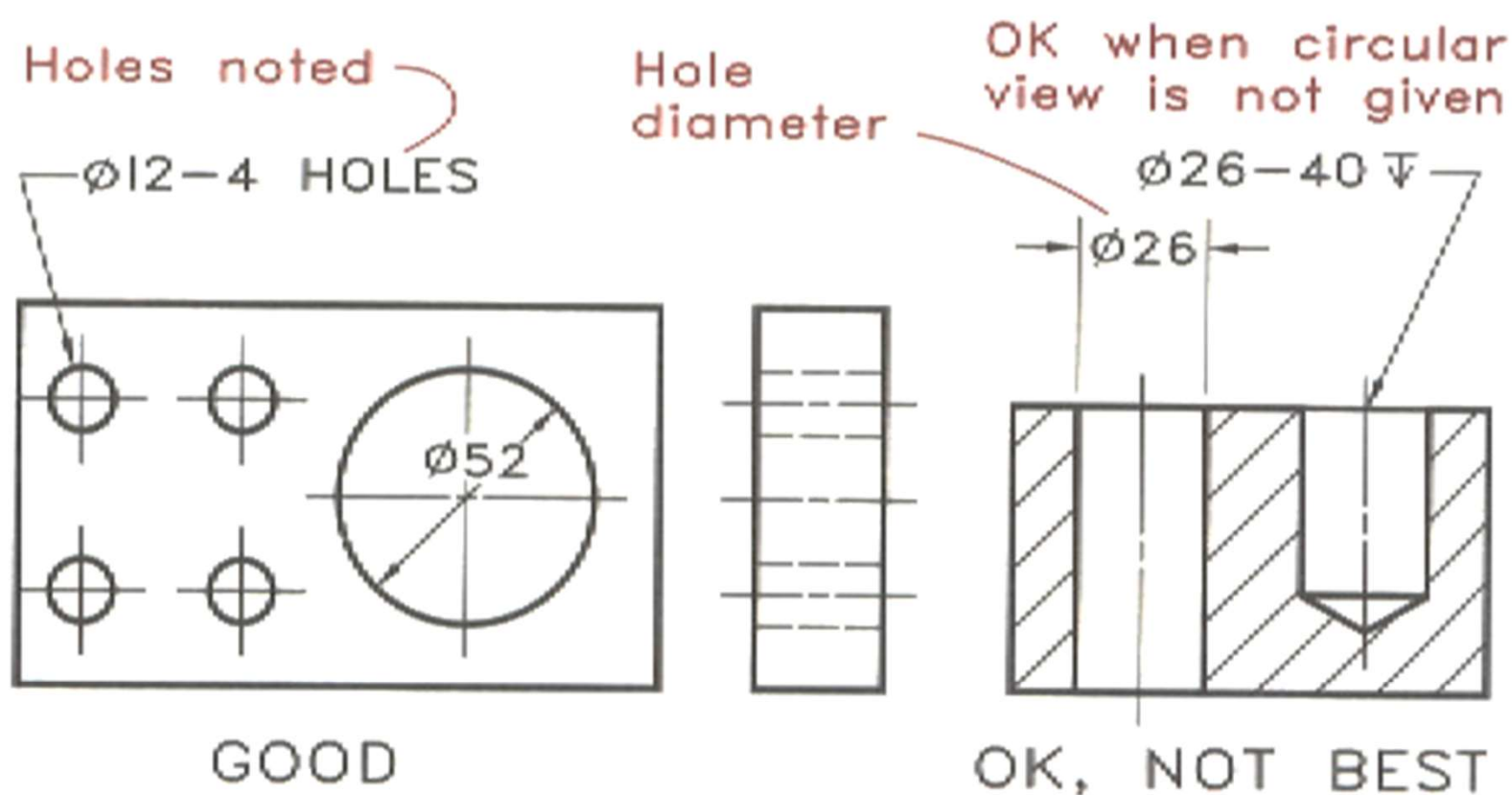
Dimension holes in their circular views with leaders. Dimension concentric cylinders with a series of diameters. Use half views to save drawing time.

RULE 14: Stagger dimension numerals to prevent crowding



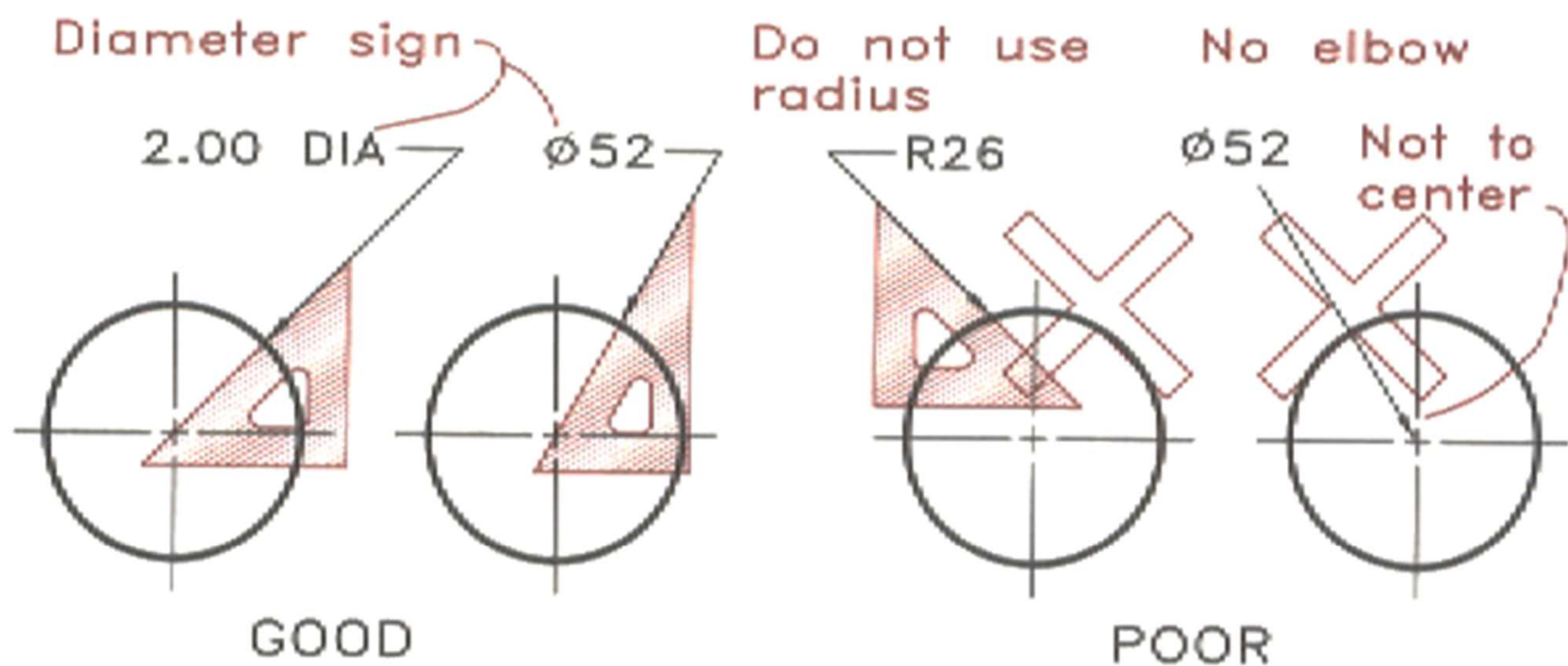
Dimensions on concentric cylinders are easier to read if they are staggered within their dimension lines.

RULE 15: Hole sizes are best given as diameters with leaders in circular views



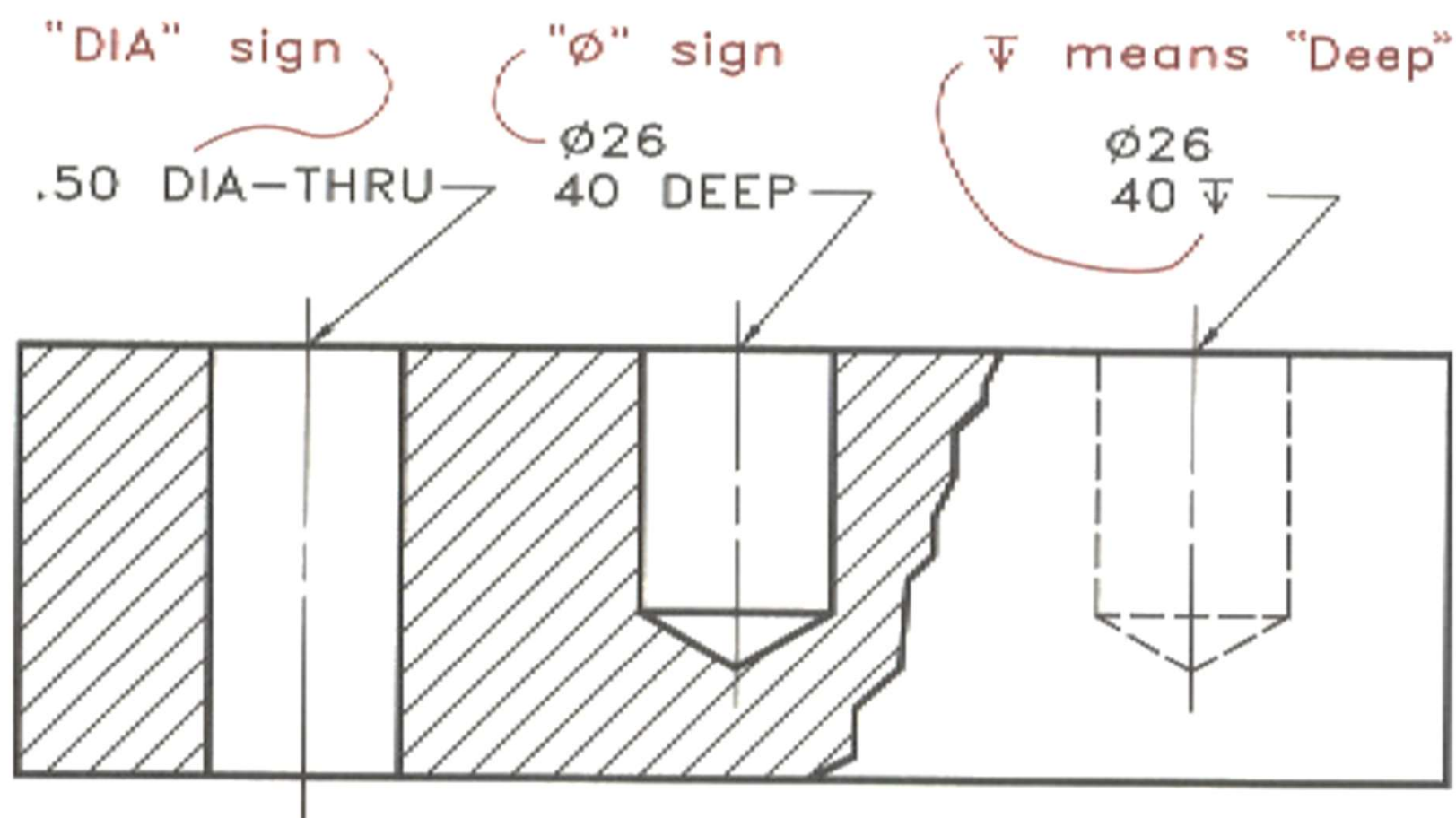
Dimension holes in their circular view with leaders whenever possible, but dimension them in their rectangular views if necessary.

RULE 16: Leaders should have horizontal elbows and point toward the hole centers



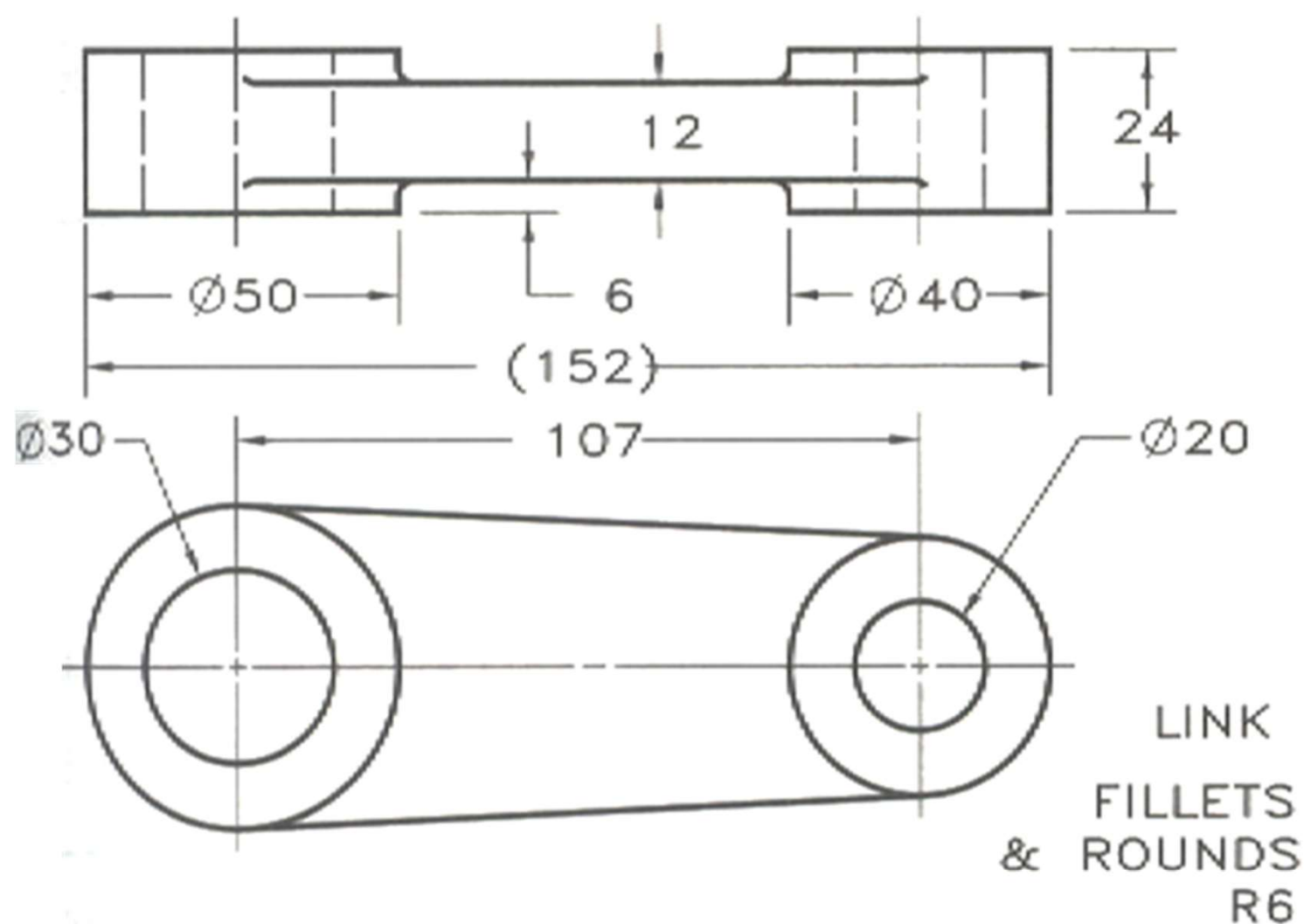
Draw leaders pointing toward the centers of holes.

NOTES ON HOLES



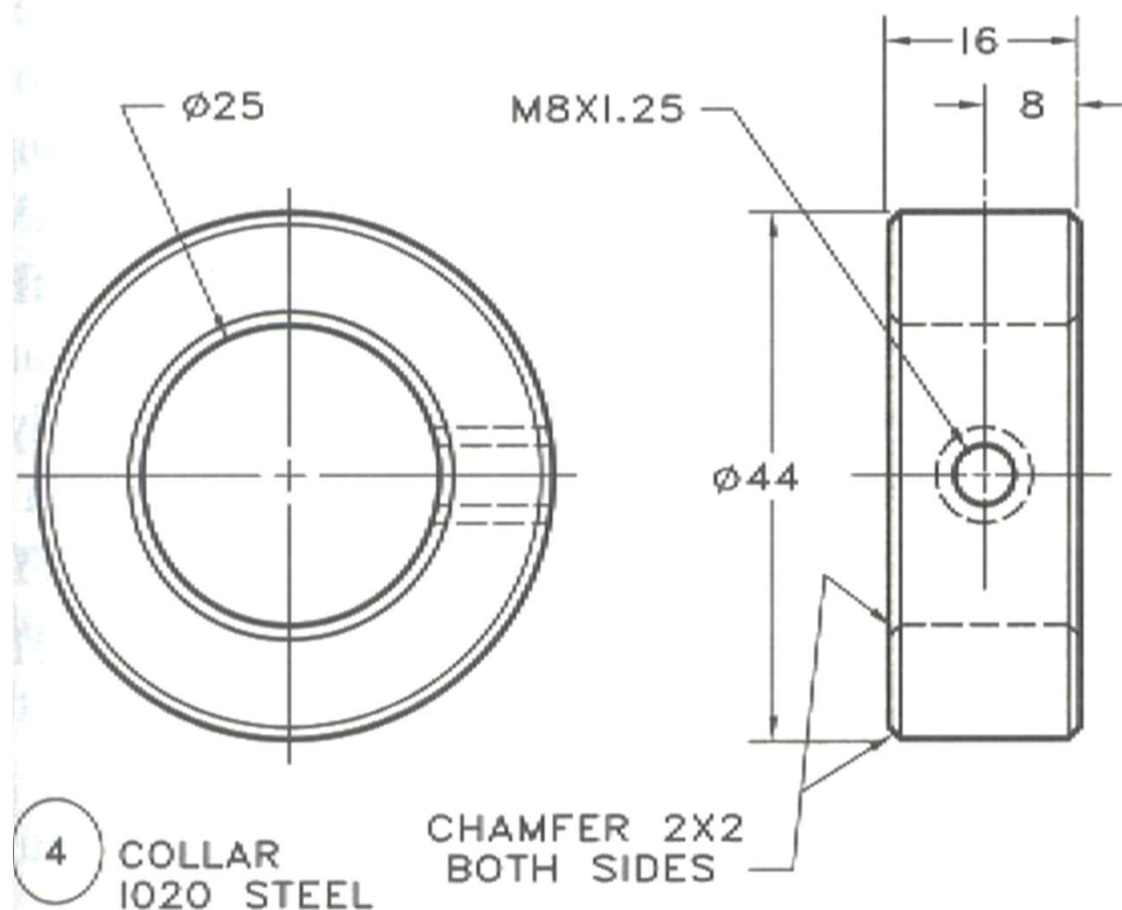
These are examples of holes noted in their rectangular views.

CYLINDRICAL FEATURES



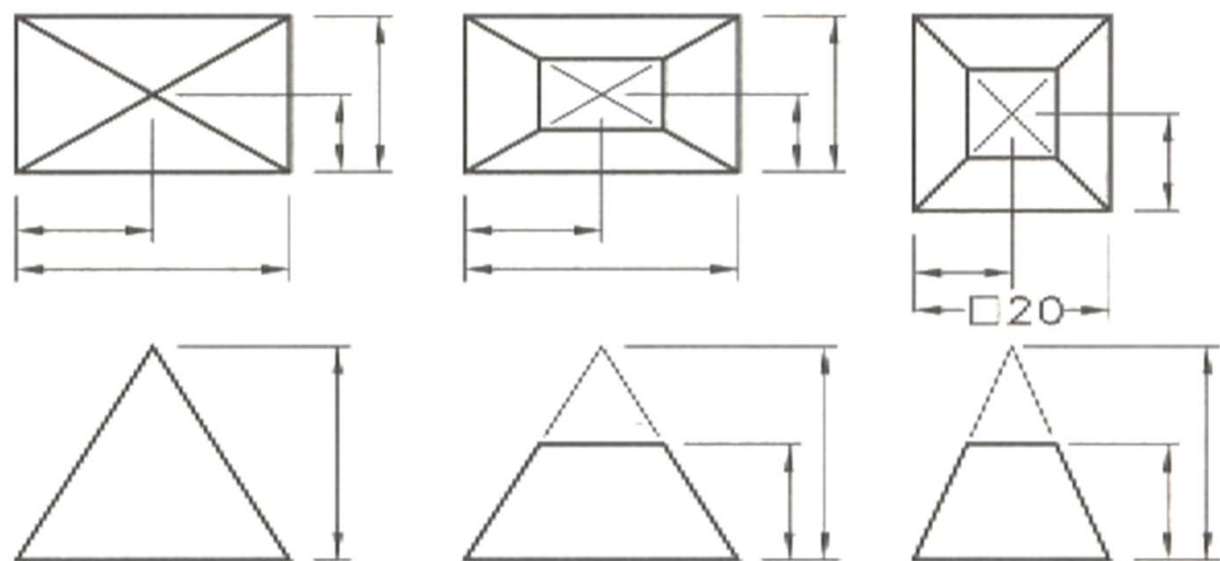
This drawing illustrates the application of dimensions to cylindrical features. (F&R R6 means that fillets and rounds have a 6 mm radius.)

DIMENSIONING A CYLINDRICAL PART



(A) This cylindrical part, a collar, is shown

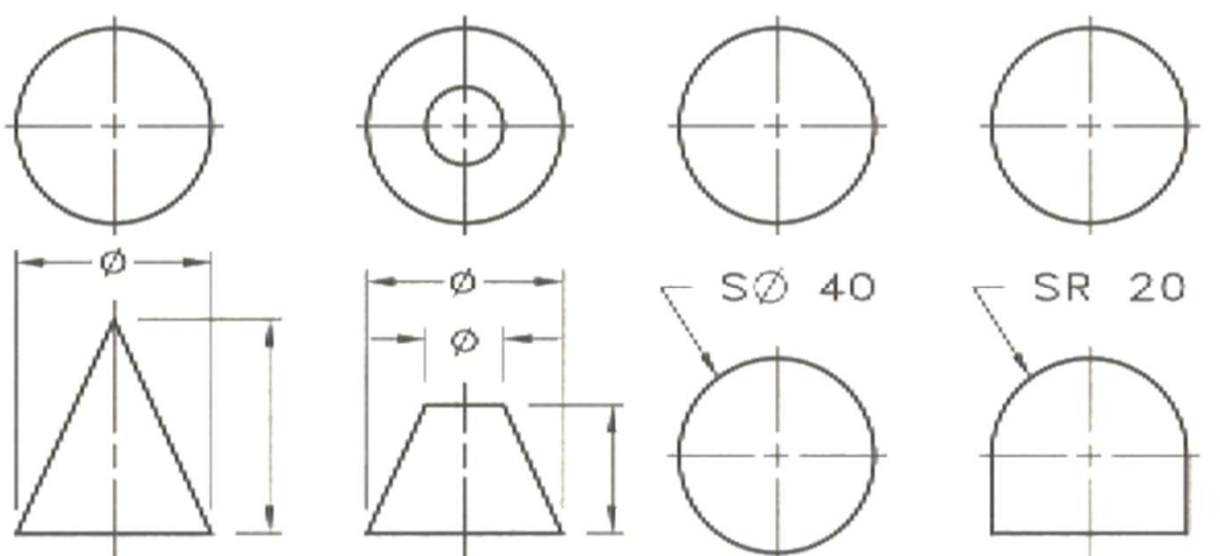
PYRAMIDS, CONES, AND SPHERES



A. PYRAMID

B. PYRAMID

C. SQ. PYRAMID



D. CONE

E. CONE

F. SPHERE

G. SPH END

This drawing shows the proper way to dimension pyramids, cones, and spheres.

RULE 17: Extend leaders from the first or the last of a note



GOOD

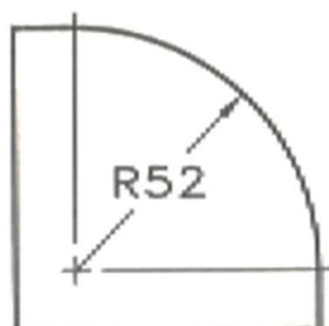


POOR

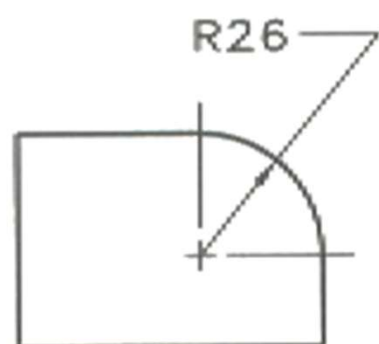
Extend leaders from the first or the last word of a note with a horizontal elbow.

RULE 18: Dimensions arcs (less than 180°)
with radii

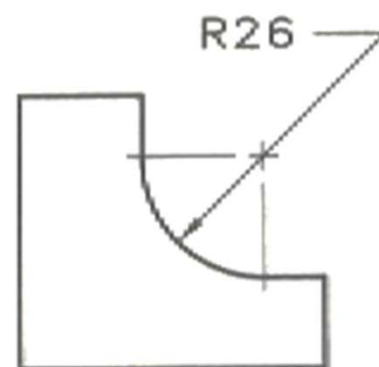
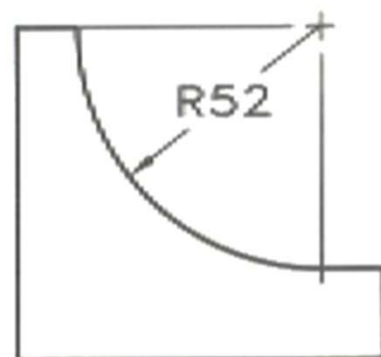
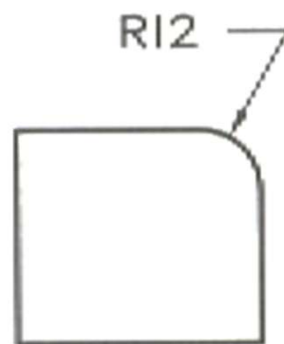
Arrow and
dim. inside



Arrow inside
dim. outside

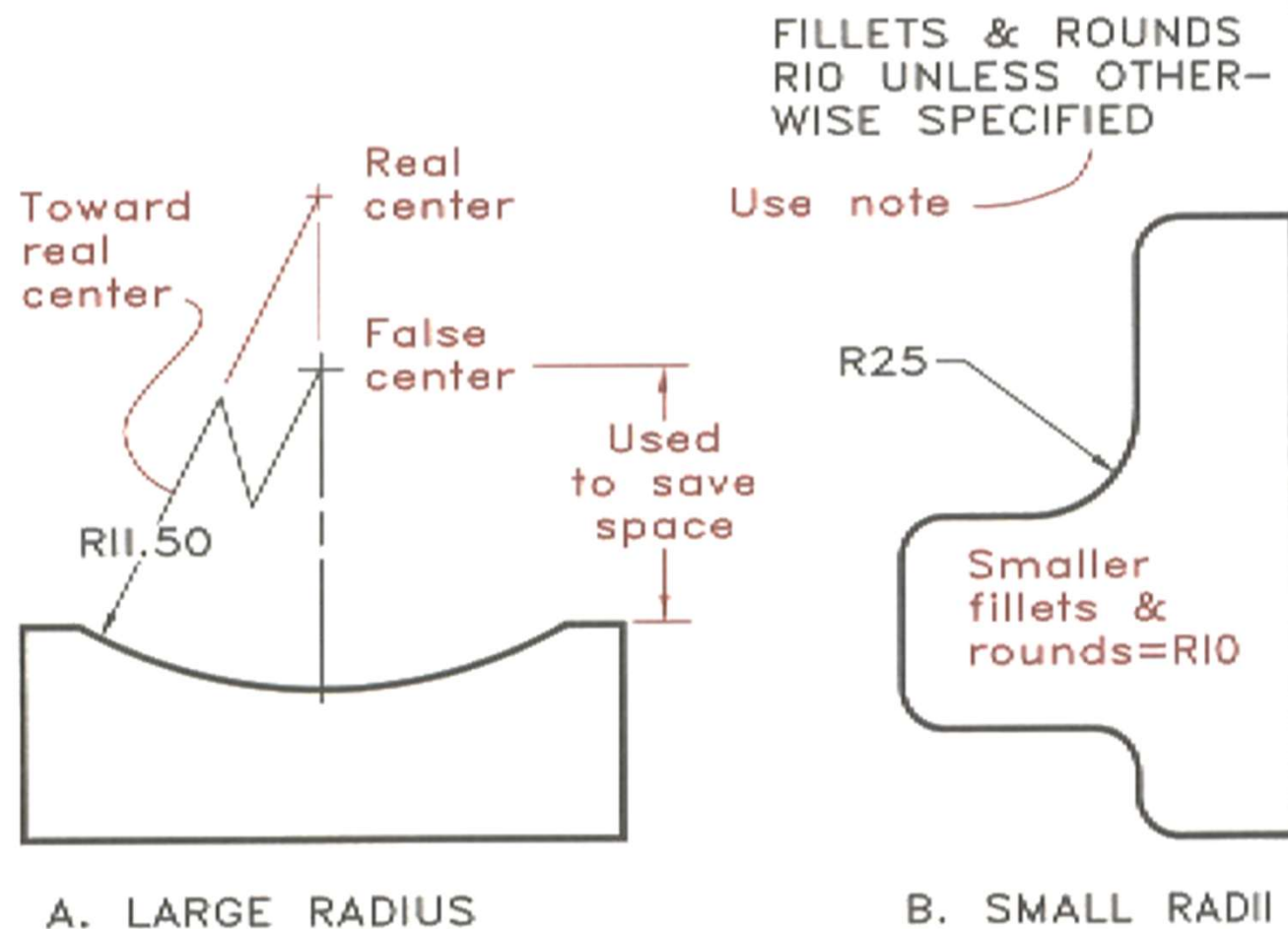


Arrow and
dim. outside



When space permits, place dimensions and arrows between the center and the arc. When space is not available for the number, place the arrow between the center and the arc number outside. If there is no space for the arrow inside, place both the dimension and arrow outside the arc with a leader.

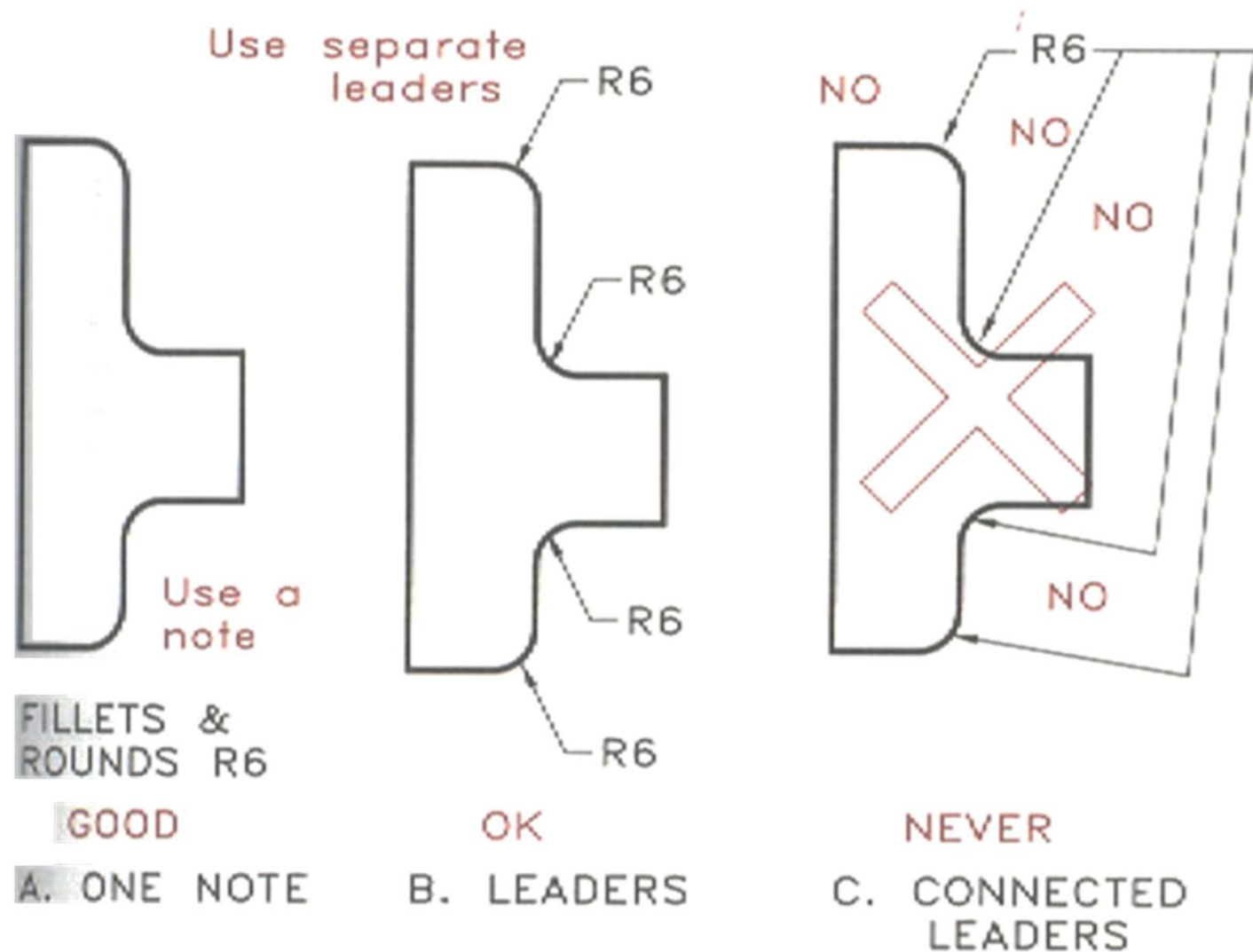
LARGE AND SMALL RADII



A Show a long radius with a false radius (a line with a zigzag) to indicate that it is not true length. Show its false center on the centerline of the true center.

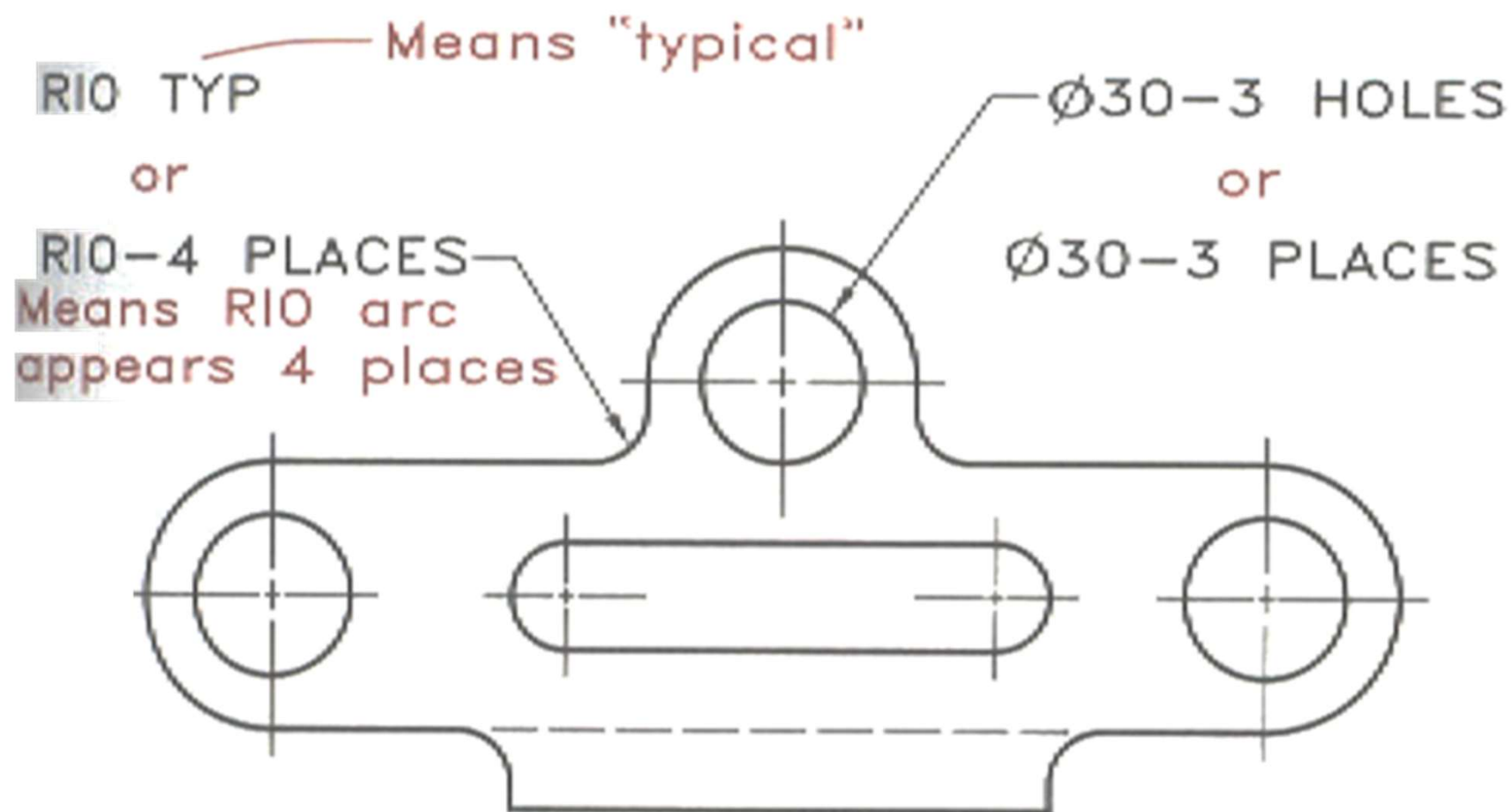
B Specify fillets and rounds with a note to reduce repetitive dimensions of small arcs.

FILLETS AND ROUNDS



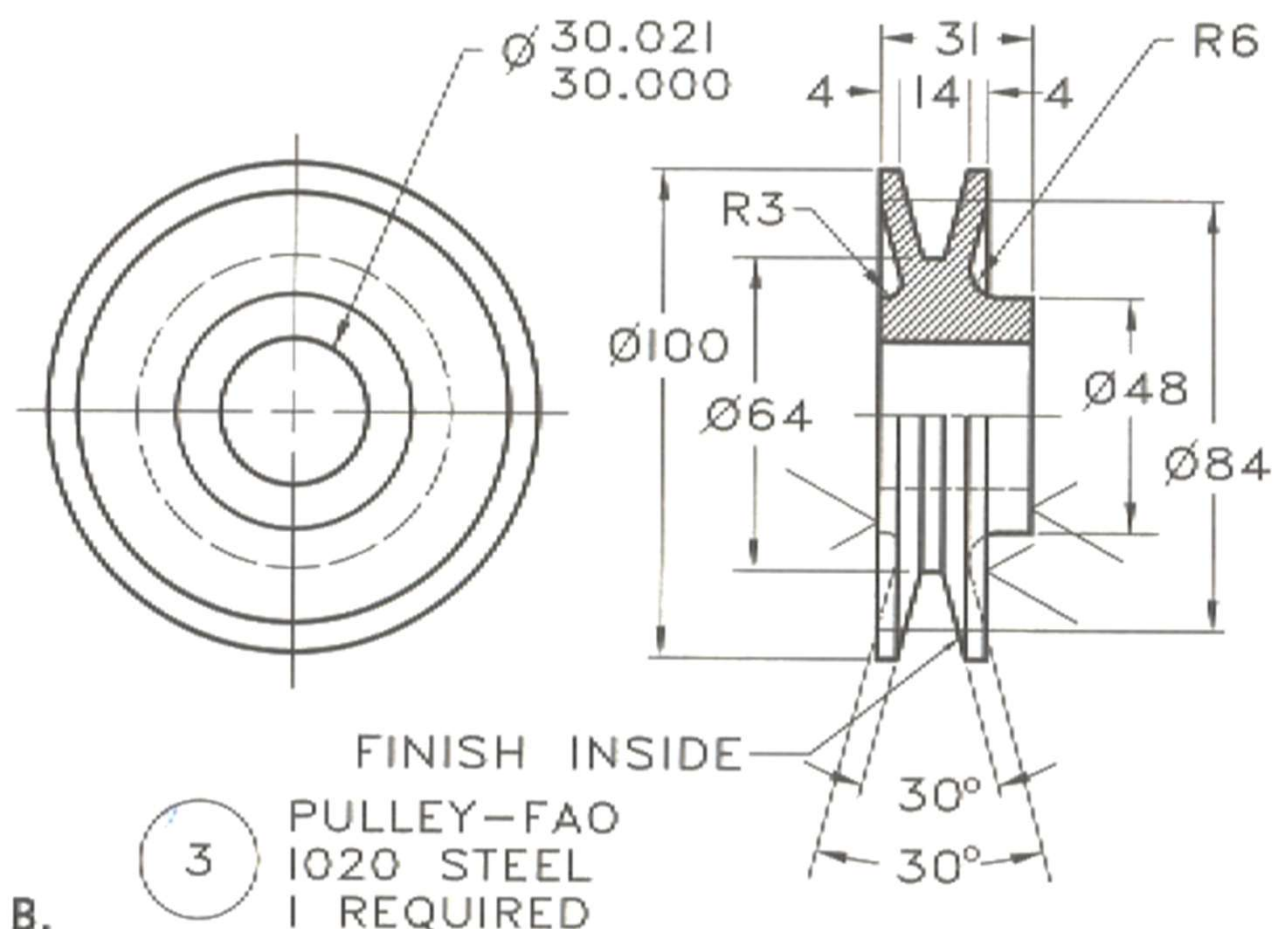
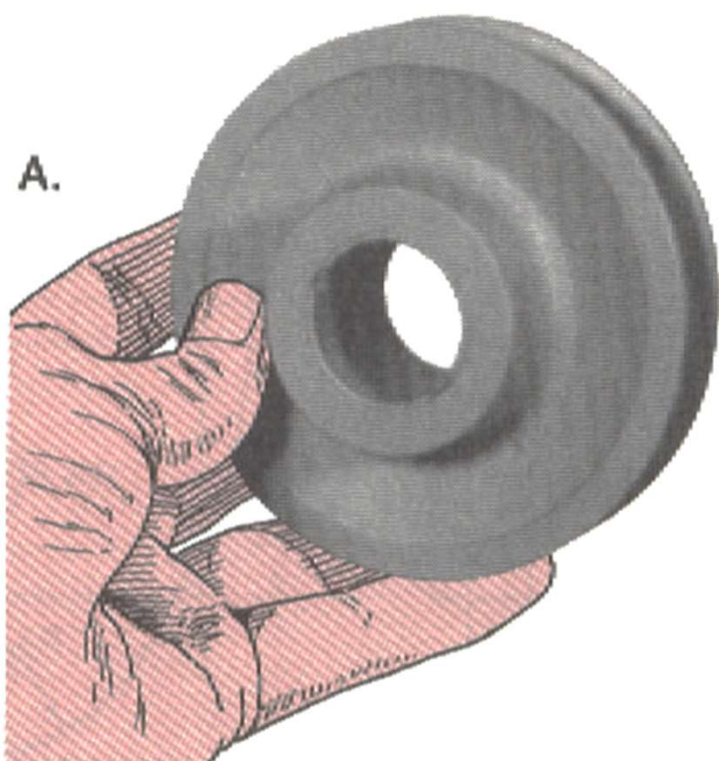
Indicate fillets and rounds by (A) notes or (B) separate leaders and dimensions. Never use confusing leaders (C).

NOTES ON PARTS



Use notes to indicate that identical features and dimensions are repeated on drawings to avoid having to dimension them individually.

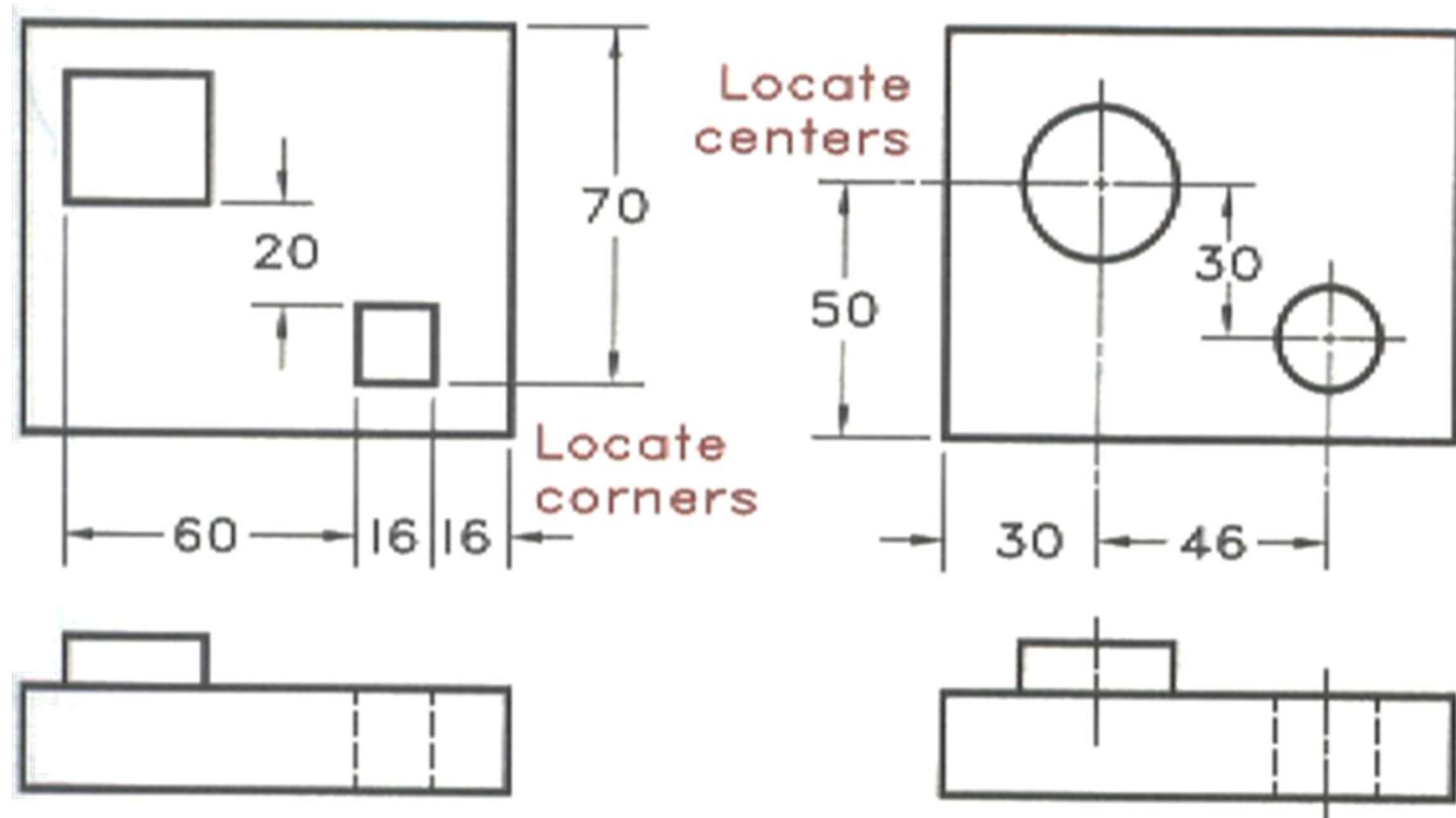
ARCS AND CYLINDERS



A This pulley comprises arcs and circles.

B This drawing illustrates how to dimension the part.

LOCATION DIMENSIONS

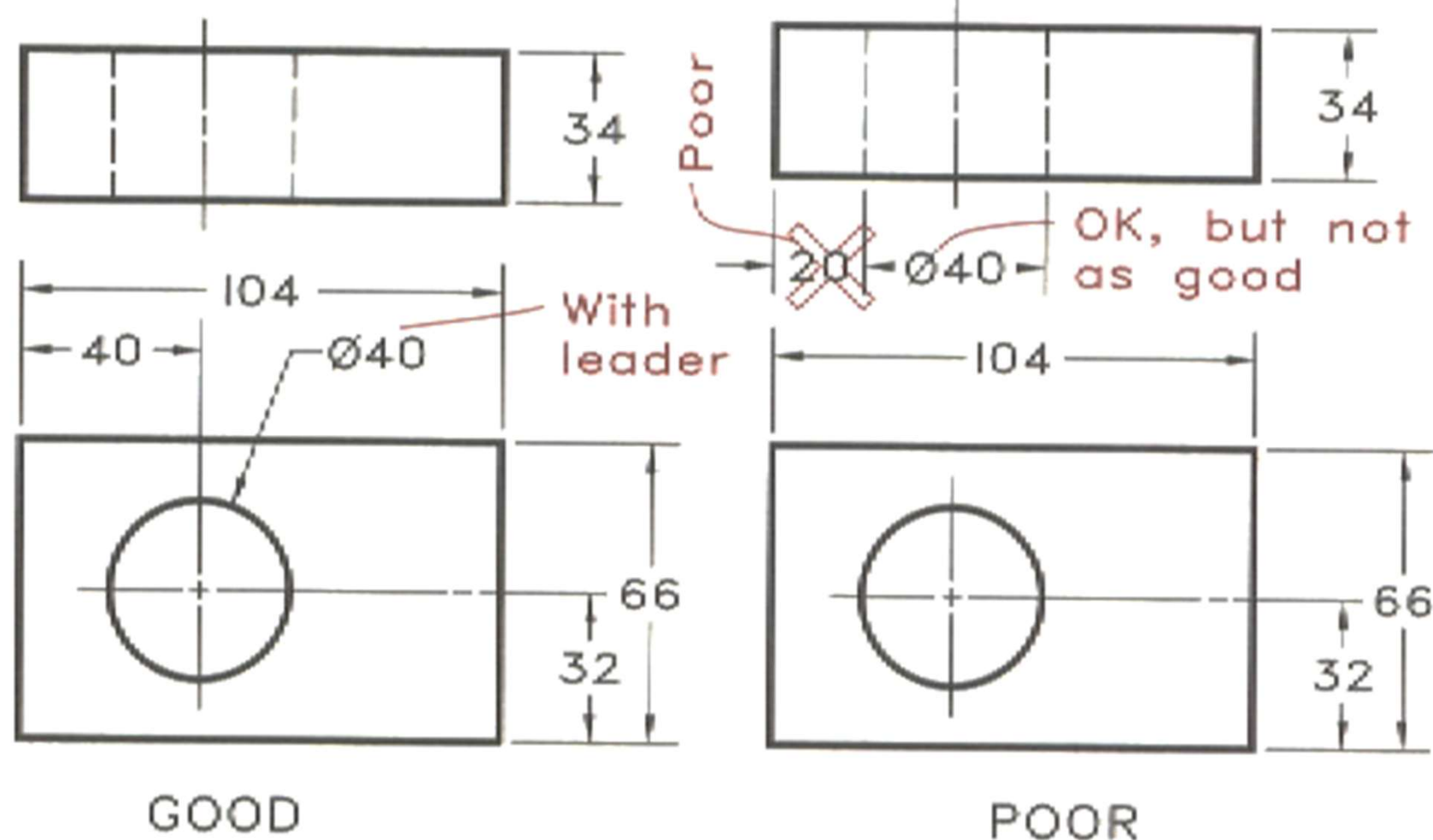


A. RECTANGLES

B. CYLINDERS

Location dimensions give the positions of geometric features with respect to other geometric features, but not their sizes.

RULE 19: Locate holes in circular views and dimension diameters with a leader



Locate cylindrical holes in their circular views by coordinates to their centers.

FINISH MARKS

H=Letter height

Draw with
F pencil

Place finish marks on all
finished surfaces, visible
or hidden, in their edge views

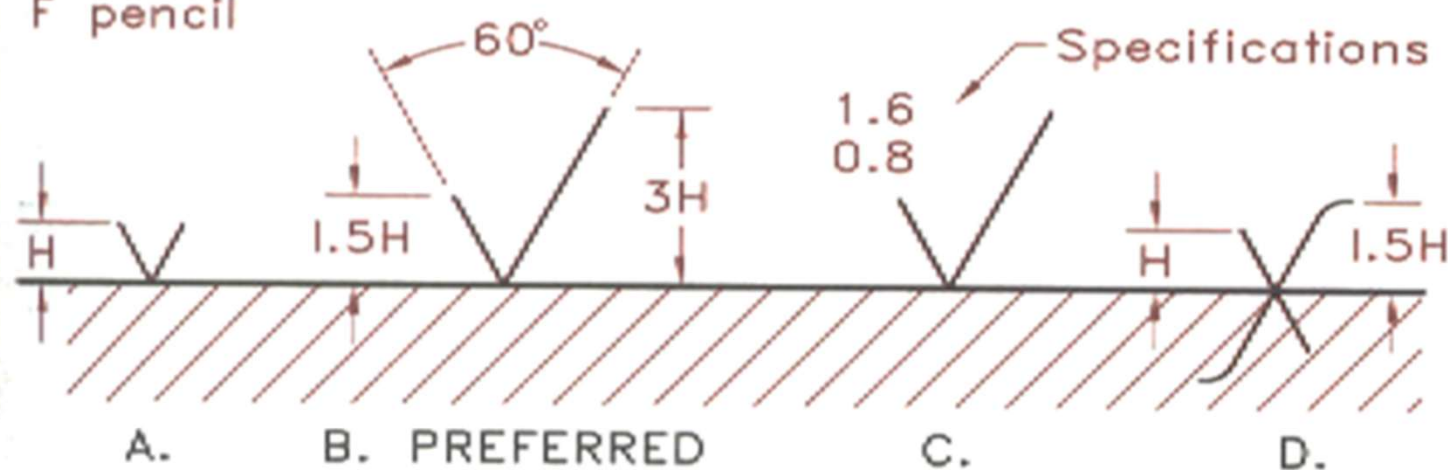


Figure 13.51 Finish marks indicate that a surface is to be machined to a smooth surface.

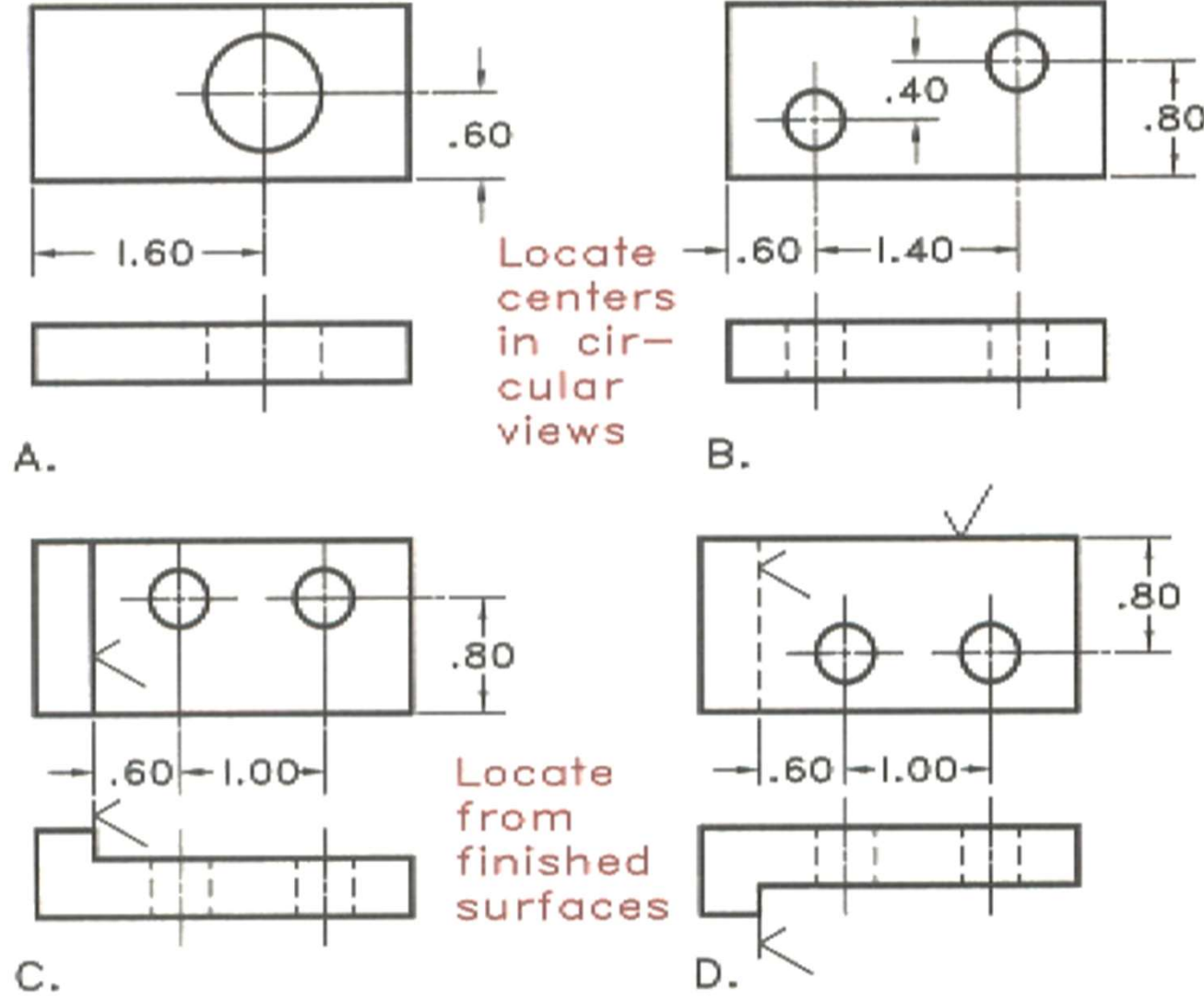
A The traditional V can be used for general applications.

B The unequal finish mark is the best one for general applications.

C Where surface texture must be specified, this finish mark is used with texture values.

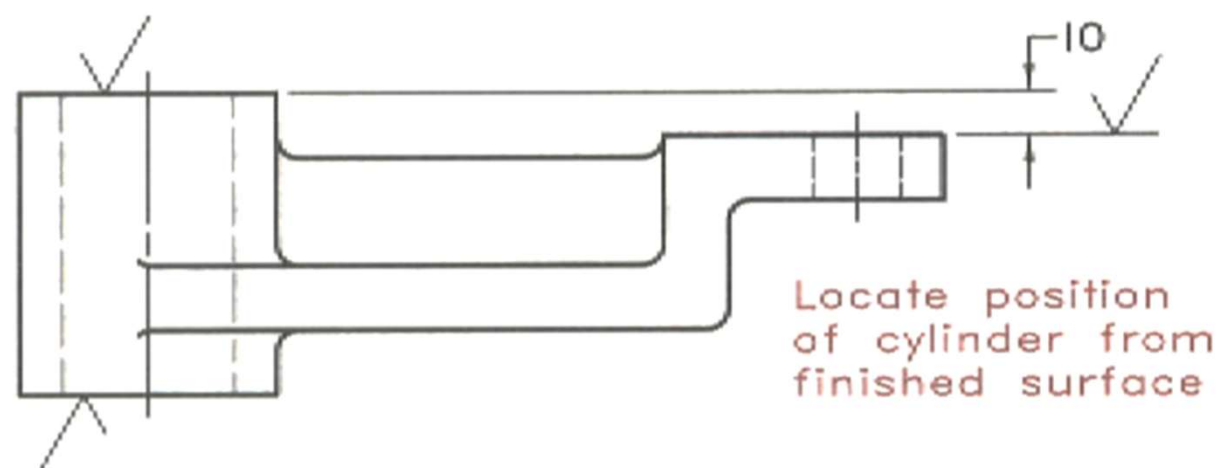
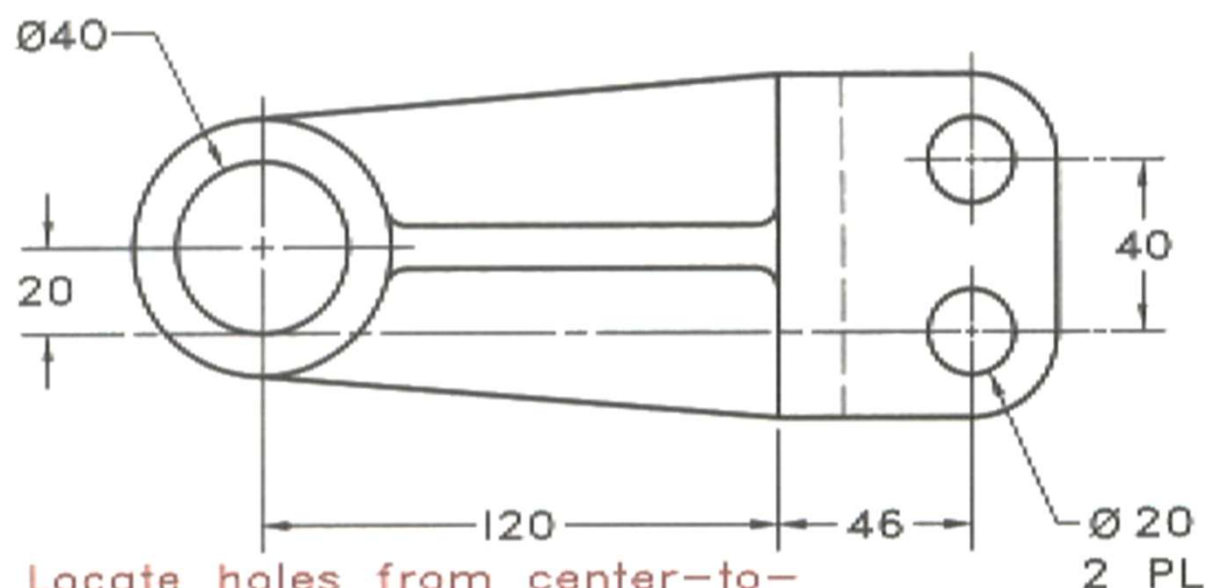
D The f-mark is the least used symbol.

LOCATING CYLINDERS



- A** Locate cylindrical holes in their circular views from two surfaces of the object.
- B** Locate multiple holes from center to center.
- C** Locate holes from finished surfaces, even if the finished surfaces are hidden as in (D).

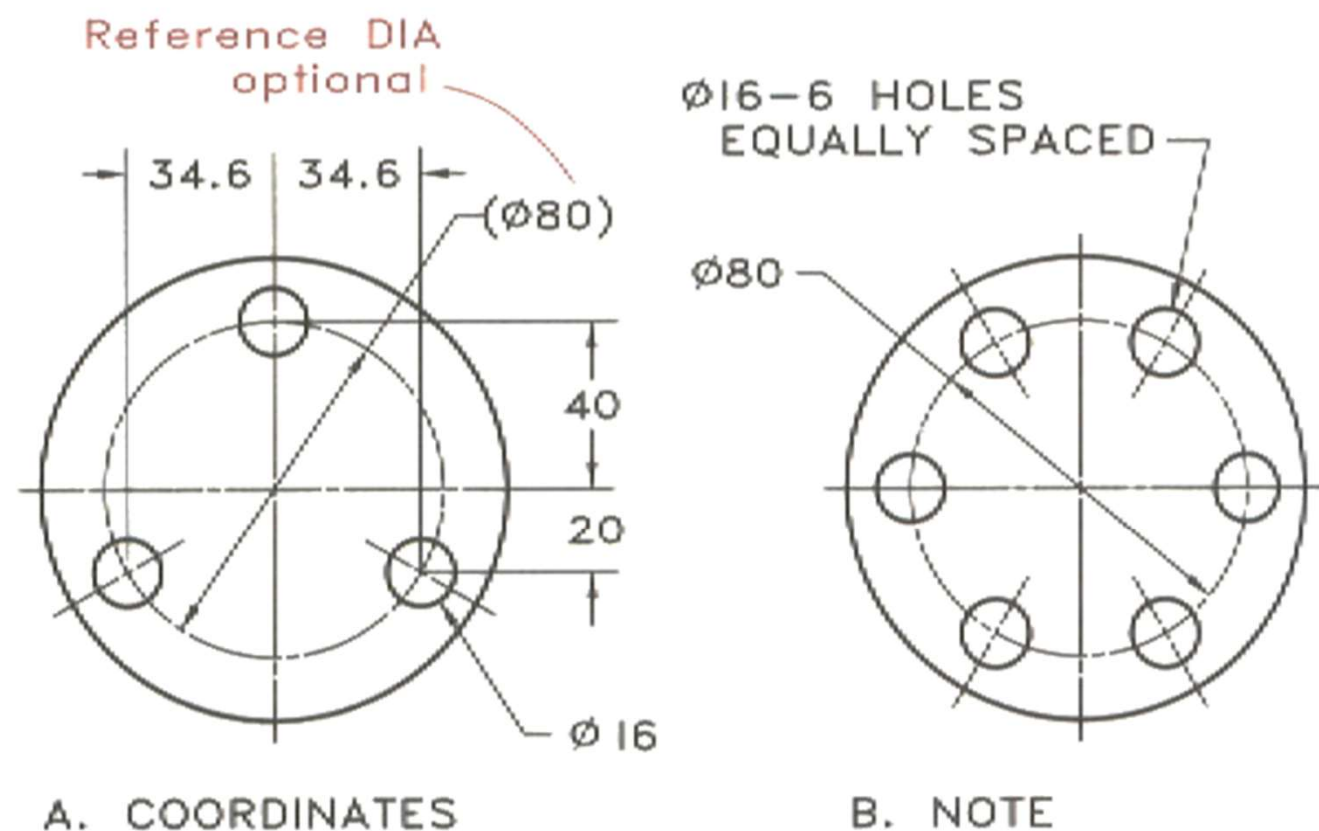
LOCATION DIMENSIONS



This example of a dimensioned shaft arm shows the application of location dimensions, with sizes omitted.

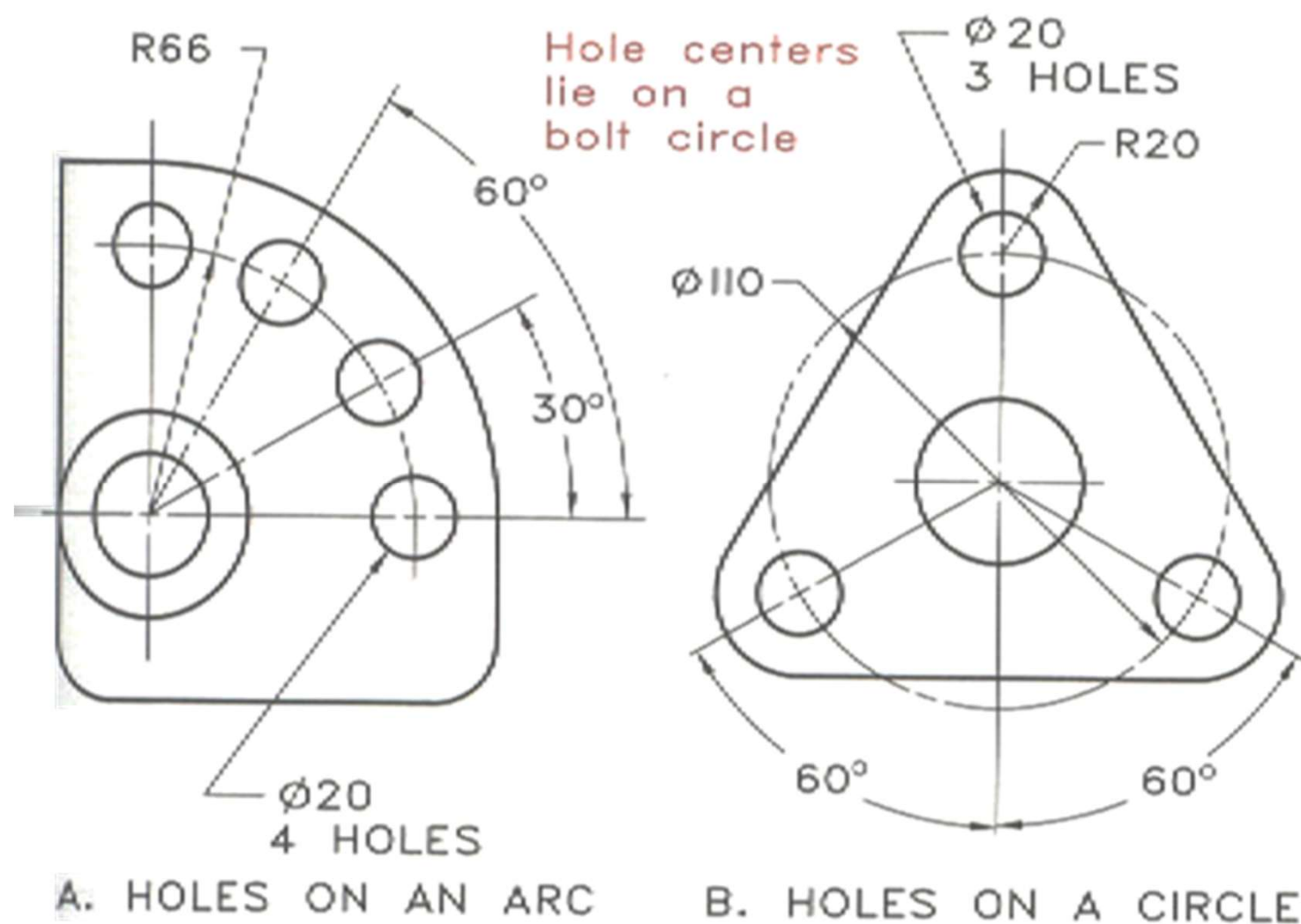
Measuring holes from two datum planes is the most accurate way to locate them and reduces the accumulation of errors possible in chain dimensioning.

BOLT CIRCLE



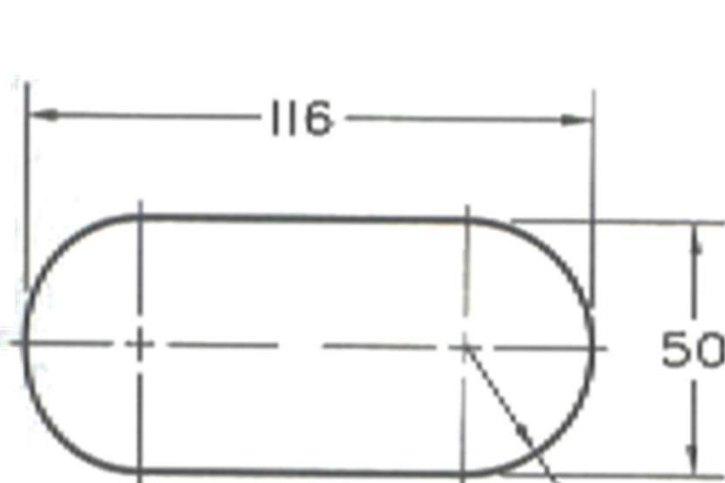
Locate holes on a bolt circle by using (A) coordinates or (B) notes.

CIRCLE OF CENTERS



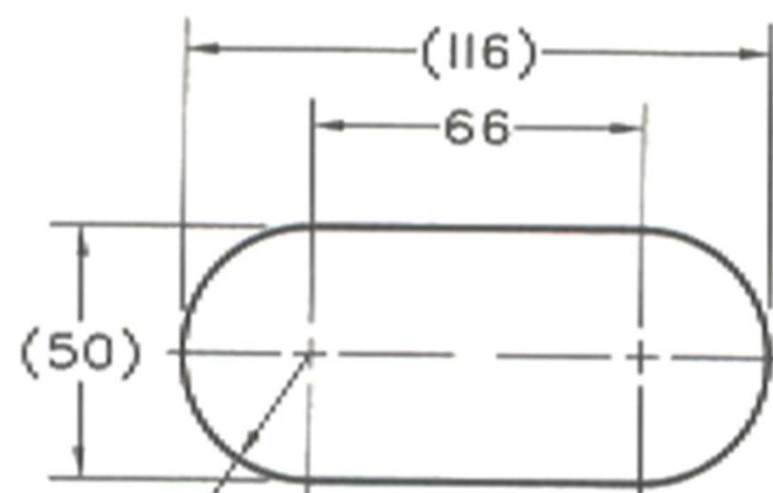
Locate centers of holes by using (A) a combination of radii and degrees or (B) a circle of centers.

OBJECTS WITH ROUNDED ENDS



R with no R
number

A. OVERALL LENGTH
PREFERRED



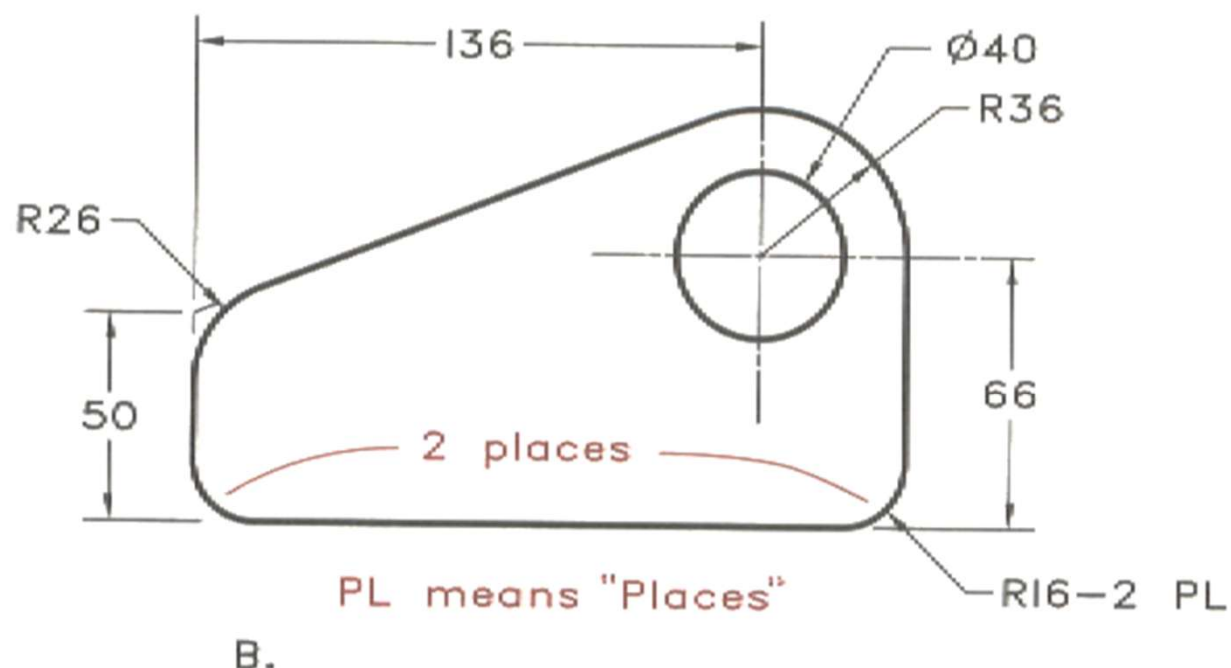
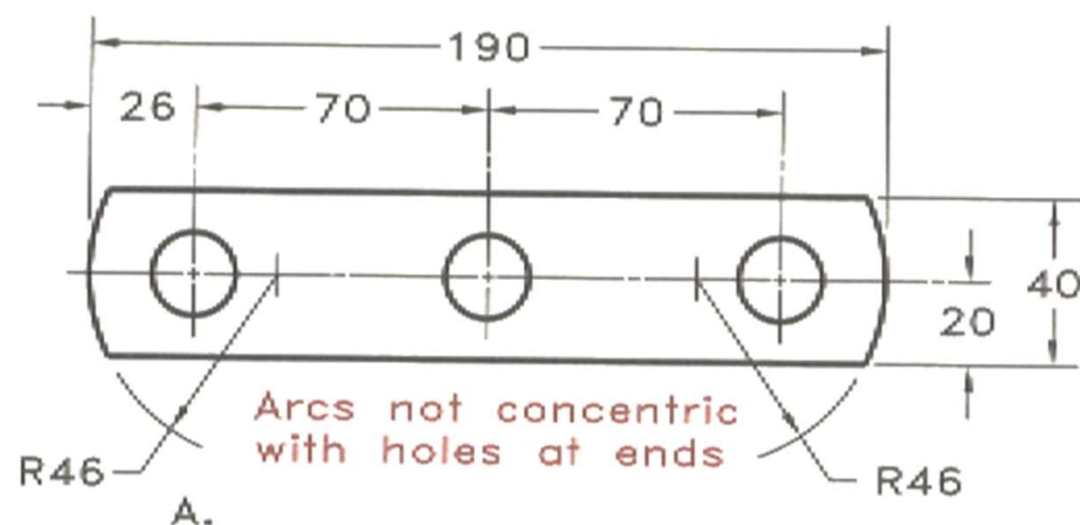
R25 R=25mm

B. CENTER-TO-CENTER
ACCEPTABLE

A Dimension objects having rounded ends from end-to-end and give the height.

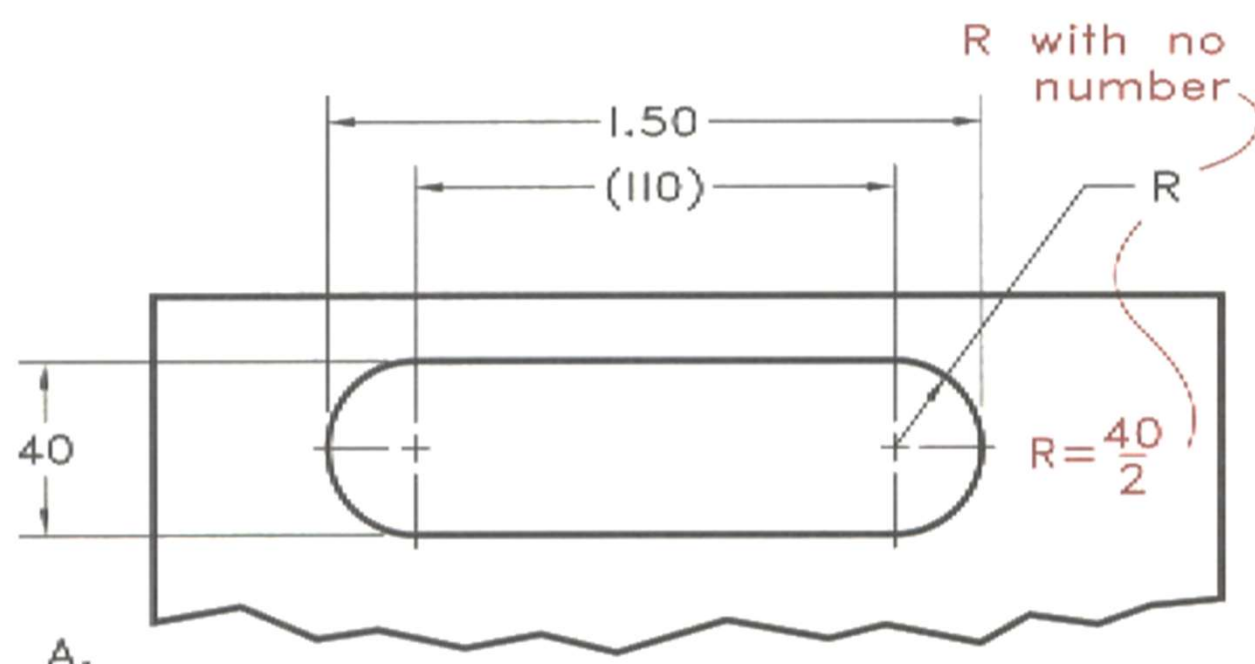
B A less desirable choice is to dimension the rounded ends from center-to-center and give the radius.

ROUNDED FEATURES

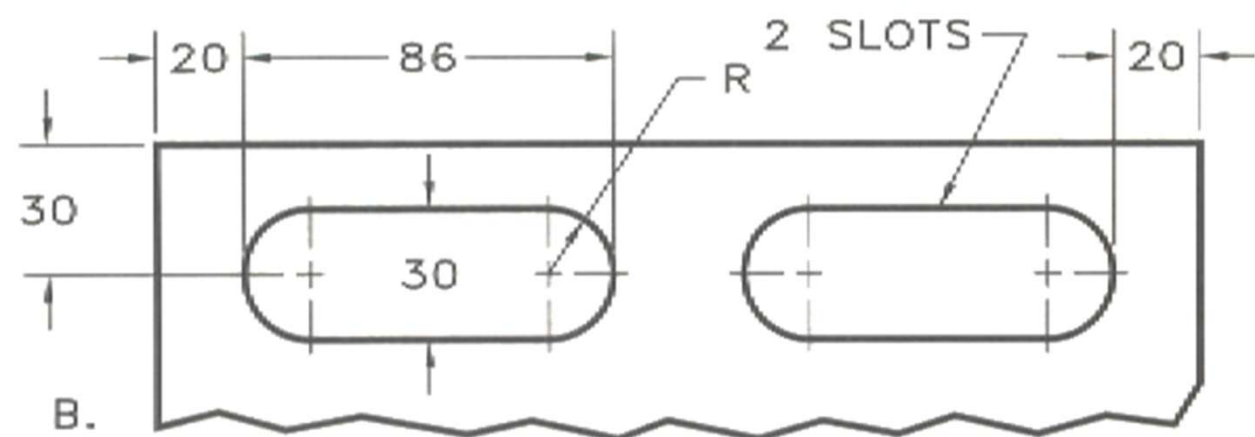


These drawings show how to dimension parts having (A) rounded ends that are not concentric with the holes and (B) rounds and cylinders.

DIMENSIONING SLOTS

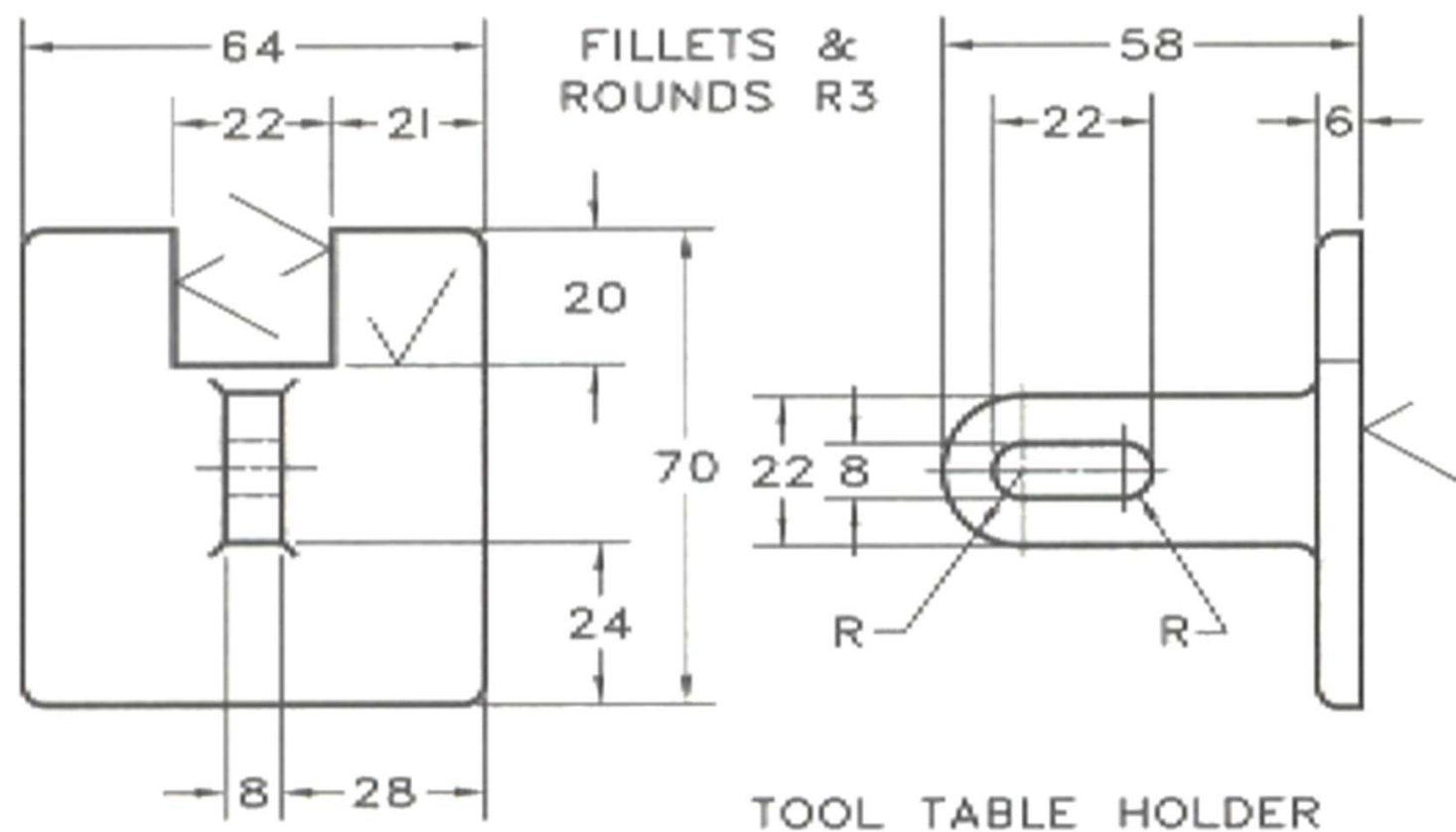


Dimension only one slot



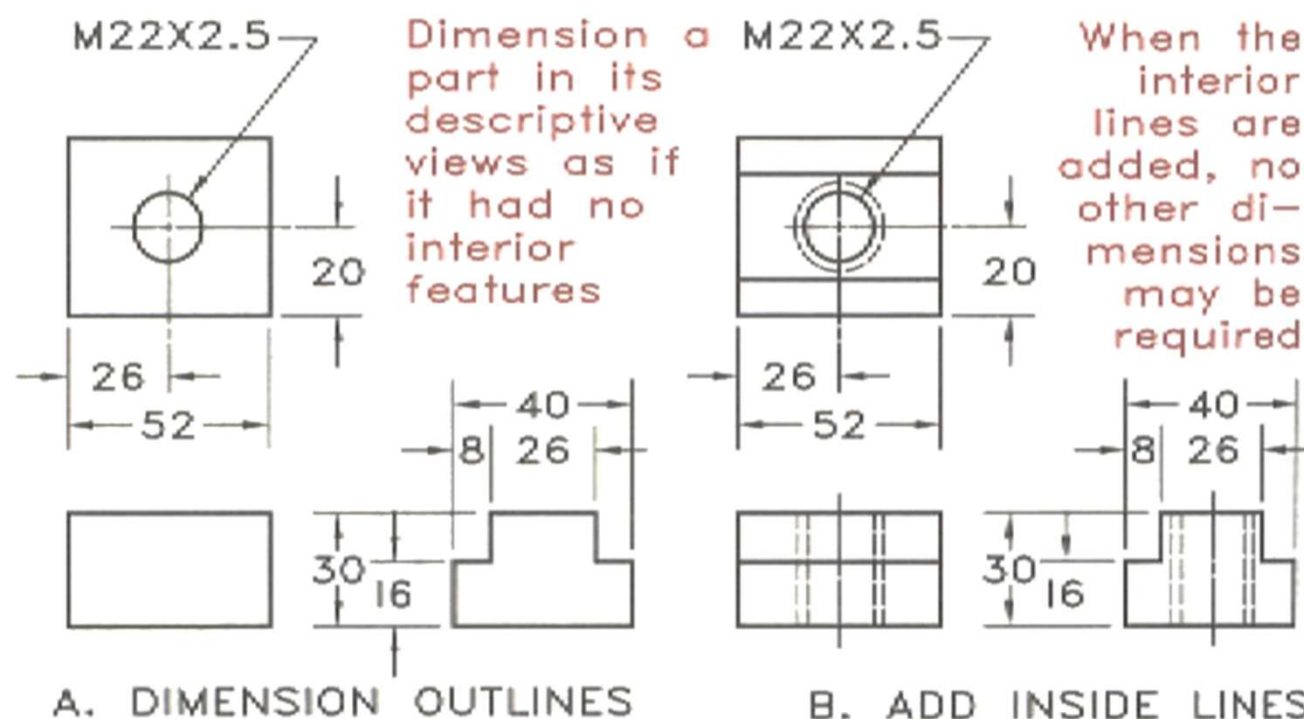
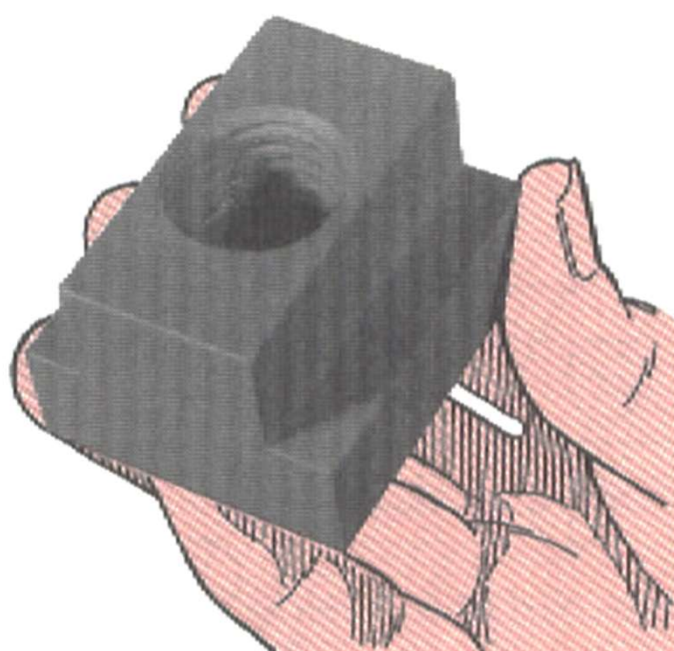
These drawings illustrate methods of dimensioning parts that have (A) one slot and (B) more than one slot.

DIMENSIONING ARCS AND SLOTS



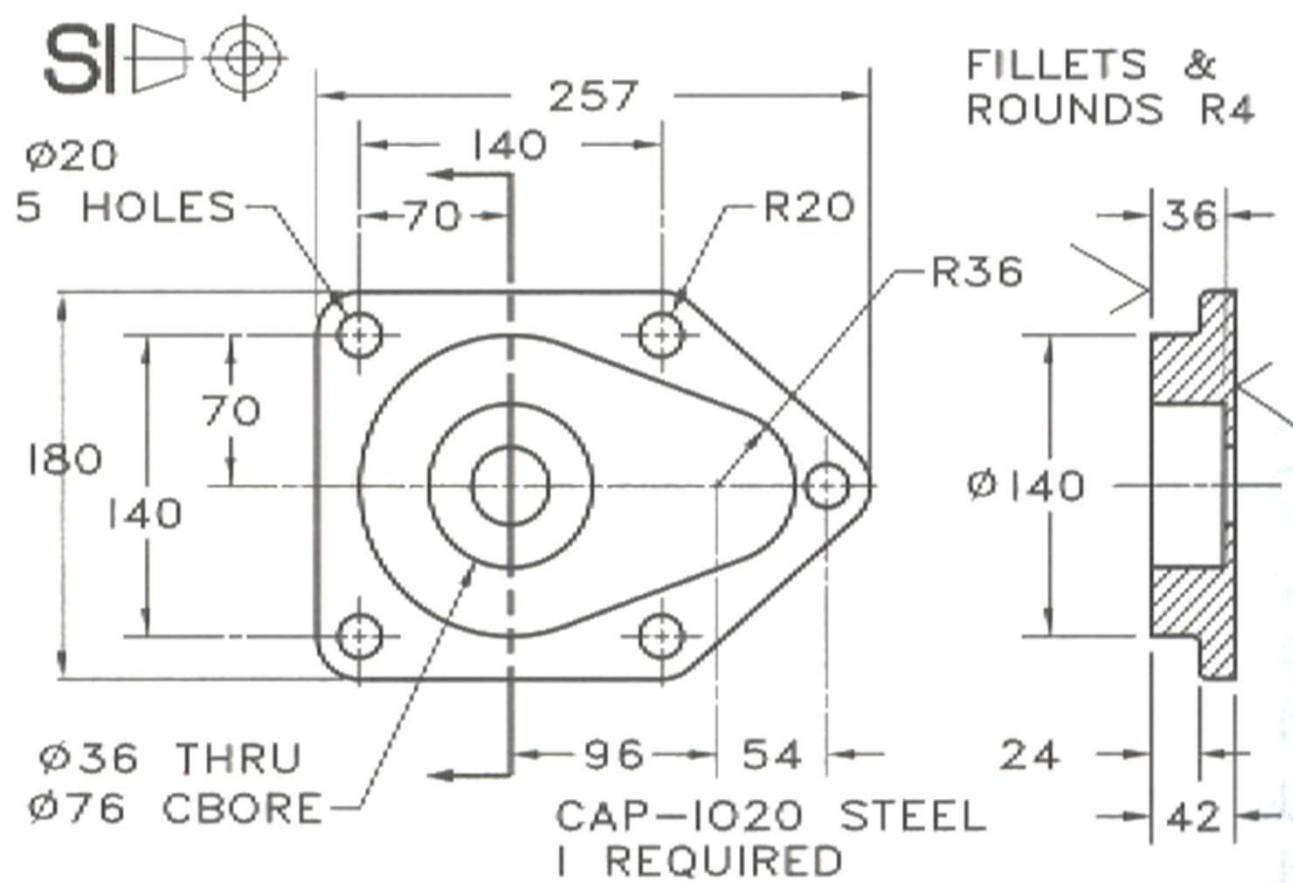
This dimensioned part has both slots and arcs.

OUTLINE DIMENSIONING



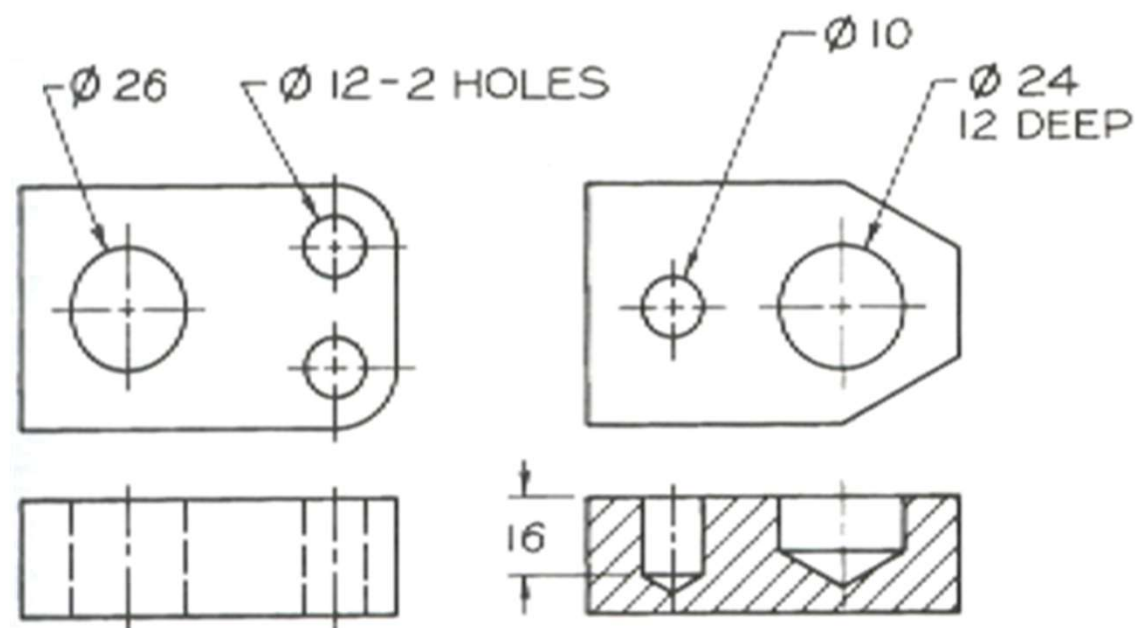
Outline dimensioning is the placement of dimensions on views as if they had no internal lines. Practice in applying this concept will help you place dimensions on their most descriptive views.

DIMENSIONED PART



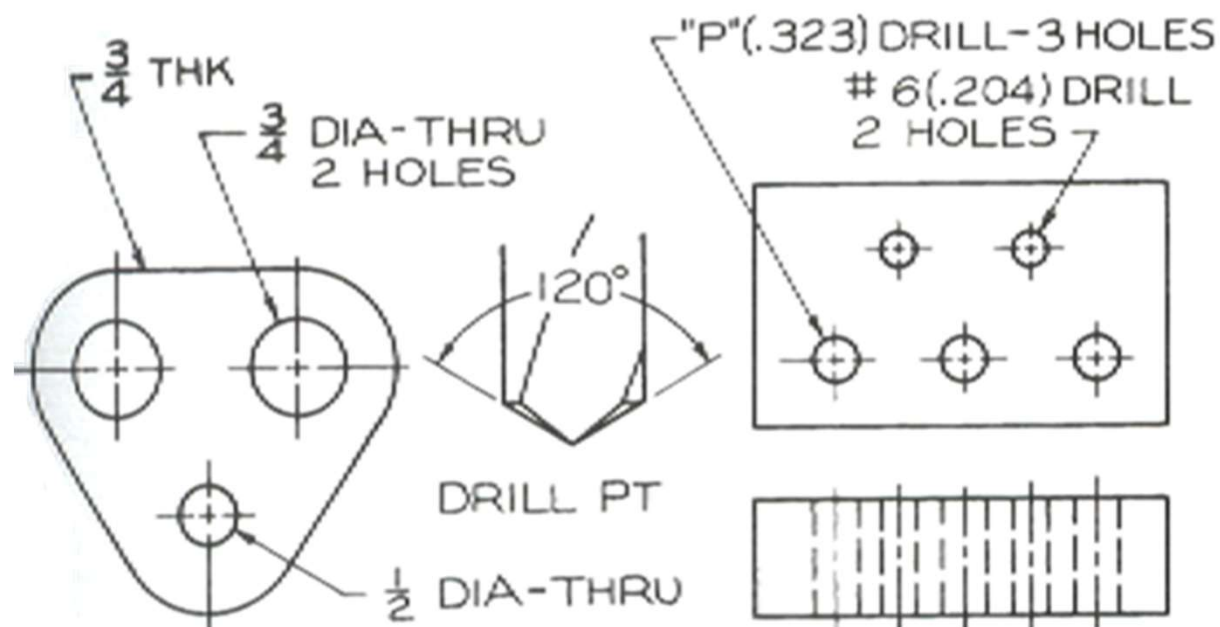
This drawing illustrates use of the outline method to dimension the cap in its most descriptive views.

DRILLED HOLES



A.

B.



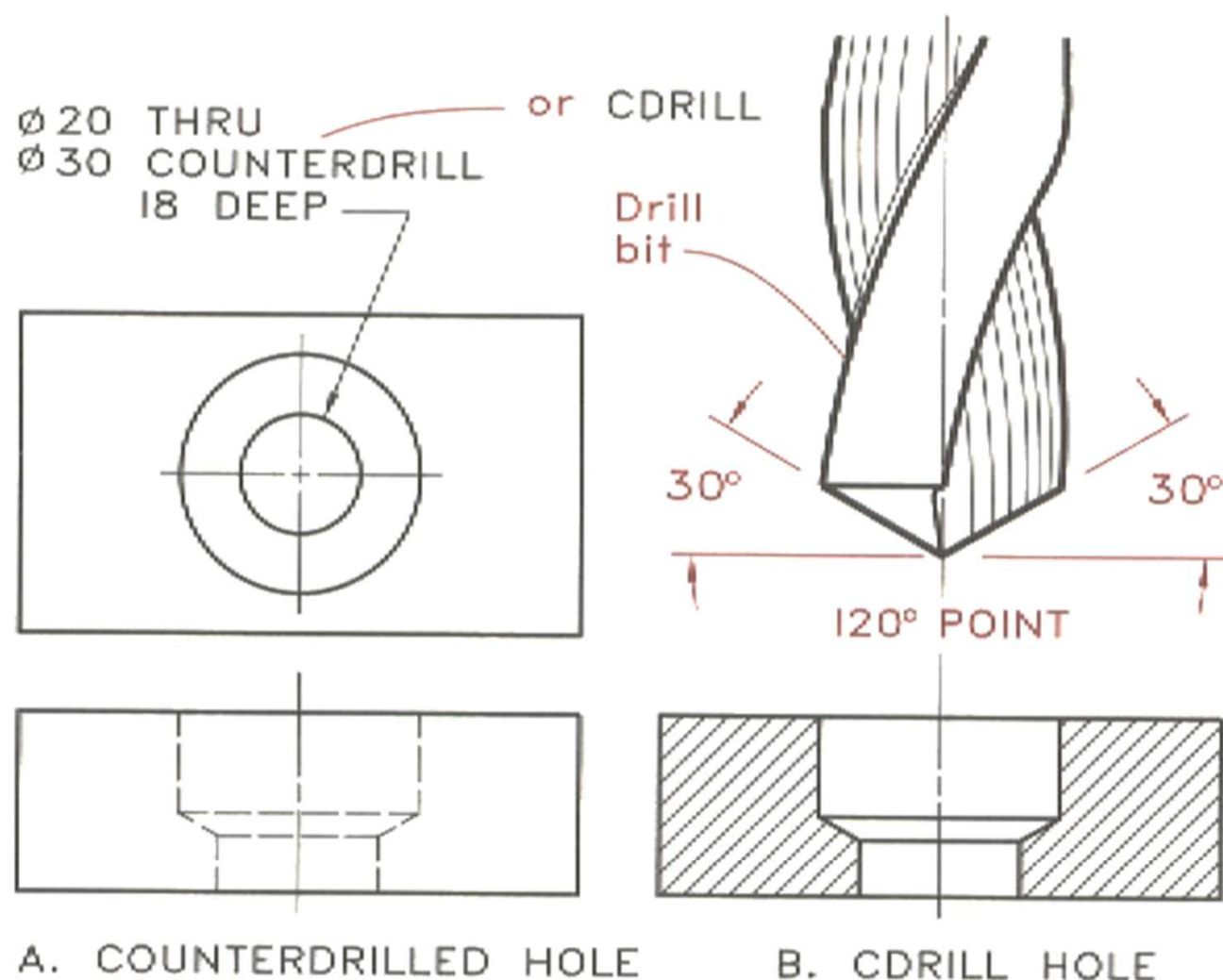
C.

D.

A and B You may dimension cylindrical holes by either of these methods.

C and D When you use only one view, you have to note the THRU holes or specify their depths.

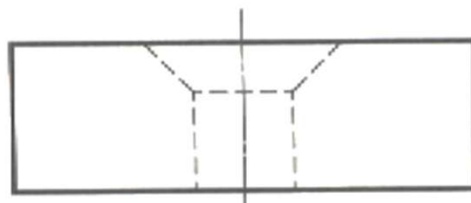
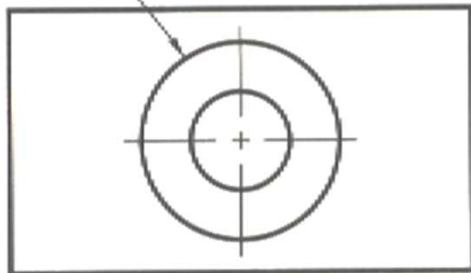
COUNTERDRILLED HOLES



Counterdrilling notes give the specifications for drilling a larger hole inside a smaller hole. Do not dimension the 120° angle, because it is a byproduct of the drill point. Noting the counterdrill with a leader from the circular view is preferable.

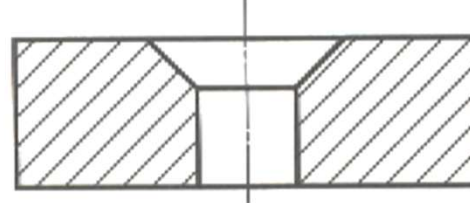
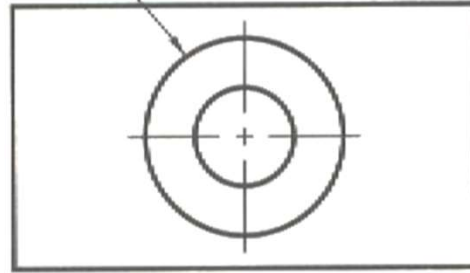
COUNTERSINK NOTES

Ø 32-82° COUNTER-
SINK TO Ø 56



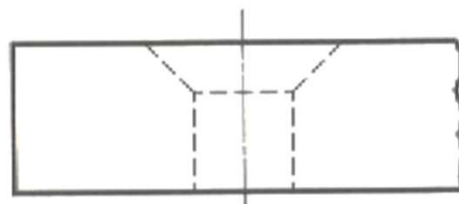
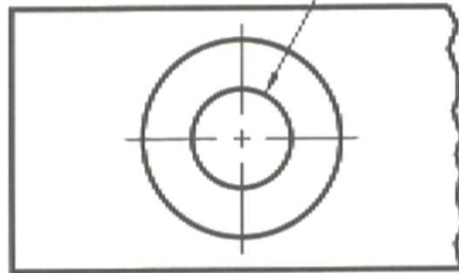
A. BY NOTE

Ø 20 THRU √ 82°
TO Ø 36 X 90°

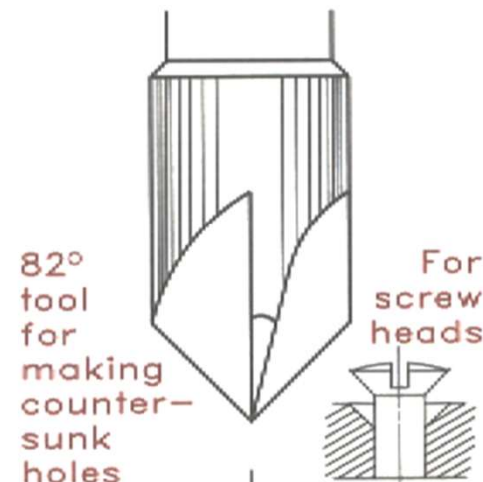


B. BY NOTE & SYMBOL

Ø .50-82° CSK
Ø 1.00-3 HOLES
EQ SPACED



C. MULTIPLE HOLES



D. COUNTERSINK TOOL

SPOTFACED HOLES

A.

Ø20 THRU
Ø38 SPOTFACE

or SF

Noted
by
symbols

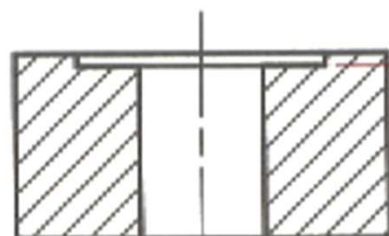
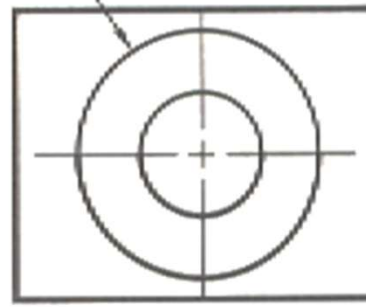
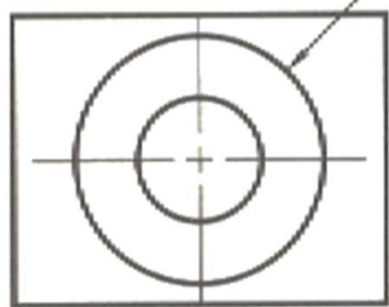
Ø16 THRU
└─┘ Ø34 ▽ 3

Noted
by
words

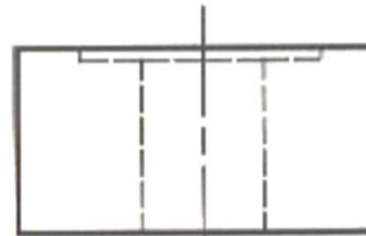


Spotfacing
tool

Depth left to
shop; purpose
is to smooth
the surface

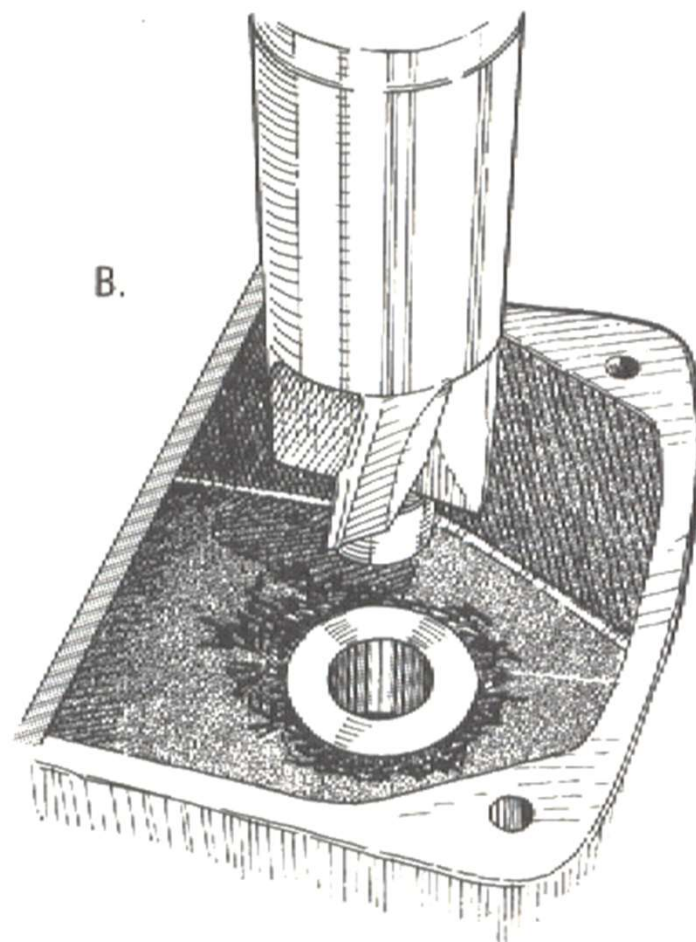


SECTION



VIEW

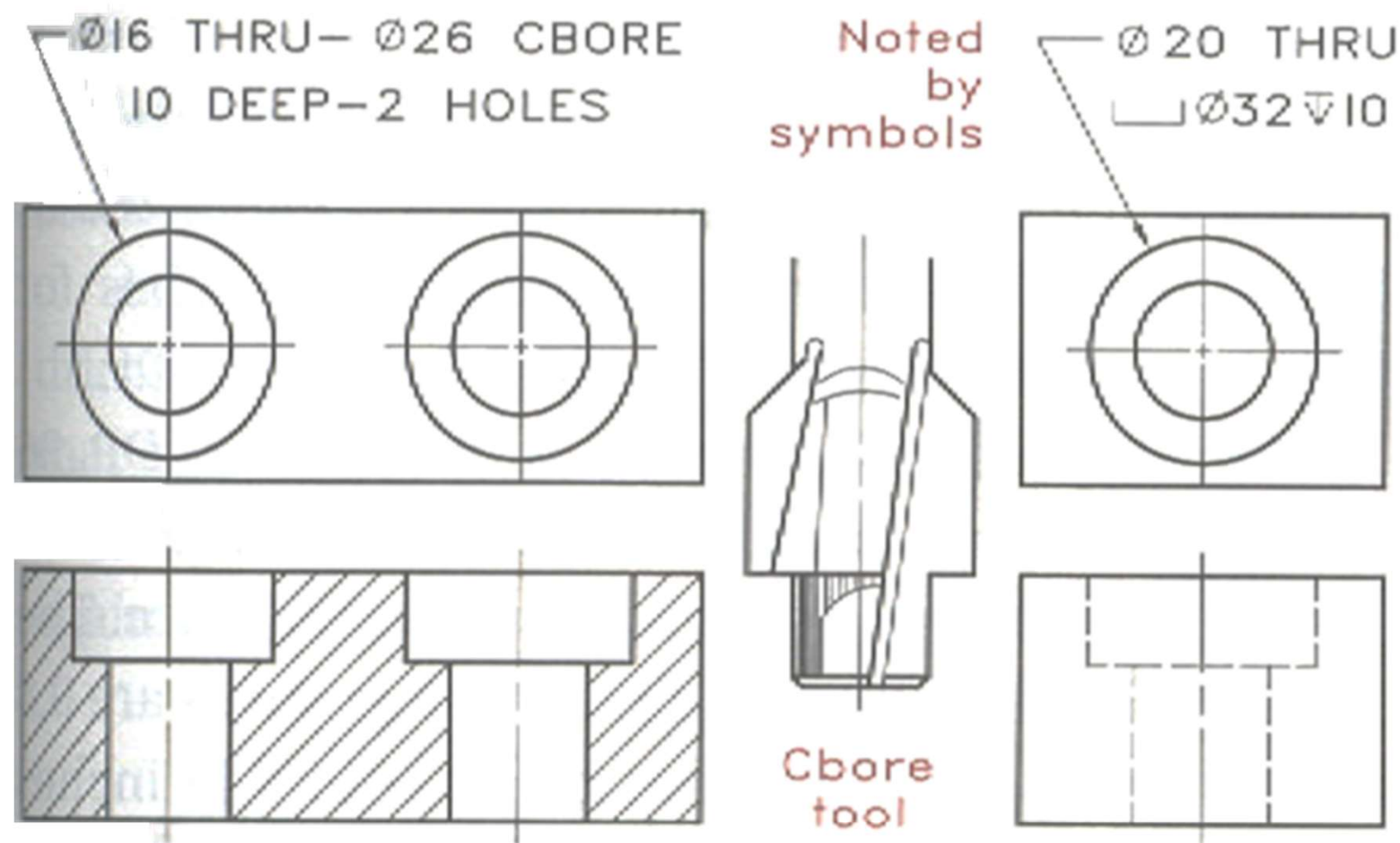
B.



A Spotfacing is the smoothing of the surface for contact with a washer or a nut or bolt; dimension for spotfacing as shown.

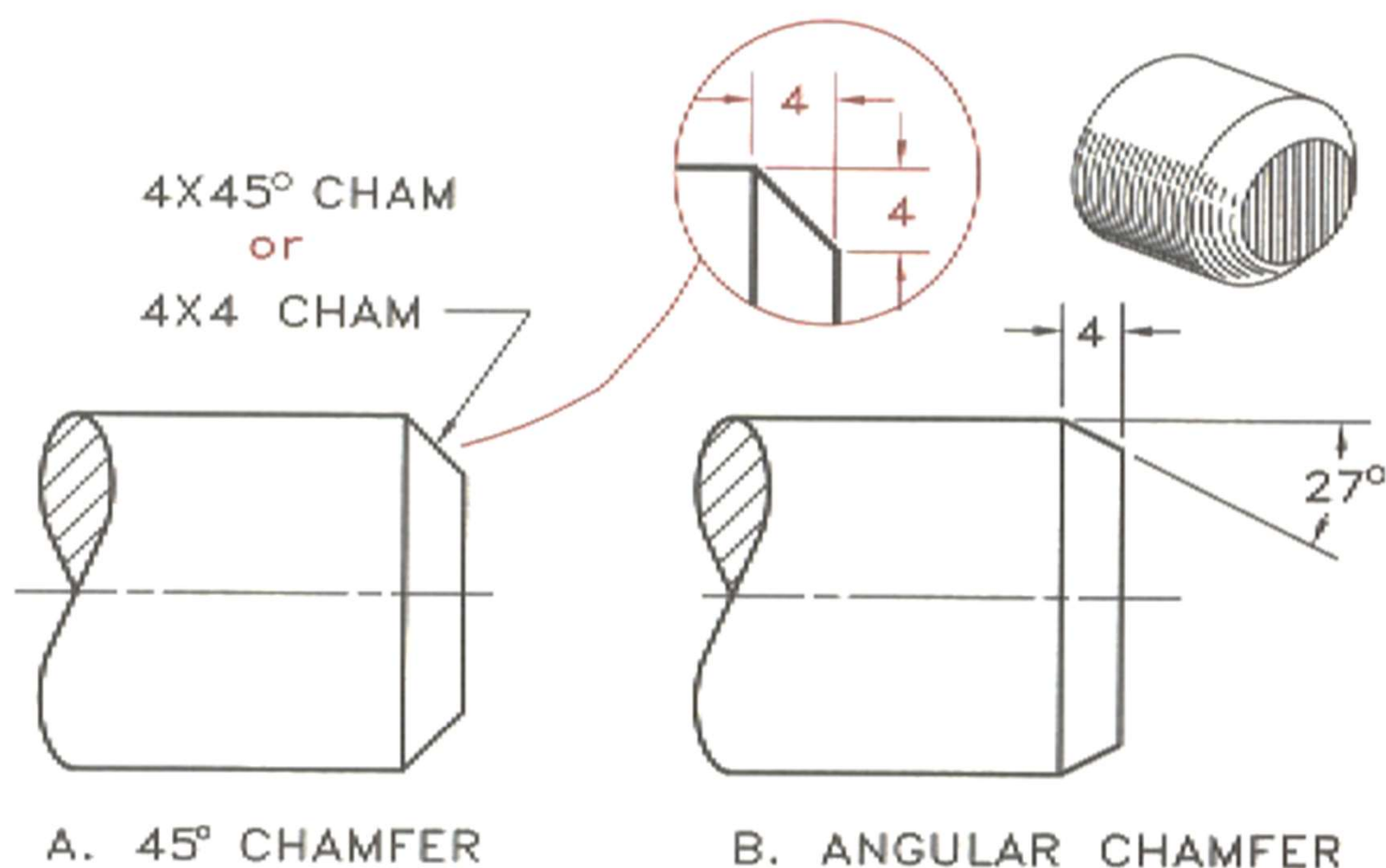
B This spotfacing tool is used to finish the cylindrical boss to provide a smooth seat for a bolt head.

COUNTERBORED HOLES



Counterbored holes are similar to counter-drilled holes but have flat bottoms instead of tapered sides. Dimension them as shown.

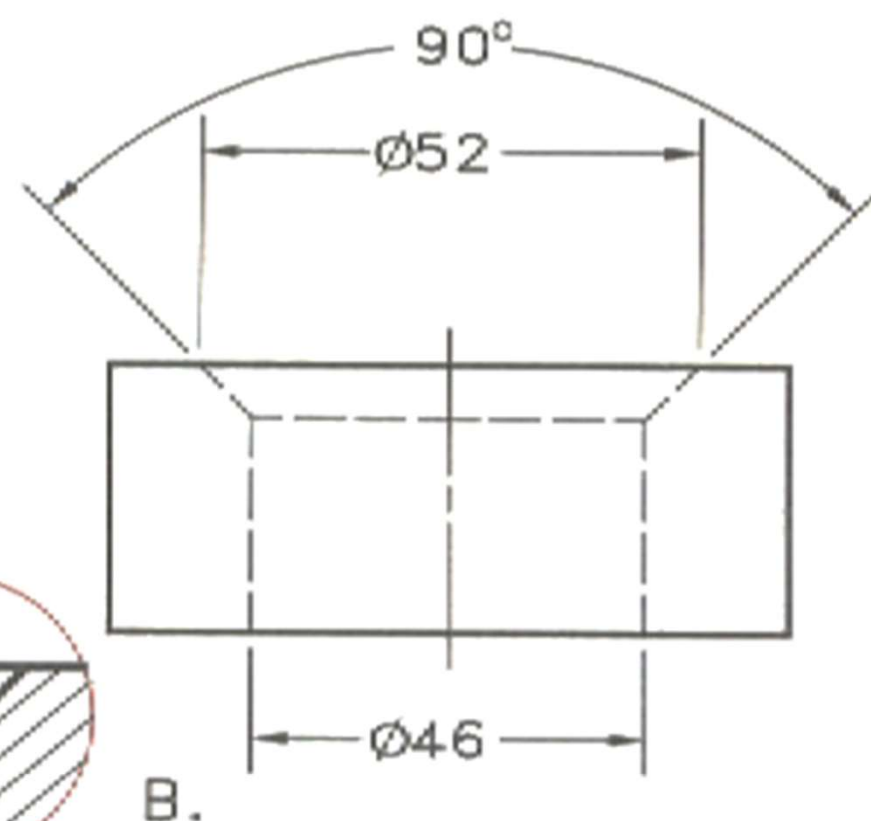
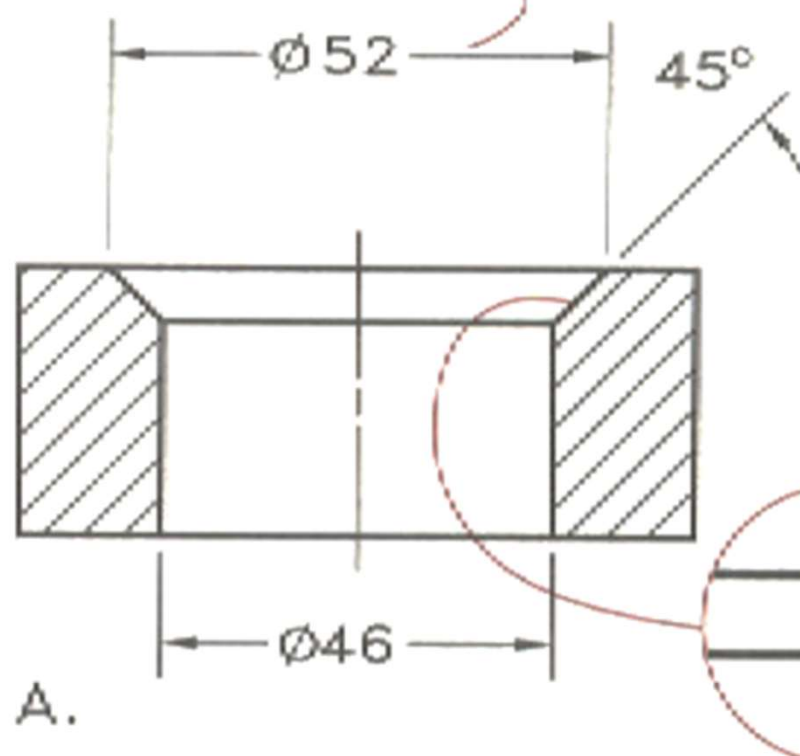
CHAMFERS



- A** Dimension 45° chamfers by one of the methods shown.
- B** Dimension chamfers of all other angles as shown.

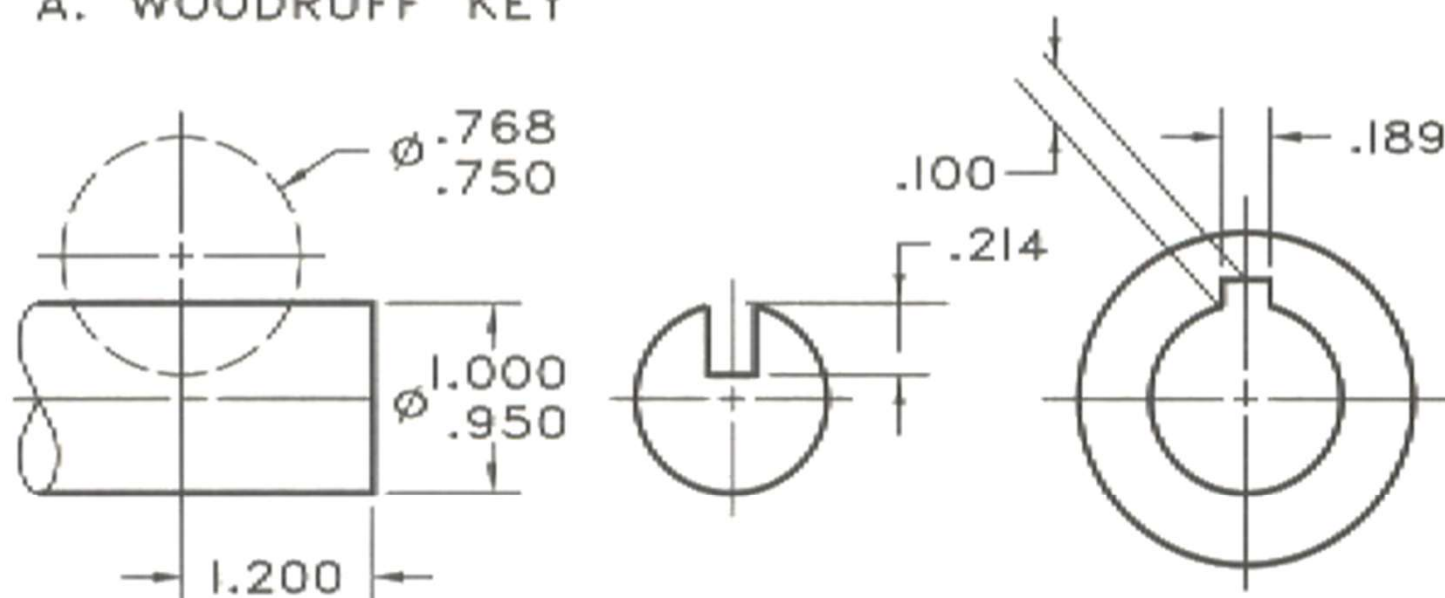
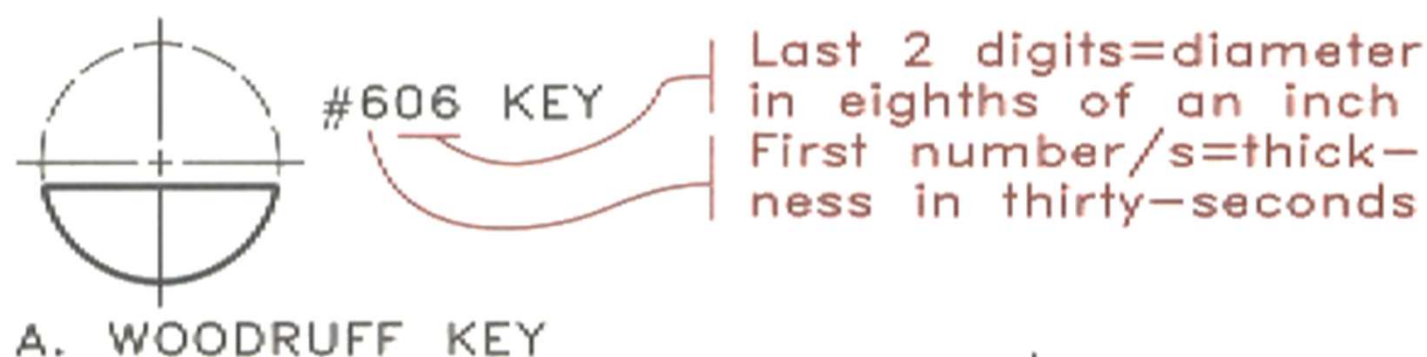
INSIDE CHAMFERS

Dimension DIA of
large circle



Dimension chamfers on the insides of cylinders as shown.

WOODRUFF KEYS



B. KEYSEATS: WOODRUFF #606 KEY

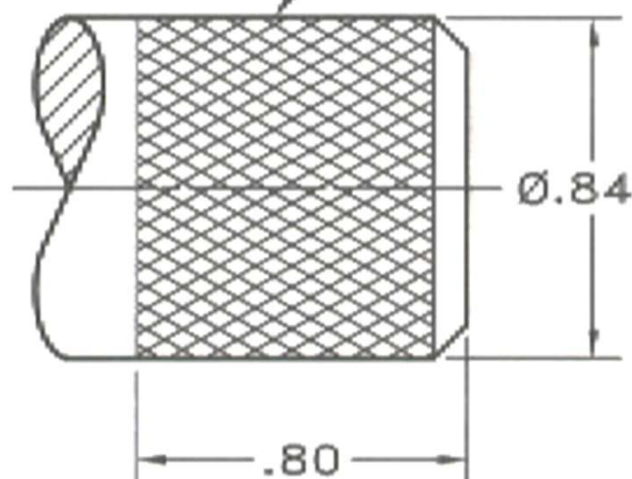
These drawings show methods of dimensioning (A) Woodruff keys and (B) keyways used to hold a part on a shaft. Appendix 29 gives their tables of sizes.

KNURL NOTES

DP=Diametral Pitch
A 64 DP on a 1" DIA
will have 64 teeth on
the circumference

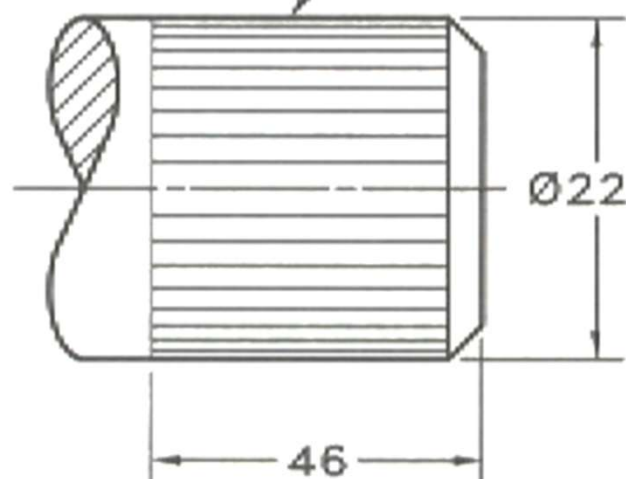
P=pitch=0.8 mm
(distance between
grooves)

96 DP DIAMOND
KNURL
(inches)



A. DIAMOND KNURL

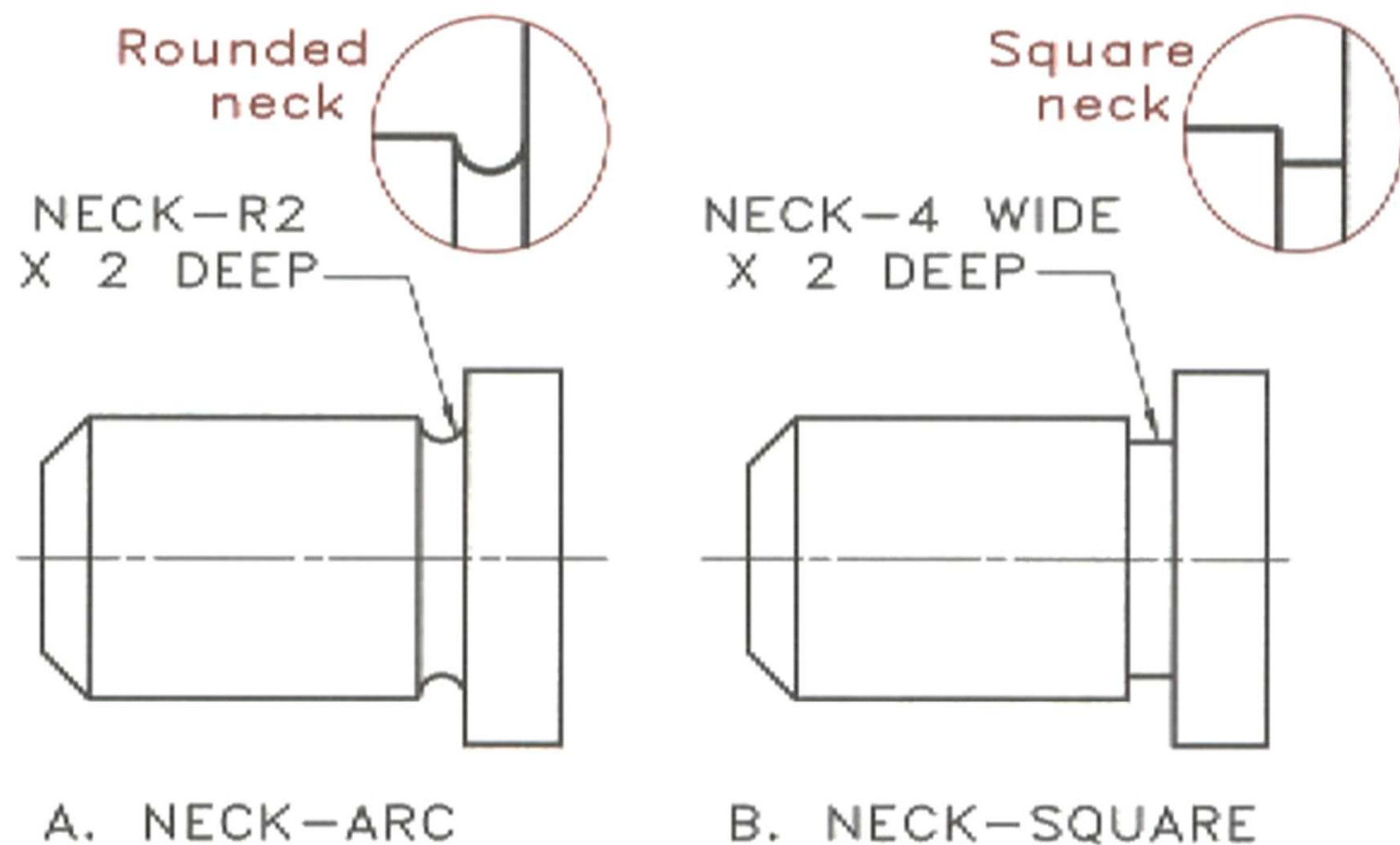
P0.8 STRAIGHT
KNURL
(millimeters)



B. STRAIGHT KNURL

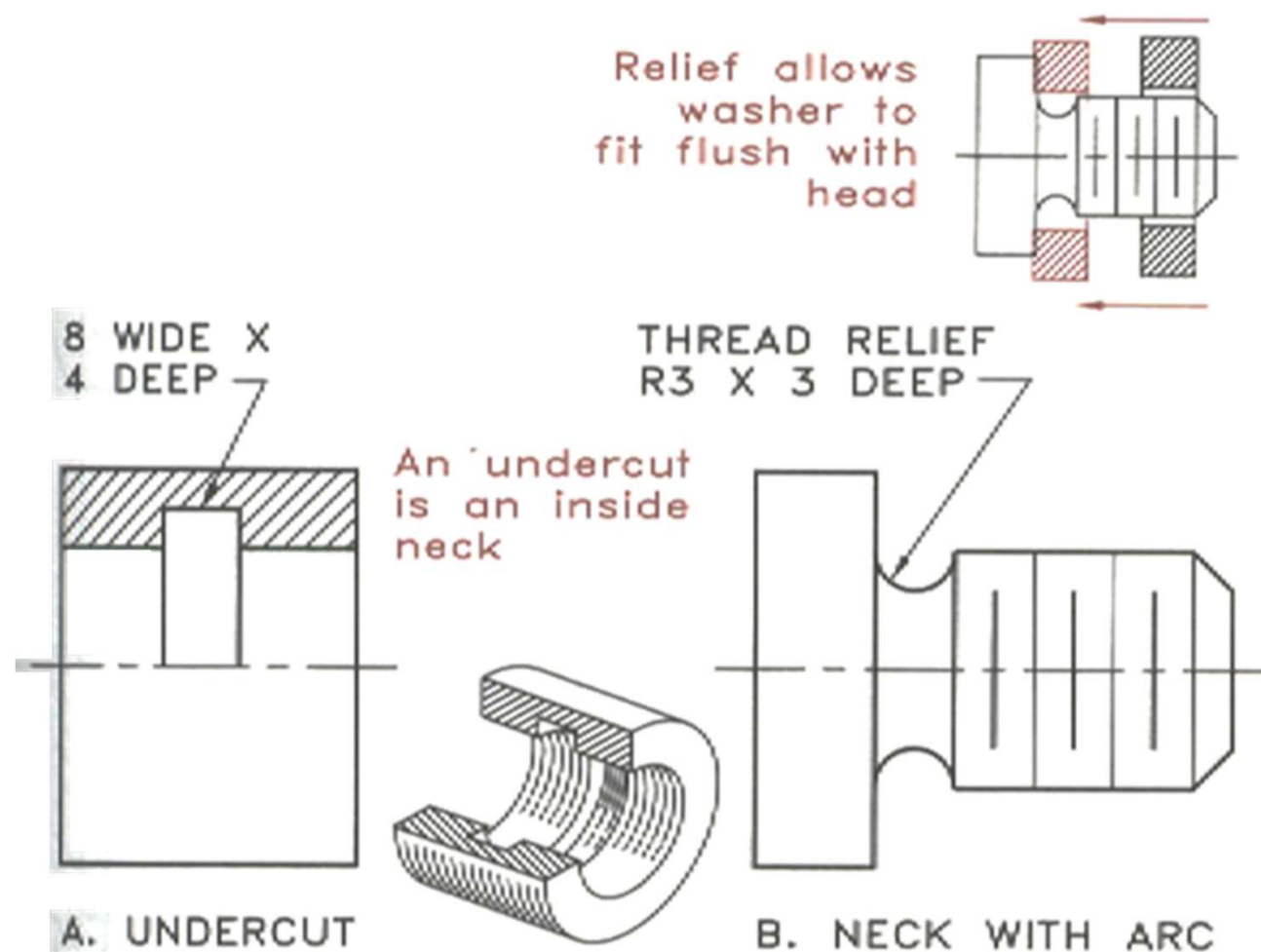
The diamond knurl has a diametral pitch, DP, of 96 and the straight knurl has a linear pitch, P, of 0.8 mm. Pitch is the distance between the grooves on the circumference.

NECKS



Necks are recesses cut in cylinders, with rounded or square bottoms, usually at the intersections of concentric cylinders. Dimension necks as shown.

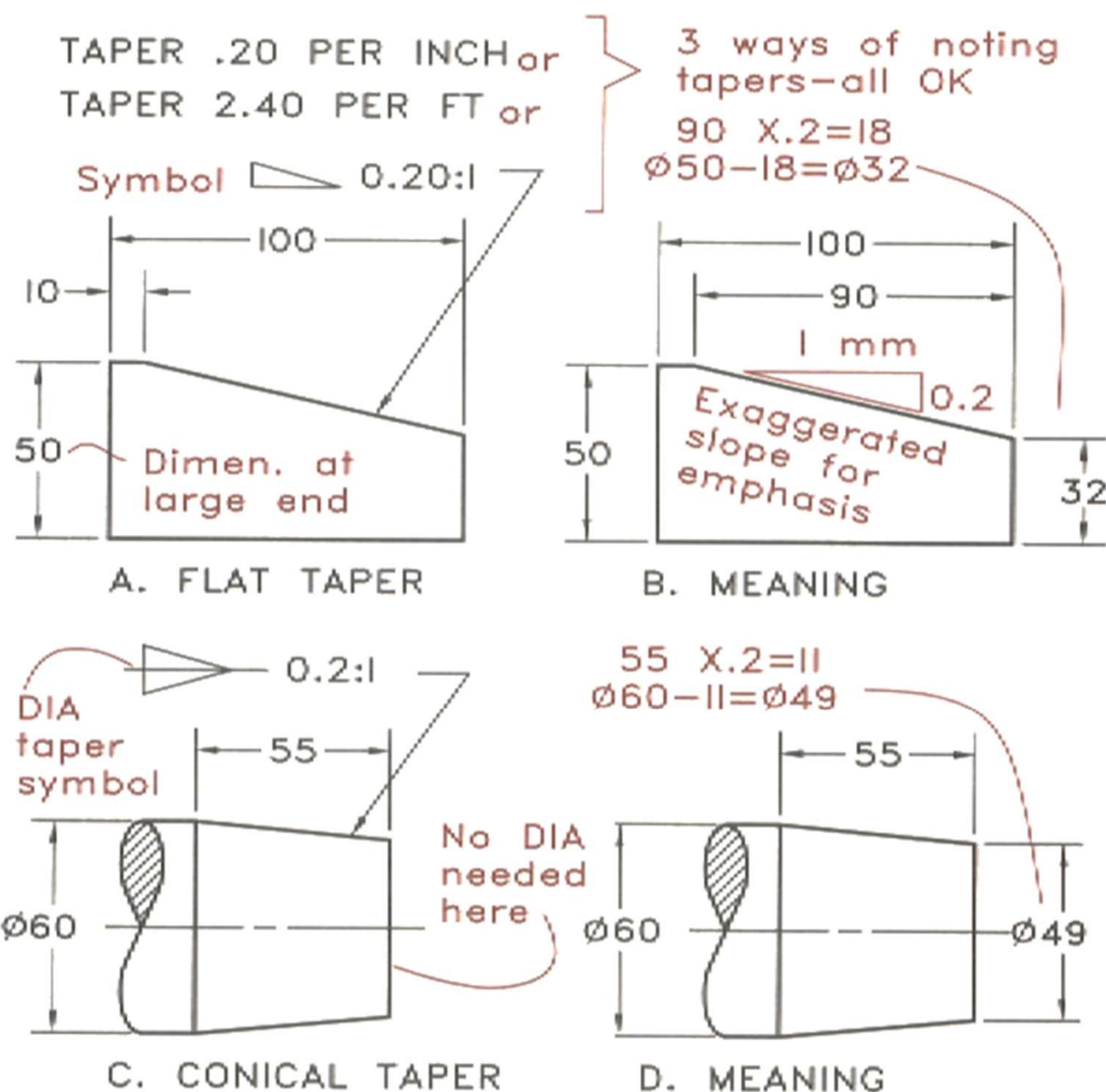
UNDERCUT AND THREAD RELIEF NECKS



A An undercut is a groove cut inside a cylinder.

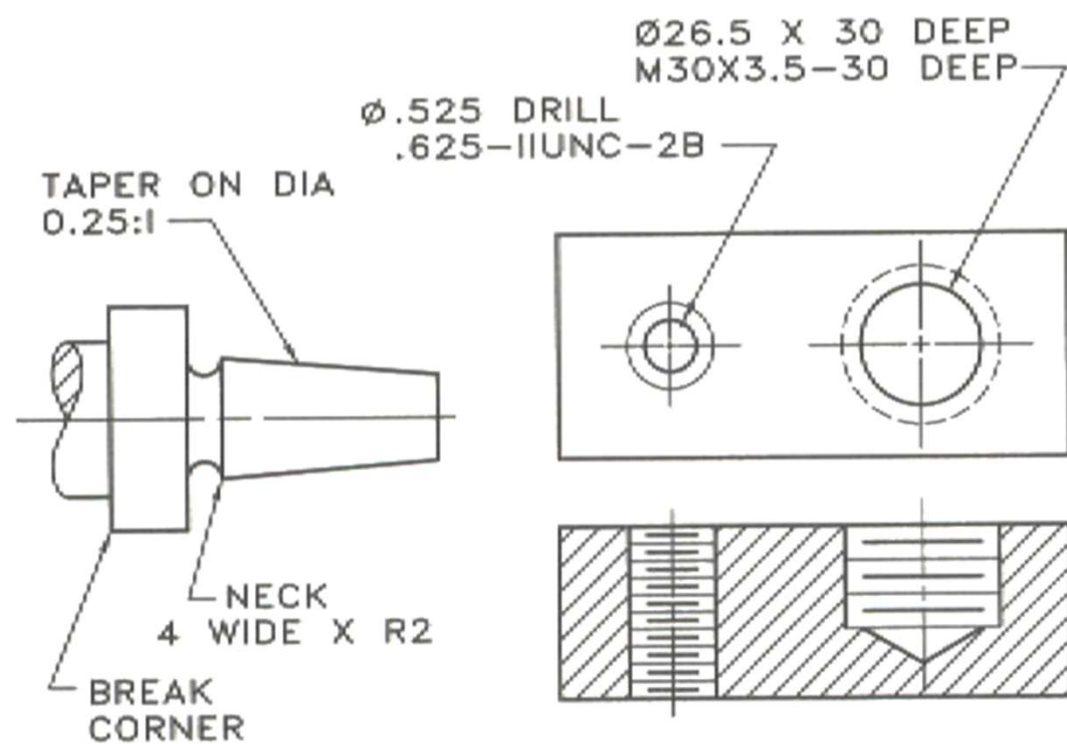
B A thread relief is a groove cut at the end of a thread to improve the screw's assembly. Dimension both types of necks as shown.

TAPER NOTES



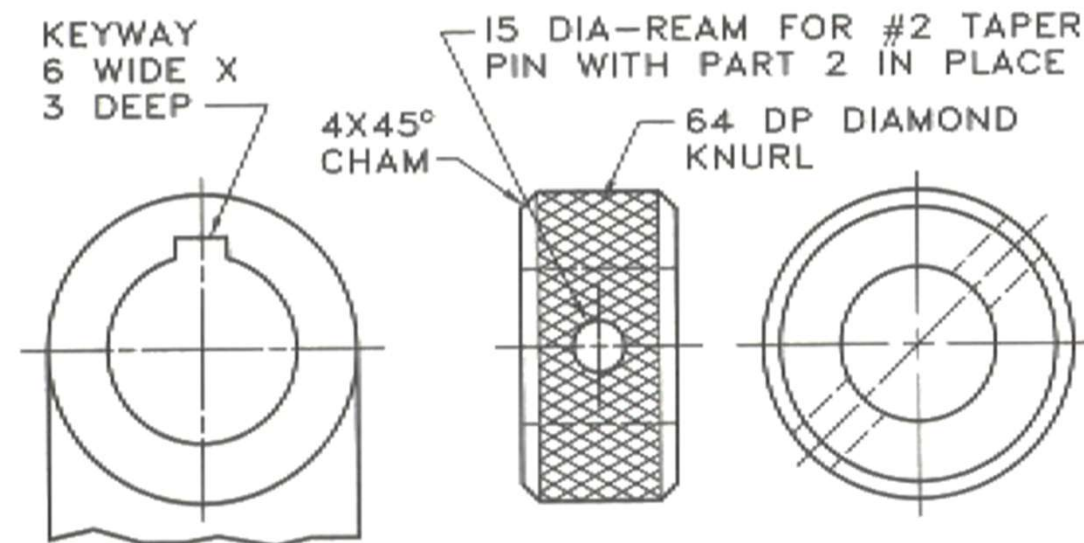
Tapers may be specified for either flat or conical surfaces: dimensioning and interpretation for (A and B) a flat taper and (C and D) for a conical taper.

MISCELLANEOUS NOTES



A. GENERAL NOTES

B. TAP DRILL & THREADS



C. KEYWAY NOTE

D. TAPER, KNURL & CHAMFER NOTES

A The notes for this part indicate a neck, a taper, and a break corner, which is a slight round to remove sharpness from a corner.

B Threaded holes are sometimes dimensioned by giving the tap drill size in addition to the thread specifications, but selection of the tap drill size usually is left to the shop.

C This note is for dimensioning a keyway.

D The notes for this collar call for knurling, chamfering, and drilling for a #2 taper pin.